Name: .... Date: .... Period: ....

Additional Practice ACC7.4.12

1. Select *all* of the ordered pairs (x, y) that are solutions to the linear equation $3x - 2y = 4$.

☑ **A.** $(0, -2)$

☐ **B.** $(2, -1)$

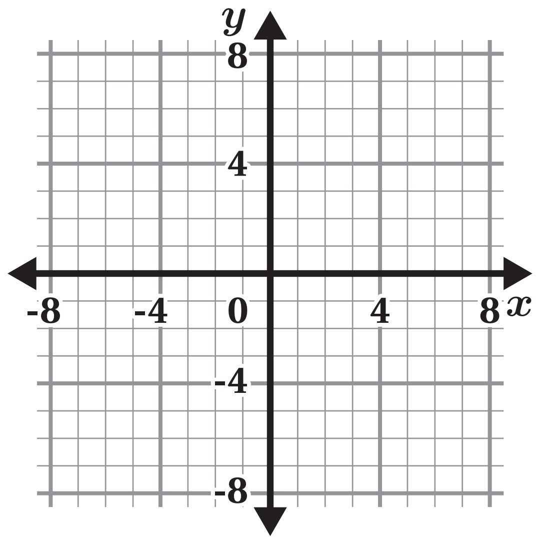
☑ **C.** $(-2, -5)$

☐ **D.** $(-4, 6)$

☑ **E.** $(4, 4)$

2. The graphs of a linear equation passes through the points $(-1, 2)$ and $(2, 8)$.

Select all the points that are also solutions to this equation. Use the graph if it helps with your thinking.



☐ **A.** $(0, -4)$

☑ **B.** $(1, 6)$

☑ **C.** $(-1, 2)$

☐ **D.** $(-2, 3)$

☑ **E.** $(-4, -4)$

3. Match each equation with its three solutions.

Equation	Solutions
a. $2x + y = 5$	c $(9, 3), (5, -1), (8, 2)$
b. $y = \frac{1}{2}x - 1$	d $(3, 2), (0, 0), (-6, -4)$
c. $x - y = 6$	a $(1, 3), (-2, 9), (3, -1)$
d. $2x = 3y$	e $(-4, -1), (4, 1), (16, 4)$
e. $y = \frac{1}{4}x$	b $(2, 0), (4, 1), (-2, -3)$

Name: .... Date: .... Period: ....

4. Kiran is determining if the ordered pair $(2, -1)$ is a solution to the equation $3x + y = 5$. His work is shown. Is he correct? Explain your thinking.

Kiran's work:

$3x + y = 5$

$3(-1) + 2 = 5$

$-3 + 2 = 5$

$-1 \neq 5$

The point $(2, -1)$ is *not* a solution to the equation.

Kiran's work is incorrect. *Explanations vary.* He substituted the x-value of 2 in for the y-value of the equation and the y-value of -1 into the x-value of the equation.

5. Determine whether the following statement is *true* or *false*. Explain your thinking.

The ordered pairs $(10, 38)$, $(-12, -17)$, and $(14, 45)$ all lie on the line whose equation is $y = \frac{5}{2}x + 13$.

False. *Explanations vary.* Substituting $(14, 45)$ into the equation makes a false statement.

$38 = \frac{5}{2}(10) + 13$	$-17 = \frac{5}{2}(-12) + 13$	$45 = \frac{5}{2}(14) + 13$
$38 = 25 + 13$	$-17 = -30 + 13$	$45 = 35 + 13$
$38 = 38$ True	$-17 = -17$ True	$45 = 48$ False

Problems 6–7: Complete each table.

6. $y = -\frac{4}{5}x$

x	y
-10	**8**
-5	4

7. $x + 2y = 7$

x	y
-1	**4**
3	2

Additional Practice | Answer Key

Name: ________ Date: ________ Period: ________

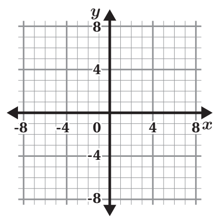
Additional Practice ACC7.4.12

1. Select *all* of the ordered pairs (x, y) that are solutions to the linear equation $3x - 2y = 4$.

- ☑ **A.** $(0, -2)$
- ☐ **B.** $(2, -1)$
- ☑ **C.** $(-2, -5)$
- ☐ **D.** $(-4, 6)$
- ☑ **E.** $(4, 4)$

2. The graphs of a linear equation passes through the points $(-1, 2)$ and $(2, 8)$.

Select all the points that are also solutions to this equation. Use the graph if it helps with your thinking.



- ☐ **A.** $(0, -4)$
- ☑ **B.** $(1, 6)$
- ☑ **C.** $(-1, 2)$
- ☐ **D.** $(-2, 3)$
- ☑ **E.** $(-4, -4)$

3. Match each equation with its three solutions.

Equation	Solutions
a. $2x + y = 5$	c $(9, 3), (5, -1), (8, 2)$
b. $y = \frac{1}{2}x - 1$	d $(3, 2), (0, 0), (-6, -4)$
c. $x - y = 6$	a $(1, 3), (-2, 9), (3, -1)$
d. $2x = 3y$	e $(-4, -1), (4, 1), (16, 4)$
e. $y = \frac{1}{4}x$	b $(2, 0), (4, 1), (-2, -3)$

Name: ________ Date: ________ Period: ________

4. Kiran is determining if the ordered pair $(2, -1)$ is a solution to the equation $3x + y = 5$. His work is shown. Is he correct? Explain your thinking.

Kiran's work:

$3x + y = 5$

$3(-1) + 2 = 5$

$-3 + 2 = 5$

$-1 \neq 5$

The point $(2, -1)$ is *not* a solution to the equation.

Kiran's work is incorrect. *Explanations vary.* He substituted the x-value of 2 in for the y-value of the equation and the y-value of -1 into the x-value of the equation.

5. Determine whether the following statement is *true* or *false*. Explain your thinking.

The ordered pairs $(10, 38)$, $(-12, -17)$, and $(14, 45)$ all lie on the line whose equation is $y = \frac{5}{2}x + 13$.

False. *Explanations vary.* Substituting $(14, 45)$ into the equation makes a false statement.

$38 = \frac{5}{2}(10) + 13$	$-17 = \frac{5}{2}(-12) + 13$	$45 = \frac{5}{2}(14) + 13$
$38 = 25 + 13$	$-17 = -30 + 13$	$45 = 35 + 13$
$38 = 38$ True	$-17 = -17$ True	$45 = 48$ False

Problems 6–7: Complete each table.

6. $y = -\frac{4}{5}x$

x	y
-10	**8**
-5	4

7. $x + 2y = 7$

x	y
-1	**4**
3	2

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.A
2	2	8.EE.C.8.A
3	1	8.EE.C.8.A
4	2	8.EE.C.8.A
5	1	8.EE.C.8.A
6	1	8.EE.C.8.A
7	1	8.EE.C.8.A

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.4.13

Problems 1–5: Maria has $\$48$ to spend on flowers for her garden. A pack of pansy seeds costs $\$4$ each and a pack of petunia seeds costs $\$6$ each.

1. Complete the table to show some possible combinations of packs of pansies and packs of petunias that will cost a total of $\$48$.

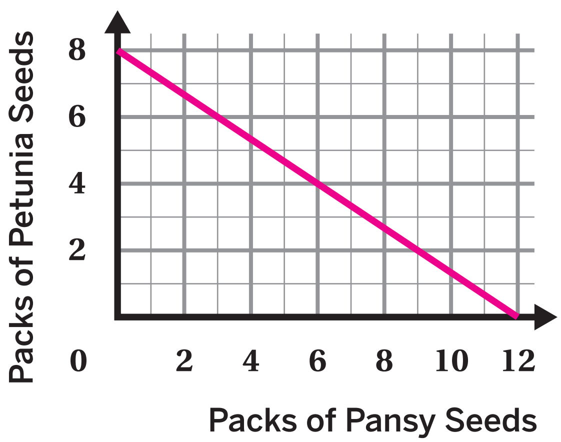
***Responses vary.* Sample shown in table.**

Number of Packs of Pansy Seeds	Number of Packs of Petunia Seeds
12	0
3	6
6	4

2. Write an equation that represents the relationship between the number of packs of pansy seeds, x, and the number of packs of petunia seeds, y, that Maria can buy.

$4x + 6y = 48$ (or equivalent)

3. Graph all possible combinations of packs of pansy seeds and packs of petunia seeds that will cost a total of $\$48$.



4. What is the slope of the line? What does it tell you about this situation?

$-\frac{2}{3}$. *Responses vary.* It means for every 2 packs of petunia seeds taken away, she can buy 3 packs of pansy seeds.

5. What are the x- and y-intercepts of the line? What do they represent in this situation?

The x-intercept is $(12, 0)$ and it represents the number of packs of pansy seeds Maria can buy for $\$48$ if she doesn't buy any packs of petunia seeds. The y-intercept is $(0, 8)$, and it represents the number of packs of petunia seeds Maria can buy for $\$48$ if she doesn't buy any packs of pansy seeds.

Problems 6–8: Jada made bracelets and necklaces to sell at the craft show. She sold her bracelets for \$3 each and her necklaces for \$6 each. She earned \$96 in sales at the craft show.

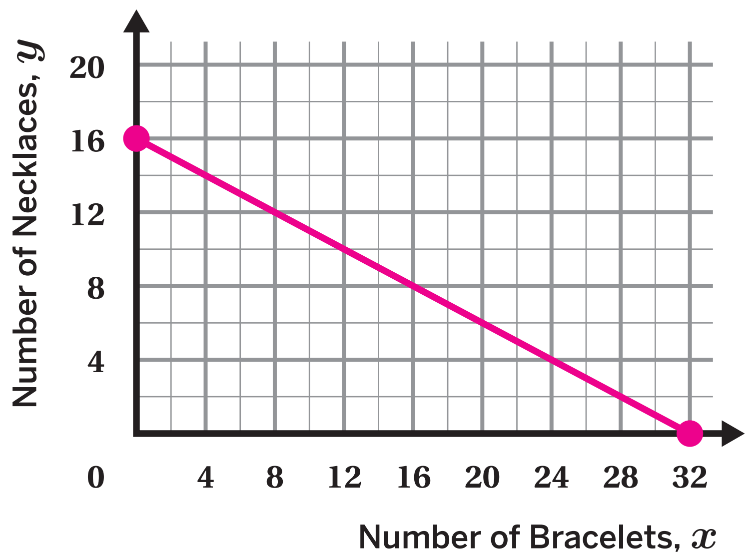
6. What are two possible combinations of the number of bracelets and necklaces sold that earn a total of \$96 in sales?

Responses vary. **Jada can sell 12 bracelets and 10 necklaces to earn \$96. Or, she can sell 10 bracelets and 11 necklaces to earn \$96.**

7. Write an equation that represents the relationship between the number of bracelets, x, and the number of necklaces, y, that Jada sold.

$3x + 6y = 96$ (or equivalent)

8. Graph this relationship on the coordinate plane.



9. What are the x- and y-intercepts of the line? What do they represent in this situation?

The x-intercept is $(32, 0)$ and it represents the number of bracelets Jada can sell to earn \$96 if she doesn't sell any necklaces. The y-intercept is $(0, 16)$, and it represents the number of necklaces Jada can sell to earn \$96 if she doesn't sell any bracelets.

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.4.13

Problems 1–5: Maria has $48 to spend on flowers for her garden. A pack of pansy seeds costs $4 each and a pack of petunia seeds costs $6 each.

1. Complete the table to show some possible combinations of packs of pansies and packs of petunias that will cost a total of $48.

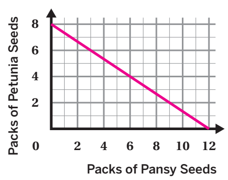
Responses vary. Sample shown in table.

Number of Packs of Pansy Seeds	Number of Packs of Petunia Seeds
12	0
3	6
6	4

2. Write an equation that represents the relationship between the number of packs of pansy seeds, x, and the number of packs of petunia seeds, y, that Maria can buy.

$4x + 6y = 48$ (or equivalent)

3. Graph all possible combinations of packs of pansy seeds and packs of petunia seeds that will cost a total of $48.



4. What is the slope of the line? What does it tell you about this situation?

$-\frac{2}{3}$. *Responses vary.* It means for every 2 packs of petunia seeds taken away, she can buy 3 packs of pansy seeds.

5. What are the x- and y-intercepts of the line? What do they represent in this situation?

The x-intercept is $(12, 0)$ and it represents the number of packs of pansy seeds Maria can buy for $48 if she doesn't buy any packs of petunia seeds. The y-intercept is $(0, 8)$, and it represents the number of packs of petunia seeds Maria can buy for $48 if she doesn't buy any packs of pansy seeds.

Name: ______ Date: ______ Period: ______

Problems 6–8: Jada made bracelets and necklaces to sell at the craft show. She sold her bracelets for $3 each and her necklaces for $6 each. She earned $96 in sales at the craft show.

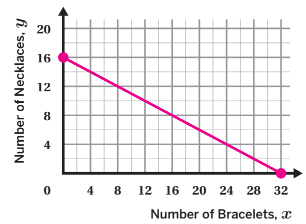
6. What are two possible combinations of the number of bracelets and necklaces sold that earn a total of $96 in sales?

Responses vary. Jada can sell 12 bracelets and 10 necklaces to earn $96. Or, she can sell 10 bracelets and 11 necklaces to earn $96.

7. Write an equation that represents the relationship between the number of bracelets, x, and the number of necklaces, y, that Jada sold.

$3x + 6y = 96$ (or equivalent)

8. Graph this relationship on the coordinate plane.



9. What are the x- and y-intercepts of the line? What do they represent in this situation?

The x-intercept is $(32, 0)$ and it represents the number of bracelets Jada can sell to earn $96 if she doesn't sell any necklaces. The y-intercept is $(0, 16)$, and it represents the number of necklaces Jada can sell to earn $96 if she doesn't sell any bracelets.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.A
2	2	8.EE.C.8.A
3	1	8.EE.C.8.A
4	2	8.EE.C.8.A
5	2	8.EE.C.8.A
6	1	8.EE.C.8.A
7	2	8.EE.C.8.A
8	1	8.EE.C.8.A
9	2	8.EE.C.8.A

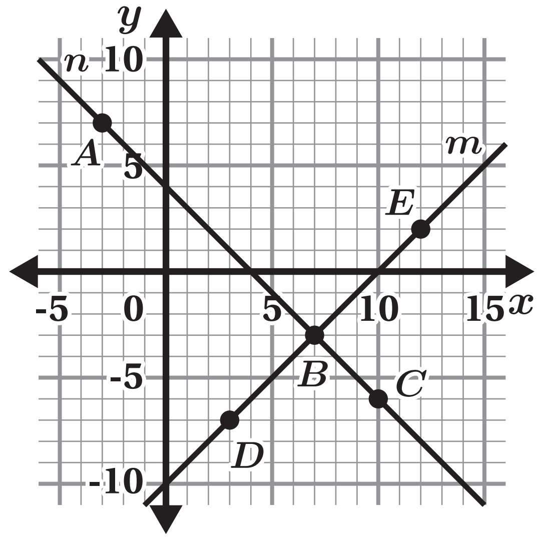
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.4.14

Problems 1–5: Here is a coordinate plane.



1. Determine which line represents this condition: The coordinates of each point have a sum of 4.

Line n

2. Determine which line represents this condition: The y-coordinate of each point is 10 less than the x-coordinate.

Line m

3. Select *all* the points whose coordinates have a sum of 4.

- ☑ **A.** Point *A*
- ☑ **B.** Point *B*
- ☑ **C.** Point *C*
- ☐ **D.** Point *D*
- ☐ **E.** Point *E*

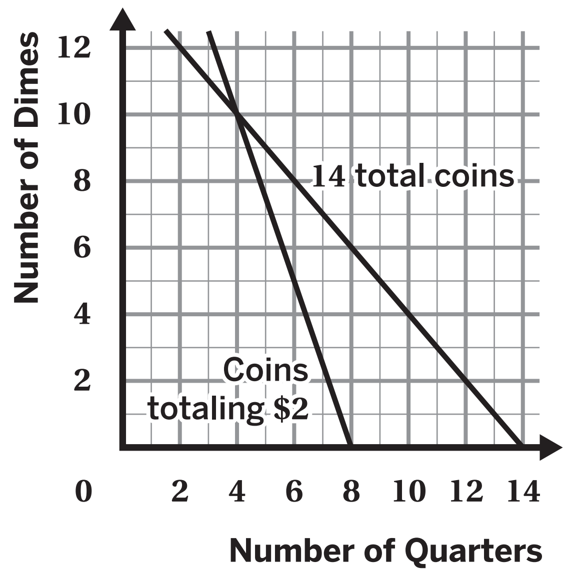
4. Select *all* the points whose y-coordinate is 10 less than the x-coordinate.

- ☐ **A.** Point *A*
- ☑ **B.** Point *B*
- ☐ **C.** Point *C*
- ☑ **D.** Point *D*
- ☑ **E.** Point *E*

5. Select *all* the points whose coordinates have a sum of 4 and whose y-coordinate is 10 less than the x-coordinate.

- ☐ **A.** Point *A*
- ☑ **B.** Point *B*
- ☐ **C.** Point *C*
- ☐ **D.** Point *D*
- ☐ **E.** Point *E*

Problems 6–8: On the coordinate plane shown, one line represents combinations of dimes and quarters that have a total value of $2. The other line represents combinations of dimes and quarters when the total number of coins is 14.



6. Select *all* the combinations of coins that have a total value of $2.

- ☐ **A.** 14 quarters and 0 dimes
- ☑ **B.** 4 quarters and 10 dimes
- ☐ **C.** 6 quarters and 8 dimes
- ☑ **D.** 8 quarters and 0 dimes
- ☐ **E.** 2 quarters and 12 dimes

7. Select *all* combinations of quarters and dimes that have a total of 14 coins.

- ☑ **A.** 10 quarters and 4 dimes
- ☐ **B.** 8 quarters and 0 dimes
- ☑ **C.** 4 quarters and 10 dimes
- ☑ **D.** 2 quarters and 12 dimes
- ☐ **E.** 4 quarters and 8 dimes

8. What combination of quarters and dimes would equal both 14 coins and total of $2?

A. 10 quarters and 4 dimes

B. 8 quarters and 0 dimes

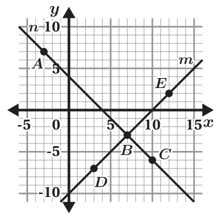
C. 2 quarters and 12 dimes

D. 4 quarters and 10 dimes

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.4.14

Problems 1–5: Here is a coordinate plane.



1. Determine which line represents this condition: The coordinates of each point have a sum of 4.
   Line n

2. Determine which line represents this condition: The y-coordinate of each point is 10 less than the x-coordinate.
   Line m

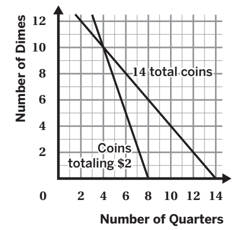
3. Select *all* the points whose coordinates have a sum of 4.
   - ☑ **A.** Point A
   - ☑ **B.** Point B
   - ☑ **C.** Point C
   - ☐ **D.** Point D
   - ☐ **E.** Point E

4. Select *all* the points whose y-coordinate is 10 less than the x-coordinate.
   - ☐ **A.** Point A
   - ☑ **B.** Point B
   - ☐ **C.** Point C
   - ☑ **D.** Point D
   - ☑ **E.** Point E

5. Select *all* the points whose coordinates have a sum of 4 and whose y-coordinate is 10 less than the x-coordinate.
   - ☐ **A.** Point A
   - ☑ **B.** Point B
   - ☐ **C.** Point C
   - ☐ **D.** Point D
   - ☐ **E.** Point E

Name: ______ Date: ______ Period: ______

Problems 6–8: On the coordinate plane shown, one line represents combinations of dimes and quarters that have a total value of \$2. The other line represents combinations of dimes and quarters when the total number of coins is 14.



6. Select *all* the combinations of coins that have a total value of \$2.
   - ☐ **A.** 14 quarters and 0 dimes
   - ☑ **B.** 4 quarters and 10 dimes
   - ☐ **C.** 6 quarters and 8 dimes
   - ☑ **D.** 8 quarters and 0 dimes
   - ☐ **E.** 2 quarters and 12 dimes

7. Select *all* combinations of quarters and dimes that have a total of 14 coins.
   - ☑ **A.** 10 quarters and 4 dimes
   - ☐ **B.** 8 quarters and 0 dimes
   - ☑ **C.** 4 quarters and 10 dimes
   - ☑ **D.** 2 quarters and 12 dimes
   - ☐ **E.** 4 quarters and 8 dimes

8. What combination of quarters and dimes would equal both 14 coins and total of \$2?
   - **A.** 10 quarters and 4 dimes
   - **B.** 8 quarters and 0 dimes
   - **C.** 2 quarters and 12 dimes
   - **D.** 4 quarters and 10 dimes

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C, 8.EE.C.8
2	1	8.EE.C, 8.EE.C.8
3	1	8.EE.C, 8.EE.C.8
4	1	8.EE.C, 8.EE.C.8
5	1	8.EE.C, 8.EE.C.8
6	1	8.EE.C, 8.EE.C.8
7	1	8.EE.C, 8.EE.C.8
8	1	8.EE.C, 8.EE.C.8

Notes:

Name: .... Date: .... Period: ....

Additional Practice

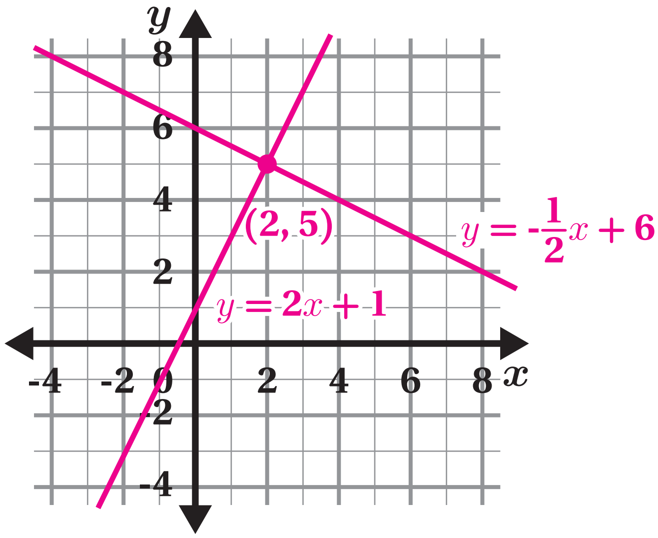
ACC7.4.15

1. Two baby elephants, Jack and Yusef, are gaining weight each day. Jack weighs 205 pounds and is gaining 2 pounds per day. Yusef weighs 195 pounds and is gaining 3 pounds per day. Is there a day when they will weigh the same amount? Explain your thinking.

Yes. *Explanations vary.* If we set the expressions $205 + 2d$ (for Jack) and $195 + 3d$ (for Yusef) equal to each other and solve for d, we will find the number of days it will take until they weigh the same amount.

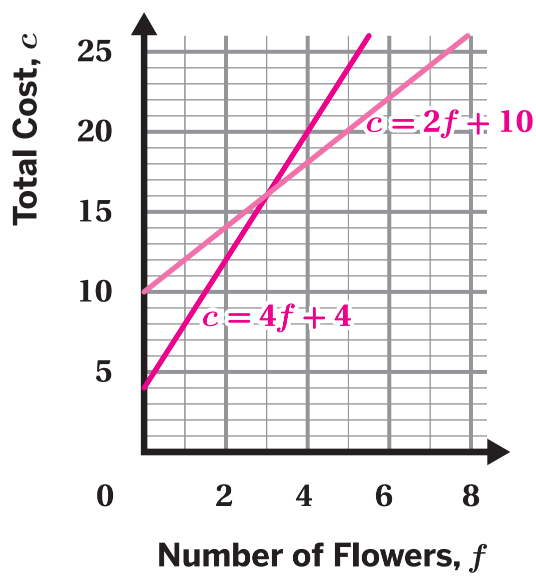
2. Draw a graph to determine x- and y-values that make both of the equations $y = 2x + 1$ and $y = -\frac{1}{2}x + 6$ true.

(2, 5)



Problems 3–4: Trevor and Angela each bought an arrangement of flowers in a vase. The cost, c, of f flowers for Trevor's arrangement is represented by the equation $c = 4f + 4$. The cost, c, of f flowers in dollars for Angela's arrangement is represented by the equation $c = 2f + 10$.

3. Graph the equations for the cost of each arrangement of flowers on the same coordinate plane.



4. Identify the point of intersection. What does the intersection point mean in this context?

The point of intersection is (3, 16). It means that if both Trevor and Angela buy 3 flowers, the total cost of their arrangements will be the same at $16 each.

Name: .... Date: .... Period: ....

Problems 5–6: Han and Elena are hosting a school bake sale.

Before the sale, Han received \$5 in donations and will earn \$2.50 per baked good sold.

The table shown represents the amount of money Elena raised, where b represents the number of baked goods sold and t represents the total amount of money raised.

Number of Baked Goods Sold (b)	Total Money Raised (t)
2	14
3	16
4	18

5. How many baked goods must be sold before they raise the same amount of money? Show or explain your thinking.

10 baked goods. *Explanations vary.*

$$2.50x + 5 = 2x + 10$$
$$0.50x + 5 = 10$$
$$0.50x = 5$$
$$x = 10$$

After selling 10 baked goods, they will have raised the same amount of money.

6. How much will each of them have raised at that time? Show or explain your thinking.

\$30. *Explanations vary.*

Han: $2.50(10) + 5 = \$30$ **Elena: $2(10) + 10 = \$30$**

They will each have raised \$30.

7. The point where the graphs of two equations intersect has a y-coordinate of -6.
One equation is $y = 2x - 4$.
Determine the other equation if its graph has a slope of -3.
Show or explain your thinking.

$y = -3x - 9$. *Explanations vary.*

Substitute -6 into the equation $y = 2x - 4$ to determine the point of intersection.

$$-6 = 2x - 4$$
$$-2 = 2x$$
$$-1 = x$$

The point of intersection is $(-1, -6)$.

Next, write the second equation in the form $y = mx + b$ and substitute the values of $x = -1$ and $y = -6$ to solve for b.

$$-6 = -3(-1) + b$$
$$-6 = 3 + b$$
$$-9 = b$$

The equation of the other line is $y = -3x - 9$.

Name: .... Date: .... Period: ....

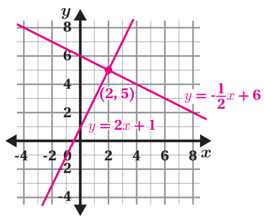
Additional Practice ACC7.4.15

1. Two baby elephants, Jack and Yusef, are gaining weight each day. Jack weighs 205 pounds and is gaining 2 pounds per day. Yusef weighs 195 pounds and is gaining 3 pounds per day. Is there a day when they will weigh the same amount? Explain your thinking.

   Yes. *Explanations vary.* If we set the expressions $205 + 2d$ (for Jack) and $195 + 3d$ (for Yusef) equal to each other and solve for d, we will find the number of days it will take until they weigh the same amount.

2. Draw a graph to determine x- and y-values that make both of the equations $y = 2x + 1$ and $y = -\frac{1}{2}x + 6$ true.

   (2, 5)

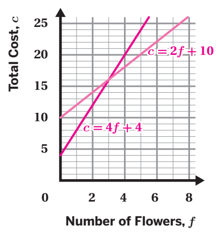


Problems 3–4: Trevor and Angela each bought an arrangement of flowers in a vase. The cost, c, of f flowers for Trevor's arrangement is represented by the equation $c = 4f + 4$. The cost, c, of f flowers in dollars for Angela's arrangement is represented by the equation $c = 2f + 10$.

3. Graph the equations for the cost of each arrangement of flowers on the same coordinate plane.

4. Identify the point of intersection. What does the intersection point mean in this context?

   The point of intersection is (3, 16). It means that if both Trevor and Angela buy 3 flowers, the total cost of their arrangements will be the same at $16 each.



Name: .... Date: .... Period: ....

Problems 5–6: Han and Elena are hosting a school bake sale.

Before the sale, Han received $5 in donations and will earn $2.50 per baked good sold.

The table shown represents the amount of money Elena raised, where b represents the number of baked goods sold and t represents the total amount of money raised.

Number of Baked Goods Sold (b)	Total Money Raised (t)
2	14
3	16
4	18

5. How many baked goods must be sold before they raise the same amount of money? Show or explain your thinking.

   10 baked goods. *Explanations vary.*

   $2.50x + 5 = 2x + 10$

   $0.50x + 5 = 10$

   $0.50x = 5$

   $x = 10$

   After selling 10 baked goods, they will have raised the same amount of money.

6. How much will each of them have raised at that time? Show or explain your thinking.

   $30. *Explanations vary.*

   Han: $2.50(10) + 5 = \$30$ Elena: $2(10) + 10 = \$30$

   They will each have raised $30.

7. The point where the graphs of two equations intersect has a y-coordinate of -6. One equation is $y = 2x - 4$. Determine the other equation if its graph has a slope of -3. Show or explain your thinking.

   $y = -3x - 9$. *Explanations vary.*

   Substitute -6 into the equation $y = 2x - 4$ to determine the point of intersection.

   $-6 = 2x - 4$

   $-2 = 2x$

   $-1 = x$ **The point of intersection is $(-1, -6)$.**

   Next, write the second equation in the form $y = mx + b$ and substitute the values of $x = -1$ and $y = -6$ to solve for b.

   $-6 = -3(-1) + b$

   $-6 = 3 + b$

   $-9 = b$ **The equation of the other line is $y = -3x - 9$.**

Practice Problem Analysis

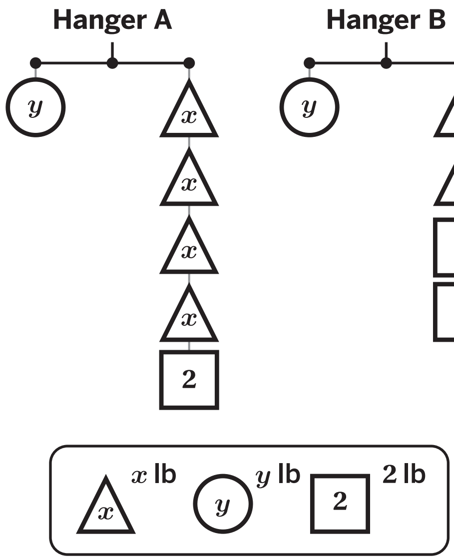
Problem	DOK	Standard(s)
1	2	8.EE.C.8.B
2	1	8.EE.C.8.A
3	1	8.F.A.3
4	2	8.EE.C.8.C
5	2	8.EE.C.8.C, 8.F.B.4
6	2	8.EE.C.8.C
7	2	8.EE.C.8.B

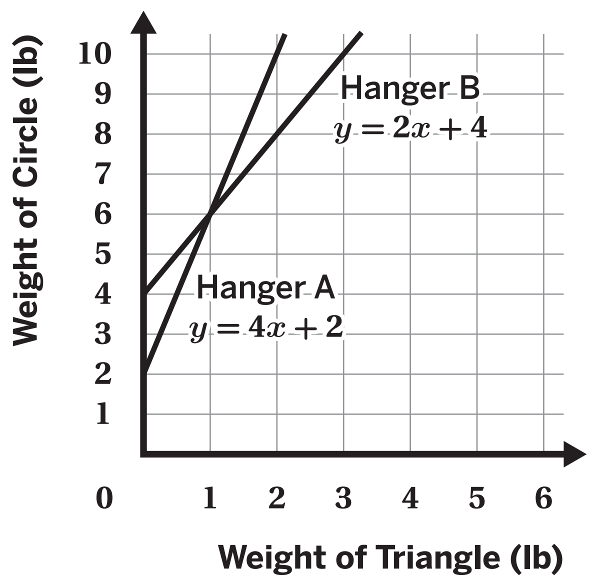
Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.4.16

Problems 1–2: The hangers and the graph represent the same system of equations.





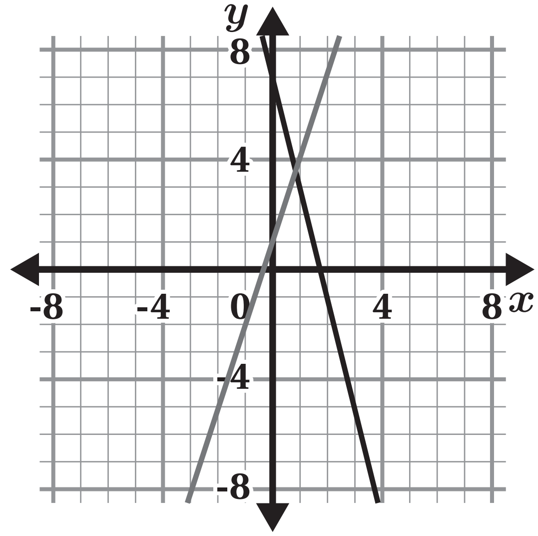
1. Determine the solution to the systems of equations.

(1, 6)

2. What does the solution tell you about the weight of a triangle and the weight of a circle that will balance the hanger?

***Responses vary.* When the triangle weighs 1 pound and the circle weighs 6 pounds, both hangers will be balanced.**

Problems 3–4: Here is a graph.



3. Write an equation that can represent each line.

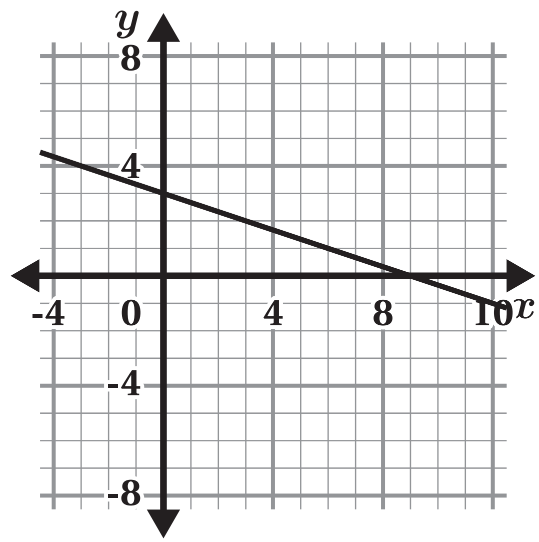
$y = 3x + 1$

$y = -4x + 7$

4. Estimate the solution to the system.

Responses vary: Approximately $(0.9, 3.5)$

Problems 5–7: Here is a graph that represents one equation in a system of equations.



5. Write a second equation for the system so that it has *infinitely many solutions*.

$y = -\frac{1}{3}x + 3$ (or equivalent)

6. Write a second equation whose graph goes through $(0, -2)$ so that the system has *no solution*.

$y = -\frac{1}{3}x - 2$ (or equivalent)

7. Write a second equation whose graph goes through $(0, -4)$ so that the system has one solution $(6, 0)$. Show or explain your thinking.

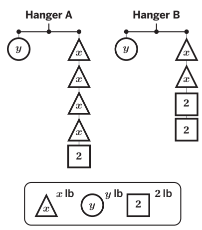
$y = \frac{2}{3}x - 4$. (or equivalent). Substituting the point $(6, 0)$ into the equation $y = mx - 4$ gives $m = \frac{2}{3}$, so the equation is $y = \frac{2}{3}x - 4$.

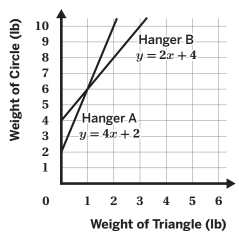
Name: ______ Date: ______ Period: ______

Additional Practice

ACC7.4.16

Problems 1–2: The hangers and the graph represent the same system of equations.





1. Determine the solution to the systems of equations.

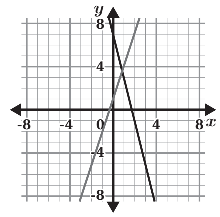
$(1, 6)$

2. What does the solution tell you about the weight of a triangle and the weight of a circle that will balance the hanger?

Responses vary. When the triangle weighs 1 pound and the circle weighs 6 pounds, both hangers will be balanced.

Name: ______ Date: ______ Period: ______

Problems 3–4: Here is a graph.



3. Write an equation that can represent each line.

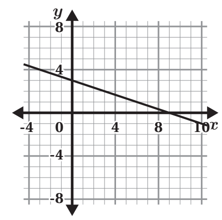
$y = 3x + 1$

$y = -4x + 7$

4. Estimate the solution to the system.

Responses vary: Approximately $(0.9, 3.5)$

Problems 5–7: Here is a graph that represents one equation in a system of equations.



5. Write a second equation for the system so that it has *infinitely many solutions.*

$y = -\frac{1}{3}x + 3$ (or equivalent)

6. Write a second equation whose graph goes through $(0, -2)$ so that the system has *no solution.*

$y = -\frac{1}{3}x - 2$ (or equivalent)

7. Write a second equation whose graph goes through $(0, -4)$ so that the system has one solution $(6, 0)$. Show or explain your thinking.

$y = \frac{2}{3}x - 4$. (or equivalent). Substituting the point $(6, 0)$ into the equation $y = mx - 4$ gives $m = \frac{2}{3}$, so the equation is $y = \frac{2}{3}x - 4$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.A
2	2	8.EE.C.8.A
3	1	8.F.B.4
4	1	8.EE.C.8.A
5	1	8.EE.C.8.B, 8.F.B.4
6	1	8.EE.C.8.B, 8.F.B.4
7	2	8.EE.C.8.A, 8.F.B.4

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.4.17

1. What is the solution to the system of equations below?

$$y = 6$$
$$y = -\frac{1}{2}x + 4$$

A. $(6, -4)$

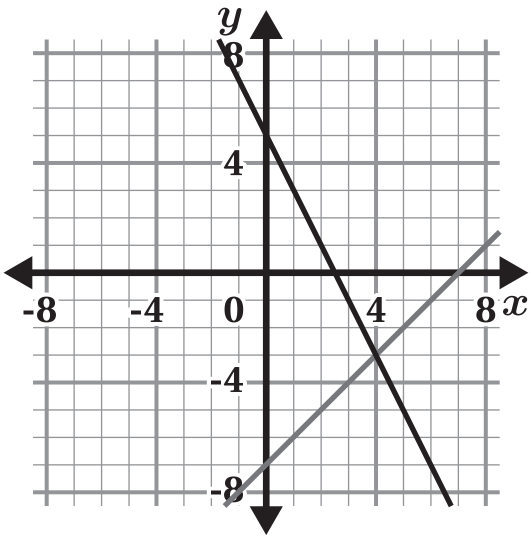
B. $(-4, -6)$

C. $(4, 6)$

D. $(-4, 6)$

Problems 2–3: Here is a graph of the systems of equations:

$$y = x - 7$$
$$y = -2x + 5$$



2. How can you determine the solution to this system of equations by looking at the graph?

Responses vary. **I can look for the point of intersection of the two lines.**

3. What is the solution to the system of equations?

$(4, -3)$

Problems 4–5: Solve each system of equations. Show your thinking.

4. $y = 2x + 5$
$x = -3$

$(-3, -1)$. *Work varies.*

$y = 2(-3) + 5$

$y = -6 + 5$

$y = -1$

The solution is $(-3, -1)$.

5. $y = 2x - 4$
$y = -2x + 12$

$(4, 4)$. *Work varies.*

$2x - 4 = -2x + 12$

$4x - 4 = 12$

$4x = 16$

$x = 4$

$y = 2(4) - 4$

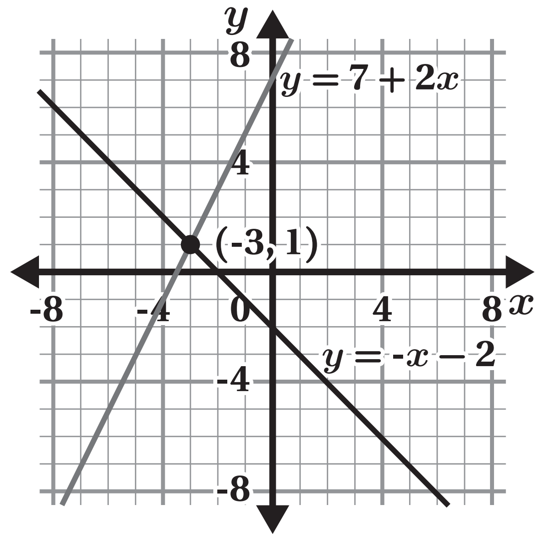
$y = 8 - 4$

$y = 4$

The solution is $(4, 4)$.

Name: .... Date: .... Period: ....

Problems 6–8: Use the lines in the graph to decide whether each statement is true or false. Show or explain your thinking.



6. The solution to the equation $7 + 2x = -x - 2$ is $x = 1$.

False. ***Explanations vary.*** **The x-value of the intersection of the lines is -3, not 1.**

7. The point $(-3, 1)$ is the solution to the following systems of equations:

$y = 7 + 2x$

$y = -x - 2$

True. ***Explanations vary.*** **This is the point of intersection of the two lines.**

8. The point $(0, -2)$ is a solution to the equation $y = -x - 2$.

True. ***Explanations vary.*** **The graph of the line $y = -x - 2$ goes through the point $(0, -2)$. Therefore, it is a solution to that equation.**

9. The solution to a system of equations is $(-1, -4)$. Select *two* equations that might make up the system. Show or explain your thinking.

- ☐ **A.** $y = 2x + 5$
- ☑ **B.** $y = x - 3$
- ☑ **C.** $y = 3x - 1$
- ☐ **D.** $y = x + 3$
- ☐ **E.** $y = -\frac{1}{2}x + 1$

Explanations vary. **I found the two equations for which $(-1, -4)$ is a solution.**

Name: ______ Date: ______ Period: ______

Additional Practice

ACC7.4.17

1. What is the solution to the system of equations below?

$$y = 6$$
$$y = -\frac{1}{2}x + 4$$

A. $(6, -4)$

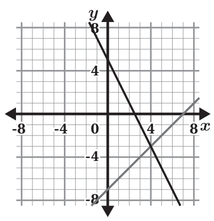
B. $(-4, -6)$

C. $(4, 6)$

D. $(-4, 6)$

Problems 2–3: Here is a graph of the systems of equations:

$$y = x - 7$$
$$y = -2x + 5$$



2. How can you determine the solution to this system of equations by looking at the graph?

Responses vary. I can look for the point of intersection of the two lines.

3. What is the solution to the system of equations?

$(4, -3)$

Problems 4–5: Solve each system of equations. Show your thinking.

4. $y = 2x + 5$
$x = -3$

$(-3, -1)$. *Work varies.*

$y = 2(-3) + 5$
$y = -6 + 5$
$y = -1$

The solution is $(-3, -1)$.

5. $y = 2x - 4$
$y = -2x + 12$

$(4, 4)$. *Work varies.*

$2x - 4 = -2x + 12$
$4x - 4 = 12$
$4x = 16$
$x = 4$

$y = 2(4) - 4$
$y = 8 - 4$
$y = 4$

The solution is $(4, 4)$.

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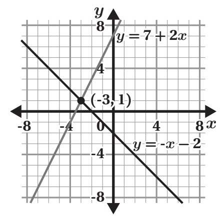
Name: ______ Date: ______ Period: ______

Problems 6–8: Use the lines in the graph to decide whether each statement is true or false. Show or explain your thinking.



6. The solution to the equation $7 + 2x = -x - 2$ is $x = 1$.

False. *Explanations vary.* The x-value of the intersection of the lines is -3, not 1.

7. The point $(-3, 1)$ is the solution to the following systems of equations:

$y = 7 + 2x$
$y = -x - 2$

True. *Explanations vary.* This is the point of intersection of the two lines.

8. The point $(0, -2)$ is a solution to the equation $y = -x - 2$.

True. *Explanations vary.* The graph of the line $y = -x - 2$ goes through the point $(0, -2)$. Therefore, it is a solution to that equation.

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- [ ] A. $y = 2x + 5$
- [x] B. $y = x - 3$
- [x] C. $y = 3x - 1$
- [ ] D. $y = x + 3$
- [ ] E. $y = -\frac{1}{2}x + 1$

Explanations vary. I found the two equations for which $(-1, -4)$ is a solution.

Unit 4 Lesson 17 150 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.B
2	1	8.EE.C.8.A
3	1	8.EE.C.8.A
4	1	8.EE.C.8.B
5	1	8.EE.C.8.B
6	1	8.EE.C.8.A
7	1	8.EE.C.8.A
8	1	8.EE.C.8.A
9	2	8.EE.C.8.A

Notes:

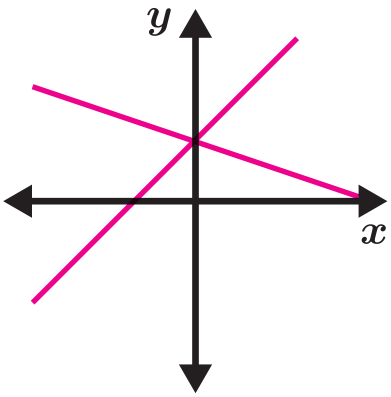
Science Mom Lesson 69

Name: .... Date: .... Period: ....

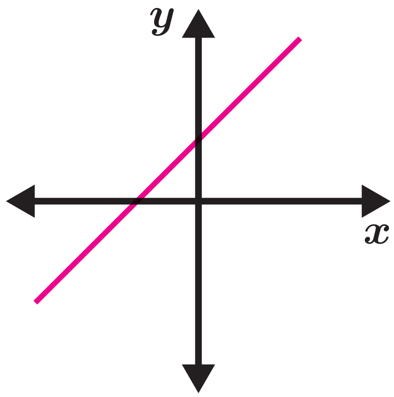
Additional Practice ACC7.4.18

Problems 1–3: Sketch two lines that match each description. Then use the following terms to describe the number of solutions for each system of equations: no solution, one solution, infinitely many solutions

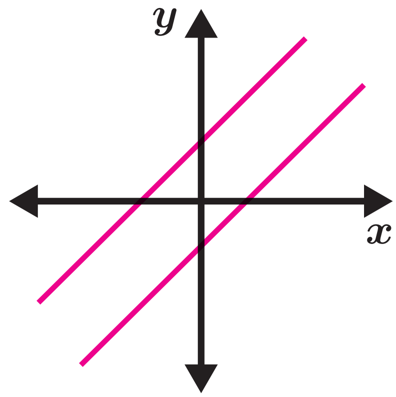
1. Two lines with different slopes but the same y-intercept.
Sketches vary.
One solution.



2. Two lines with the same slope and the same y-intercept.
Sketches vary.
Infinitely many solutions.



3. Two lines with the same slope and different y-intercepts.
Sketches vary.
No solution.



4. Which equation, together with the equation $y = -8x + 4$, creates a system with no solution? Select all that apply.

- ☐ **A.** $y = 2(4x + 2)$
- ☐ **B.** $y = 8x - 4$
- ☑ **C.** $y = -2(4x + 2)$
- ☐ **D.** $y = -8(x - \frac{1}{2})$
- ☑ **E.** $y = 4(-2x - 1)$

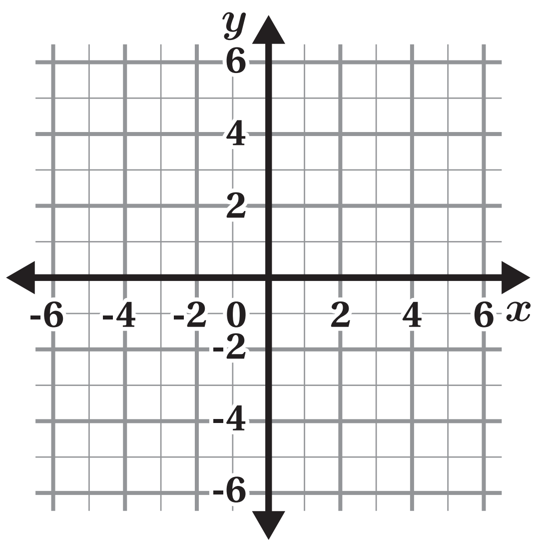
5. How many solutions does this system have? Use the graph if it helps your thinking.

$$y = \frac{2}{3}x - 4$$

$$y = \frac{1}{3}(2x - 6)$$

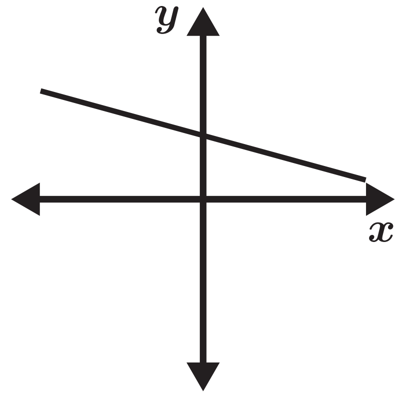
Show or explain your thinking.

No solution. ***Explanations vary.*** **These two graphs will have the same slope but different y-intercepts.**



Problems 6–7: A graph of a system of equations is provided.

6. Select *two* equations that could make up the system of equations shown on the graph.



- ☑ **A.** $y = -\frac{1}{2}x + 4$
- ☐ **B.** $y = -\frac{1}{2}(x + 4)$
- ☐ **C.** $y = -\frac{1}{2}(x + 16)$
- ☑ **D.** $y = -\frac{1}{2}(x - 8)$
- ☐ **E.** $y = -4 - \frac{1}{2}x$

7. Does this system of equations have no solution, one solution, or infinitely many solutions? Explain your thinking.

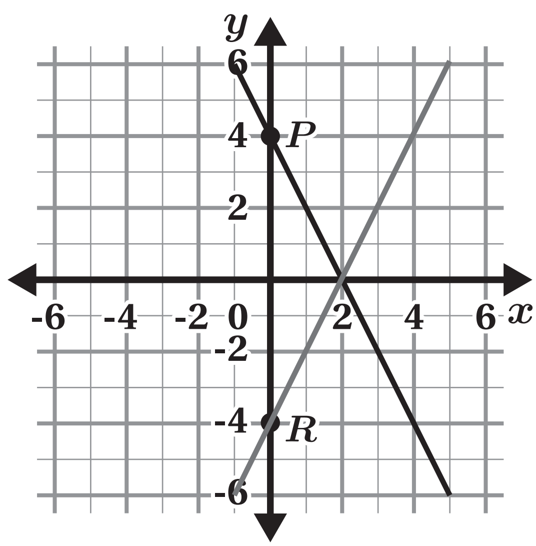
Infinitely many solutions. *Explanations vary.* There are an infinite number of solutions to this system because they are the same line with the same slope and the same y-intercept.

Problems 8–9: Alex graphed this system:

$$y = 2x - 4$$

$$y = -2x + 4$$

He marked its solutions with points P and R.



8. Which statement describes Alex's solutions?

A. His solutions are correct.

B. He marked the y-intercepts instead of the x-intercepts.

C. He marked only the y-intercepts instead of the x- and y-intercepts.

D. He marked the y-intercepts instead of the intersection point of the two lines.

9. What is the solution to the system of equations?

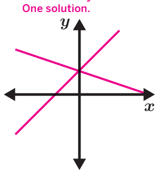
$(2, 0)$

Name: .... Date: .... Period: ....

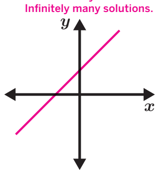
Additional Practice ACC7.4.18

Problems 1–3: Sketch two lines that match each description. Then use the following terms to describe the number of solutions for each system of equations: no solution, one solution, infinitely many solutions

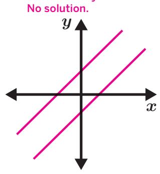
1. Two lines with different slopes but the same y-intercept.
Sketches vary.
One solution.



2. Two lines with the same slope and the same y-intercept.
Sketches vary.
Infinitely many solutions.



3. Two lines with the same slope and different y-intercepts.
Sketches vary.
No solution.



4. Which equation, together with the equation $y = -8x + 4$, creates a system with no solution? Select all that apply.

- ☐ **A.** $y = 2(4x + 2)$
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- ☐ **D.** $y = -8(x - \frac{1}{2})$
- ☑ **E.** $y = 4(-2x - 1)$

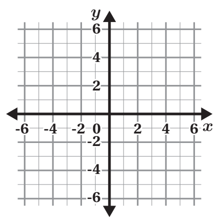
5. How many solutions does this system have? Use the graph if it helps your thinking.

$y = \frac{2}{3}x - 4$

$y = \frac{1}{3}(2x - 6)$

Show or explain your thinking.

No solution. ***Explanations vary.*** **These two graphs will have the same slope but different y-intercepts.**

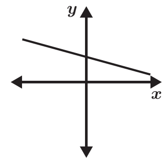


Name: .... Date: .... Period: ....

Problems 6–7: A graph of a system of equations is provided.

6. Select *two* equations that could make up the system of equations shown on the graph.

- ☑ **A.** $y = -\frac{1}{2}x + 4$
- ☐ **B.** $y = -\frac{1}{2}(x + 4)$
- ☐ **C.** $y = -\frac{1}{2}(x + 16)$
- ☑ **D.** $y = -\frac{1}{2}(x - 8)$
- ☐ **E.** $y = -4 - \frac{1}{2}x$



7. Does this system of equations have no solution, one solution, or infinitely many solutions? Explain your thinking.

Infinitely many solutions. ***Explanations vary.*** **There are an infinite number of solutions to this system because they are the same line with the same slope and the same y-intercept.**

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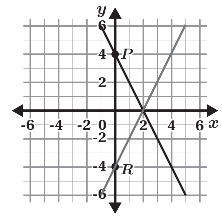

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C. He marked only the y-intercepts instead of the x- and y-intercepts.

D. He marked the y-intercepts instead of the intersection point of the two lines.

9. What is the solution to the system of equations?

(2, 0)

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.B
2	1	8.EE.C.8.B
3	1	8.EE.C.8.B
4	2	8.EE.C.8.B
5	2	8.EE.C.8.B
6	2	8.EE.C.8.B, 8.F.B.4
7	1	8.EE.C.8.B
8	2	8.EE.C.8.A
9	1	8.EE.C.8.A

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.4.19

Problems 1–6: Solve each system of equations. Show your thinking.

1. $y = -2x$
$3x + y = 10$

(10, –20). ***Work varies.***

$3x + (-2x) = 10$

$x = 10$

$y = -2(10)$

$y = -20$

2. $y = 2x$
$x = 4y - 14$

(2, 4). ***Work varies.***

$x = 4(2x) - 14$

$x = 8x - 14$

$-7x = -14$

$x = 2$

$y = 2(2)$

$y = 4$

3. $y = 6x + 10$
$y = -2x - 14$

(–3, –8). ***Work varies.***

$6x + 10 = -2x - 14$

$8x = -24$

$x = -3$

$y = 6(-3) + 10$

$y = -18 + 10$

$y = -8$

4. $y = -3x + 12$
$y = 5x - 8$

(2.5, 4.5). ***Work varies.***

$-3x + 12 = 5x - 8$

$-8x = -20$

$x = 2.5$

$y = 5(2.5) - 8$

$y = 12.5 - 8$

$y = 4.5$

5. $x = -3$
$y = 4x + 8$

(–3, –4). ***Work varies.***

$y = 4(-3) + 8$

$y = -12 + 8$

$y = -4$

6. $y = 12$
$y = \frac{1}{2}x - 4$

(32, 12). ***Work varies.***

$12 = \frac{1}{2}x - 4$

$16 = \frac{1}{2}x$

$32 = x$

Problems 7–9: Here is an incomplete system of equations.

$$y = \frac{4}{5}x + 6$$
$$y = ???$$

7. Create a second equation so that the system has no solution.

Responses vary. **Any equation with a slope of $\frac{4}{5}$ and a y-intercept not equal to 6.**

8. Create a second equation so that the system has an infinite number of solutions.

$y = \frac{4}{5}x + 6$ (or equivalent)

9. Create a second equation so that the system has a solution of $(10, 14)$.

Responses vary. **$y = 2x - 6$.**

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.4.19

Problems 1–6: Solve each system of equations. Show your thinking.

1. $y = -2x$
$3x + y = 10$

(10, −20). Work varies.
$3x + (-2x) = 10$
$x = 10$
$y = -2(10)$
$y = -20$

2. $y = 2x$
$x = 4y - 14$

(2, 4). Work varies.
$x = 4(2x) - 14$
$x = 8x - 14$
$-7x = -14$
$x = 2$
$y = 2(2)$
$y = 4$

3. $y = 6x + 10$
$y = -2x - 14$

(−3, −8). Work varies.
$6x + 10 = -2x - 14$
$8x = -24$
$x = -3$
$y = 6(-3) + 10$
$y = -18 + 10$
$y = -8$

4. $y = -3x + 12$
$y = 5x - 8$

(2.5, 4.5). Work varies.
$-3x + 12 = 5x - 8$
$-8x = -20$
$x = 2.5$
$y = 5(2.5) - 8$
$y = 12.5 - 8$
$y = 4.5$

5. $x = -3$
$y = 4x + 8$

(−3, −4). Work varies.
$y = 4(-3) + 8$
$y = -12 + 8$
$y = -4$

6. $y = 12$
$y = \frac{1}{2}x - 4$

(32, 12). Work varies.
$12 = \frac{1}{2}x - 4$
$16 = \frac{1}{2}x$
$32 = x$

Name: ______ Date: ______ Period: ______

Problems 7–9: Here is an incomplete system of equations.

$$y = \frac{4}{5}x + 6$$
$$y = ???$$

7. Create a second equation so that the system has no solution.

Responses vary. Any equation with a slope of $\frac{4}{5}$ and a y-intercept not equal to 6.

8. Create a second equation so that the system has an infinite number of solutions.

$y = \frac{4}{5}x + 6$ (or equivalent)

9. Create a second equation so that the system has a solution of (10, 14).

Responses vary. $y = 2x - 6$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.C.8.B
2	1	8.EE.C.8.B
3	1	8.EE.C.8.B
4	1	8.EE.C.8.B
5	1	8.EE.C.8.B
6	1	8.EE.C.8.B
7	2	8.EE.C.8.B
8	2	8.EE.C.8.B
9	2	8.EE.C.8.B

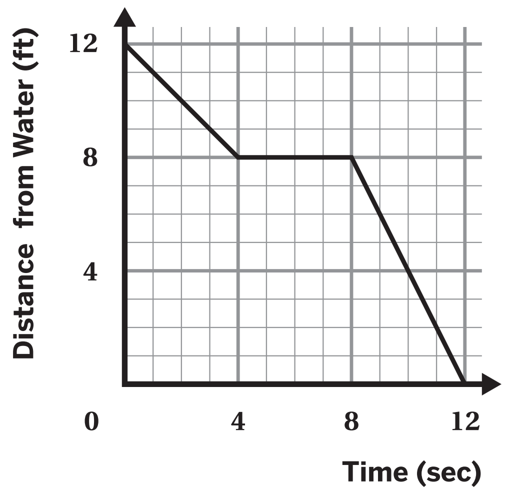
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.5.01

Problems 1–3: This graph shows a turtle's journey towards the ocean. Determine whether each statement is true or false. If the statement is false, explain your thinking.



1. The turtle was 4 feet from the water at 4 seconds.

False. ***Explanations vary*****. The turtle was 8 feet from the water at 4 seconds.**

2. The turtle was 12 feet from the water at 0 seconds.

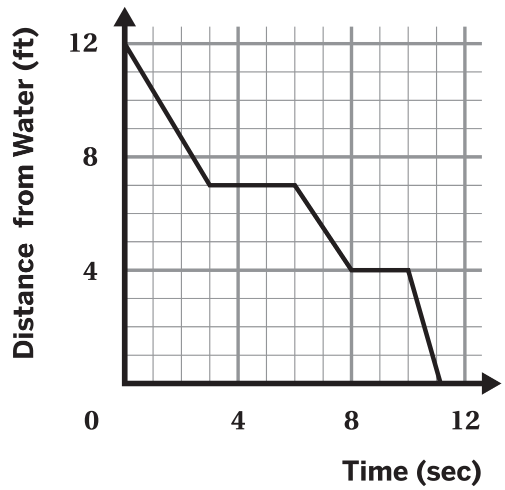
True

3. The turtle's distance from the water did not change from 4 to 9 seconds.

False. ***Explanations vary*** **The turtle's distance from the water did not change from 4 to 8 seconds.**

Name: .... Date: .... Period: ....

Problems 4–6: This graph represents another turtle walking across the sand towards the water.



4. What story does the graph tell about the turtle's journey?

***Responses vary*. The turtle started their journey 12 feet from the water. The turtle traveled for 3 seconds until they were 7 feet from the water. The turtle rested for 3 seconds. Then, the turtle traveled until they were 4 feet from the water. The turtle rested one more time for 2 seconds. The turtle reached the water 11 seconds after it started.**

5. How far was the turtle from the water after 5 seconds?

7 feet.

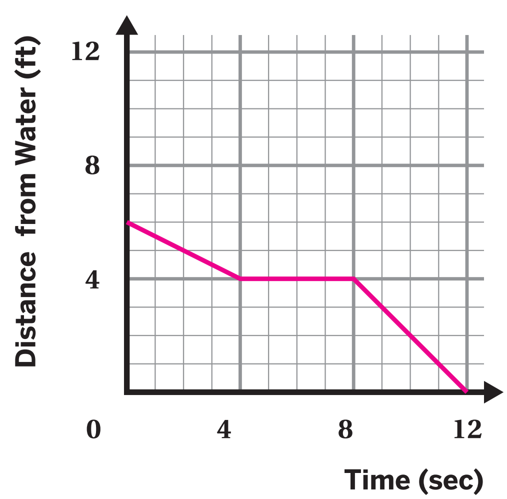
6. After how many seconds is the turtle's distance 4 feet from the water?

A. 3 seconds **B.** 6 seconds **C.** 8 seconds **D.** 11 seconds

7. A turtle started their journey towards the ocean. When the turtle started their journey, they were 6 feet from the water. The turtle traveled for 4 seconds. When the turtle was 4 feet from the water, they rested for 4 seconds. Then, the turtle kept traveling towards the water. They reached the water at 12 seconds.

Sketch a graph that could represent the turtle's distance from the water.

***Responses vary*. Sample shown on graph.**

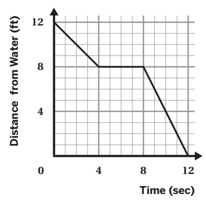


Name: ________ Date: ________ Period: ________

Additional Practice

ACC7.5.01

Problems 1–3: This graph shows a turtle's journey towards the ocean. Determine whether each statement is true or false. If the statement is false, explain your thinking.



1. The turtle was 4 feet from the water at 4 seconds.

False. ***Explanations vary.*** **The turtle was 8 feet from the water at 4 seconds.**

2. The turtle was 12 feet from the water at 0 seconds.

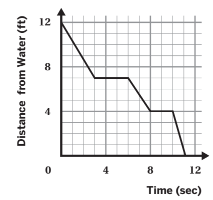
True

3. The turtle's distance from the water did not change from 4 to 9 seconds.

False. ***Explanations vary*** **The turtle's distance from the water did not change from 4 to 8 seconds.**

Name: ________ Date: ________ Period: ________

Problems 4–6: This graph represents another turtle walking across the sand towards the water.



4. What story does the graph tell about the turtle's journey?

Responses vary. **The turtle started their journey 12 feet from the water. The turtle traveled for 3 seconds until they were 7 feet from the water. The turtle rested for 3 seconds. Then, the turtle traveled until they were 4 feet from the water. The turtle rested one more time for 2 seconds. The turtle reached the water 11 seconds after it started.**

5. How far was the turtle from the water after 5 seconds?

7 feet.

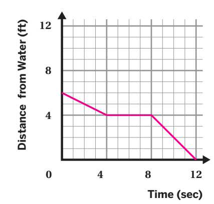
6. After how many seconds is the turtle's distance 4 feet from the water?

A. 3 seconds **B.** 6 seconds **C.** 8 seconds **D.** 11 seconds

7. A turtle started their journey towards the ocean. When the turtle started their journey, they were 6 feet from the water. The turtle traveled for 4 seconds. When the turtle was 4 feet from the water, they rested for 4 seconds. Then, the turtle kept traveling towards the water. They reached the water at 12 seconds.

Sketch a graph that could represent the turtle's distance from the water.

Responses vary. **Sample shown on graph.**



Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.1, 8.F.B.5
2	1	8.F.A.1, 8.F.B.5
3	1	8.F.A.1, 8.F.B.5
4	2	8.F.A.1, 8.F.B.5
5	1	8.F.A.1, 8.F.B.5
6	1	8.F.A.1, 8.F.B.5
7	2	8.F.A.1, 8.F.B.5

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.5.02

1. Determine whether each table could represent a function. Explain your thinking.

a

Input	Output
4	6
5	7
6	8
4	9
7	10

Not a function. ***Explanations vary.*** **There are two different output values for the same input value 4.**

b

Input	Output
-4	17
-2	5
0	0
2	5
3	17

Function. ***Explanations vary.*** **There is only one output for each given input.**

2. A birthstone is a gemstone that represents the month in which you were born. Both tables show a relationship between birthday and birthstone.

Table A

Birthdate	Gemstone
October 16	Opal
July 5	Ruby
March 20	Aquamarine
June 22	Pearl
October 26	Opal

Table B

Gemstone	Birthdate
Opal	October 16
Ruby	July 5
Aquamarine	March 20
Pearl	June 22
Opal	October 26

Which table(s) represent a function?

A. Table A **B.** Table B **C.** Neither **D.** Both

3. Kiran earns an hourly wage. Determine whether each statement is *true* or *false*. Explain your thinking.

a Kiran's earnings are a function of his hours worked.

True. *Explanations vary.* For each hour worked, Kiran earns a unique dollar amount.

b Kiran's hours worked are a function of his earnings.

True. *Explanations vary.* Each amount of earnings relates to only one number of hours worked.

4. A partially completed input-output table is shown. Complete the table so that it represents a function.

***Responses vary.* Sample shown in table.**

Input	Output
-5	25
-3	9
-1	1
3	**9**
10	**100**

5. Diego wants to know if it is possible for a function to have more than one input value, but only one output value. What would you tell Diego? Use the table if it helps your thinking.

***Responses vary.* Yes, it is possible. For example, the table shown is a function where each input has the same output of 12. This represents a function because each input has only one output.**

Input	Output
1	**12**
2	**12**
3	**12**

6. Identify a possible rule for the table shown.

***Responses vary.* If the input is a negative number, the output is 0. If not, the output is 1.**

Input	Output
1	1
2	1
0	1
-1	0
-2	0

Name: Date: Period:

Additional Practice ACC7.5.02

1. Determine whether each table could represent a function. Explain your thinking.

a

Input	Output
4	6
5	7
6	8
4	9
7	10

Not a function. *Explanations vary.* There are two different output values for the same input value 4.

b

Input	Output
−4	17
−2	5
0	0
2	5
3	17

Function. *Explanations vary.* There is only one output for each given input.

2. A birthstone is a gemstone that represents the month in which you were born. Both tables show a relationship between birthday and birthstone.

Table A

Birthdate	Gemstone
October 16	Opal
July 5	Ruby
March 20	Aquamarine
June 22	Pearl
October 26	Opal

Table B

Gemstone	Birthdate
Opal	October 16
Ruby	July 5
Aquamarine	March 20
Pearl	June 22
Opal	October 26

Which table(s) represent a function?

A. Table A B. Table B C. Neither D. Both

Name: Date: Period:

3. Kiran earns an hourly wage. Determine whether each statement is *true* or *false*. Explain your thinking.

a Kiran's earnings are a function of his hours worked.

True. *Explanations vary.* For each hour worked, Kiran earns a unique dollar amount.

b Kiran's hours worked are a function of his earnings.

True. *Explanations vary.* Each amount of earnings relates to only one number of hours worked.

4. A partially completed input-output table is shown. Complete the table so that it represents a function.

Responses vary. Sample shown in table.

Input	Output
−5	25
−3	9
−1	1
3	9
10	100

5. Diego wants to know if it is possible for a function to have more than one input value, but only one output value. What would you tell Diego? Use the table if it helps your thinking.

Responses vary. Yes, it is possible. For example, the table shown is a function where each input has the same output of 12. This represents a function because each input has only one output.

Input	Output
1	12
2	12
3	12

6. Identify a possible rule for the table shown.

Responses vary. If the input is a negative number, the output is 0. If not, the output is 1.

Input	Output
1	1
2	1
0	1
−1	0
−2	0

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.1
2	1	8.F.A.1
3	2	8.F.A.1
4	2	8.F.A.1
5	2	8.F.A.1
6	3	8.F.A.1

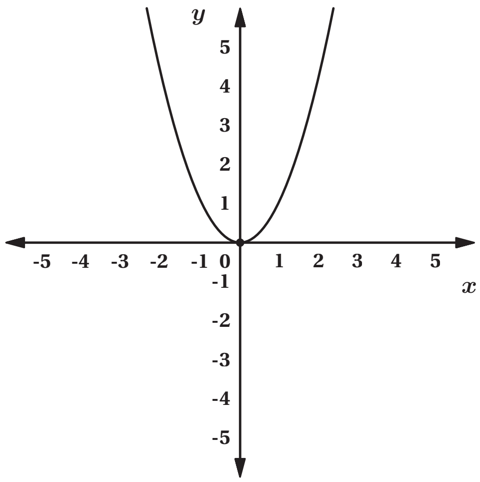
Notes:

Name: .... Date: .... Period: ....

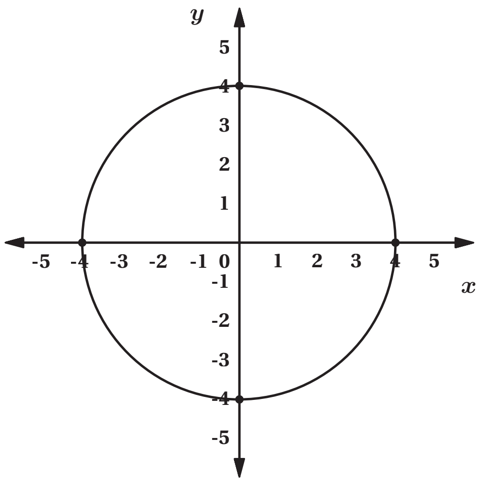
Additional Practice ACC7.5.03

1. Which graph represents y as a function of x?

A.

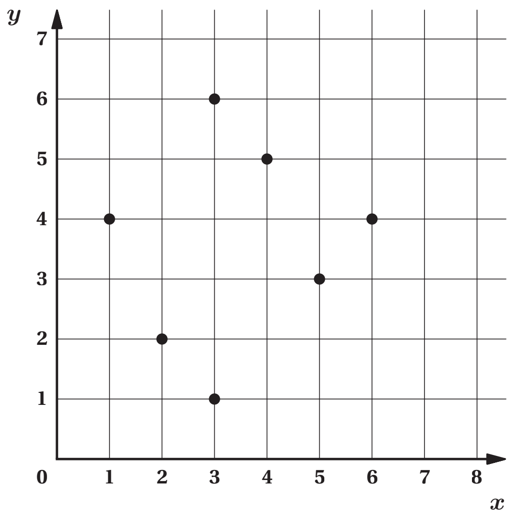


B.

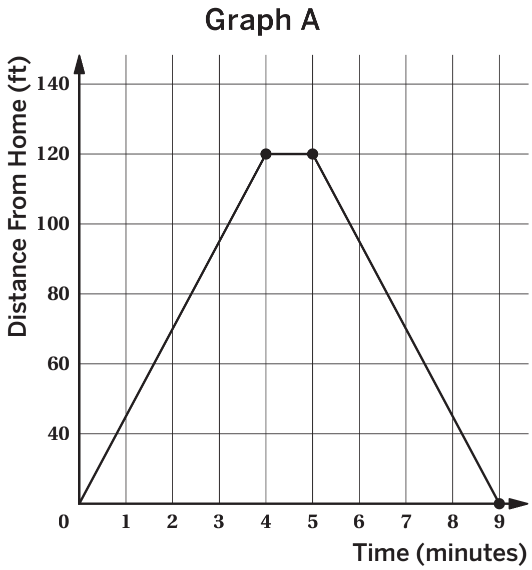


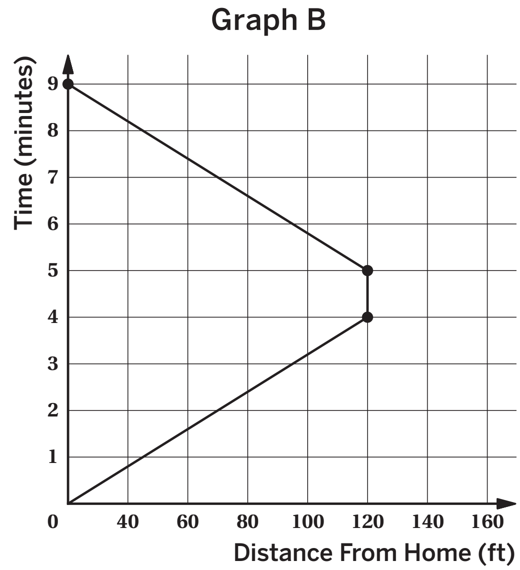
2. Which point on the graph could you remove so that y is a function of x?

(3, 1) or (3, 6)



3. Clare rode her bike to her friend's house and back. Both graphs show the relationship between her time and her distance from home.





Which graph represents a function?

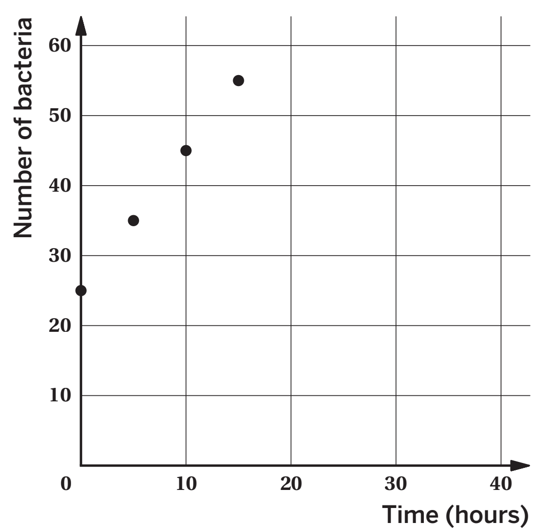
A. Graph A **B.** Graph B **C.** Both graphs **D.** Neither graph

Explain your thinking.

***Explanations vary*. Graph A represents a function because each input has only one output.**

4. A scientist graphs the growth of a strain of bacteria. The graph represents a function. Suppose there was the same number of bacteria at 20 and 22 hours. Would the graph still be a function? Explain your thinking.

Yes. *Explanations vary*. There still would be one input value paired with only one output value.

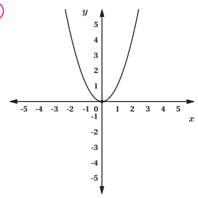


Name: ________ Date: ________ Period: ________

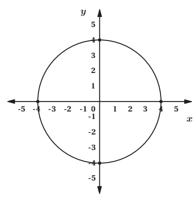
Additional Practice ACC7.5.03

1. Which graph represents y as a function of x?

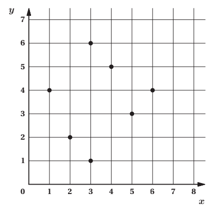
A. (circled)



B.



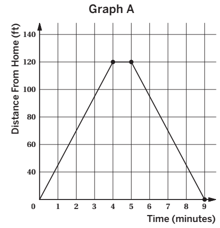
2. Which point on the graph could you remove so that y is a function of x?
(3, 1) or (3, 6)



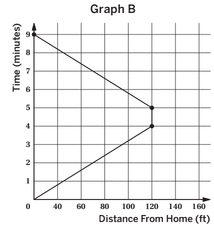
Name: ________ Date: ________ Period: ________

3. Clare rode her bike to her friend's house and back. Both graphs show the relationship between her time and her distance from home.

Graph A



Graph B

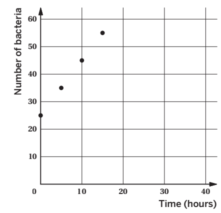


Which graph represents a function?

A. Graph A B. Graph B C. Both graphs D. Neither graph

Explain your thinking.
Explanations vary. Graph A represents a function because each input has only one output.

4. A scientist graphs the growth of a strain of bacteria. The graph represents a function. Suppose there was the same number of bacteria at 20 and 22 hours. Would the graph still be a function? Explain your thinking.
Yes. *Explanations vary.* There still would be one input value paired with only one output value.



Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.1
2	2	8.F.A.1
3	2	8.F.A.1
4	2	8.F.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.5.04

1. Which equation expresses the output as a function of the input for the following scenario?

a The amount your school charges, c, for t tickets to the school play that cost \$4 each.

A. $c = \frac{1}{4}t$

B. $t = \frac{1}{4}c$

C. $c = 4t$

D. $t = 4c$

b Identify the independent variable and dependent variable for you answer in part a.

a amount your school charges, c

Dependent variable

b tickets, t

Independent variable

Problems 2–4: For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.

2. The circumference, C, of a circle with diameter, d.

Equation: $C = \pi \cdot d$ **(or equivalent)**

Independent variable: **diameter of the circle, d**

Dependent variable: **circumference of the circle, C**

3. The selling price p, after a markup of 15% is applied to the original price of an item, r.

Equation: $p = 1.15r$ **(or equivalent)**

Independent variable: **original price, r**

Dependent variable: **selling price, p**

4. The area, A, of a square with a side length, s.

Equation: $A = s^2$ **(or equivalent)**

Independent variable: **side length of the square, s**

Dependent variable: **area of the square, A**

5. Han's family car averages 25 miles per gallon when driven. Han writes the equation $y = 25x$ to represent the distance, in miles, his family has traveled on a certain number of gallons of gasoline.

a What do x and y represent in this situation?

x represents the input, or the number of gallons of gasoline.

y represents the output, or distance, in miles, Han's family drives on x gallons of gasoline.

b Which variable represents the independent variable? Which variable represents the dependent variable?

x represents the independent variable.

y represents the dependent variable.

6. Mai is buying vegetable plants for her garden. Tomato plants cost \$3 each and pepper plants cost \$1 each. Mai has \$24 to spend on these plants. Let t represent the number of tomato plants Mai buys and p represent the number of pepper plants Mai buys.

a Write an equation relating the two variables.

$3t + p = 24$ (or equivalent)

b Rewrite the equation so that it expresses p as the dependent variable in terms of t as the independent variable.

$p = 24 - 3t$ (or equivalent)

c Rewrite the equation so that it expresses t as the dependent variable in terms of p as the independent variable.

$t = 8 - \frac{1}{3}p$ (or equivalent)

7. To determine the number of degrees Fahrenheit of a temperature given in degrees Celsius, you multiply the degrees Celsius by $\frac{9}{5}$ and add 32. For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.

a The temperature in degrees Fahrenheit, F, based on the temperature in degrees Celsius, C.

$F = \frac{9}{5}C + 32$; Independent variable: temperature in degrees Celsius or C; Dependent variable: temperature in degrees Fahrenheit or F

b The temperature in degrees Celsius, C, based on the temperature in degrees Fahrenheit, F.

$C = \frac{5}{9}(F - 32)$; Independent variable: temperature in degrees Fahrenheit or F; Dependent variable: temperature in degrees Celsius or C

Name: .... Date: .... Period: ....

Additional Practice ACC7.5.04

1. Which equation expresses the output as a function of the input for the following scenario?

a. The amount your school charges, c, for t tickets to the school play that cost $4 each.

A. $c = \frac{1}{4}t$　　**B.** $t = \frac{1}{4}c$

C. $c = 4t$　　**D.** $t = 4c$

b. Identify the independent variable and dependent variable for you answer in part a.

a. amount your school charges, c
Dependent variable

b. tickets, t
Independent variable

Problems 2–4: For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.

2. The circumference, C, of a circle with diameter, d.

Equation: $C = \pi \cdot d$ **(or equivalent)**
Independent variable: **diameter of the circle, d**
Dependent variable: **circumference of the circle, C**

3. The selling price p, after a markup of 15% is applied to the original price of an item, r.

Equation: $p = 1.15r$ **(or equivalent)**
Independent variable: **original price, r**
Dependent variable: **selling price, p**

4. The area, A, of a square with a side length, s.

Equation: $A = s^2$ **(or equivalent)**
Independent variable: **side length of the square, s**
Dependent variable: **area of the square, A**

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Name: .... Date: .... Period: ....

5. Han's family car averages 25 miles per gallon when driven. Han writes the equation $y = 25x$ to represent the distance, in miles, his family has traveled on a certain number of gallons of gasoline.

a. What do x and y represent in this situation?
x represents the input, or the number of gallons of gasoline.
y represents the output, or distance, in miles, Han's family drives on x gallons of gasoline.

b. Which variable represents the independent variable? Which variable represents the dependent variable?
x represents the independent variable.
y represents the dependent variable.

6. Mai is buying vegetable plants for her garden. Tomato plants cost $3 each and pepper plants cost $1 each. Mai has $24 to spend on these plants. Let t represent the number of tomato plants Mai buys and p represent the number of pepper plants Mai buys.

a. Write an equation relating the two variables.
$3t + p = 24$ **(or equivalent)**

b. Rewrite the equation so that it expresses p as the dependent variable in terms of t as the independent variable.
$p = 24 - 3t$ **(or equivalent)**

c. Rewrite the equation so that it expresses t as the dependent variable in terms of p as the independent variable.
$t = 8 - \frac{1}{3}p$ **(or equivalent)**

7. To determine the number of degrees Fahrenheit of a temperature given in degrees Celsius, you multiply the degrees Celsius by $\frac{9}{5}$ and add 32. For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.

a. The temperature in degrees Fahrenheit, F, based on the temperature in degrees Celsius, C.
$F = \frac{9}{5}C + 32$; Independent variable: temperature in degrees Celsius or C; Dependent variable: temperature in degrees Fahrenheit or F

b. The temperature in degrees Celsius, C, based on the temperature in degrees Fahrenheit, F.
$C = \frac{5}{9}(F - 32)$; Independent variable: temperature in degrees Fahrenheit or F; Dependent variable: temperature in degrees Celsius or C

Unit 5 Lesson 4　164　Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.2
2–4	2	8.F.A.2
5	2	8.F.A.2
6	2	8.F.A.2
7	3	8.F.A.2

Notes:

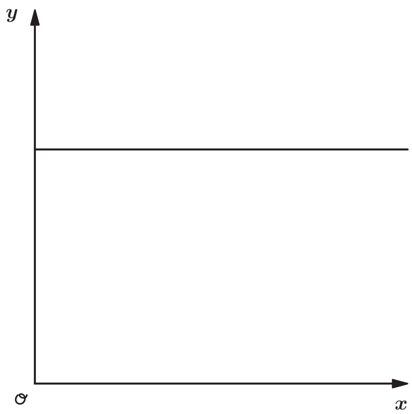
Name: .... Date: .... Period: ....

Additional Practice ACC7.5.05

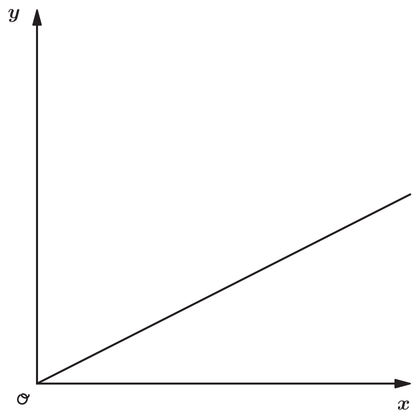
Problems 1–2: Noah walks from his home to the park at a steady pace without stopping.

1. Which graph represents the relationship between Noah's distance from home and time?

Graph A



Graph B



2. Indicate whether the variable described is the possible *independent* or *dependent* variable.

a Distance from home

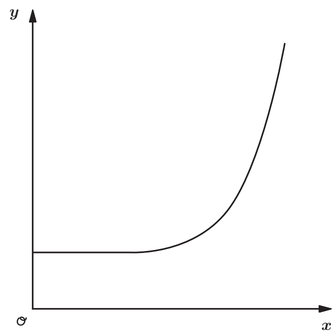
Dependent variable

b Time

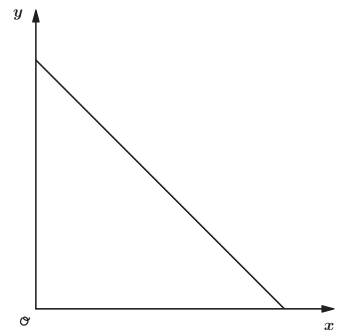
Independent variable

3. For each scenario, determine which graph best represents it. Then identify the possible independent and dependent variables and how you would label the axes.

Graph 1



Graph 2



a Mai has some money saved and starts spending the same amount each week.

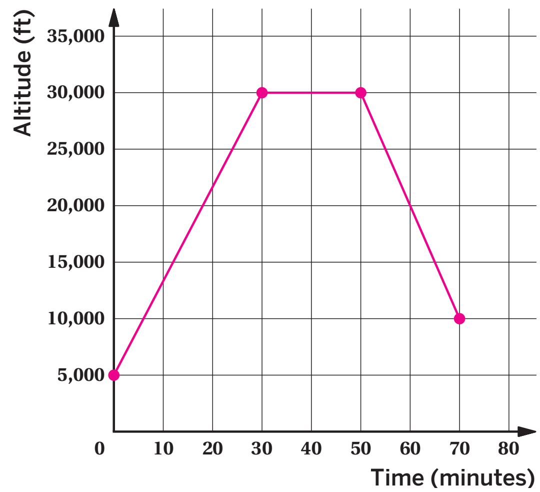
Graph 2. Independent variable and horizontal axis label: Time (weeks). Dependent variable and vertical axis label: Money saved ($).

b The day started cold, but then it got warmer.

Graph 1. Independent variable and horizontal axis label: Time (hours). Dependent variable and vertical axis label: Temperature (°F).

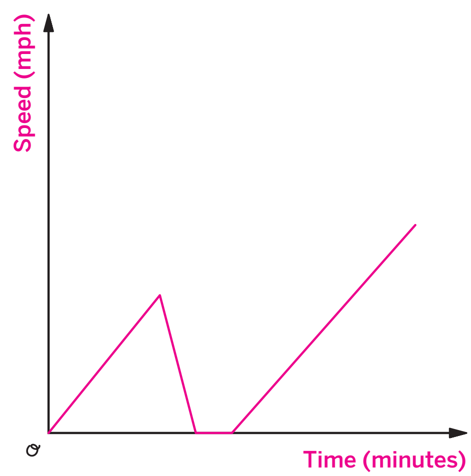
4. An airplane was at an altitude of 5,000 feet. It took the airplane 30 minutes to go from an altitude of 5,000 feet to 30,000 feet. Once at that altitude, the airplane flew for 20 minutes. Then it took the plane 20 minutes to reach an altitude of 10,000 feet.

   Sketch a graph that represents this situation.
   Sample response shown.



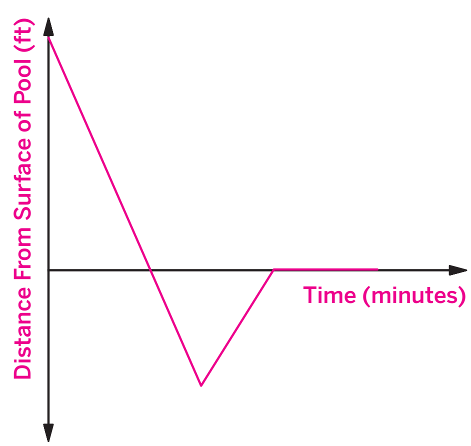
5. Tyler got on his rollerblades. He was increasing his speed but then fell down. He got up and continued rollerblading, gradually increasing his speed.

   Sketch a graph that shows Tyler's speed, in miles per hour, after several minutes. Be sure to label the axes.
   Sample response shown.



6. Kiran was on a water slide. He slid down the slide and dove into the water. He was below the surface water and then swam back to the surface of the water. He swam at the surface of the water for a few minutes.

   Sketch a graph that shows Kiran's distance from the surface of the water as a function of time in minutes. Be sure to label the axes.
   Sample response shown.



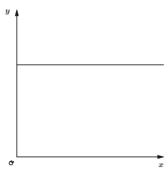
Name: .... Date: .... Period: ....

Additional Practice ACC7.5.05

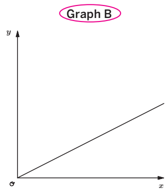
Problems 1–2: Noah walks from his home to the park at a steady pace without stopping.

1. Which graph represents the relationship between Noah's distance from home and time?

Graph A



Graph B



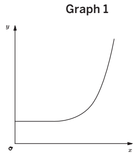
2. Indicate whether the variable described is the possible *independent* or *dependent* variable.

a. Distance from home
Dependent variable

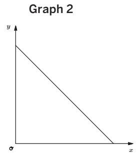
b. Time
Independent variable

3. For each scenario, determine which graph best represents it. Then identify the possible independent and dependent variables and how you would label the axes.

Graph 1



Graph 2



a. Mai has some money saved and starts spending the same amount each week.
Graph 2. Independent variable and horizontal axis label: Time (weeks). Dependent variable and vertical axis label: Money saved ($).

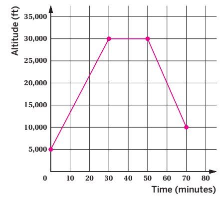
b. The day started cold, but then it got warmer.
Graph 1. Independent variable and horizontal axis label: Time (hours). Dependent variable and vertical axis label: Temperature (°F).

Name: .... Date: .... Period: ....

4. An airplane was at an altitude of 5,000 feet. It took the airplane 30 minutes to go from an altitude of 5,000 feet to 30,000 feet. Once at that altitude, the airplane flew for 20 minutes. Then it took the plane 20 minutes to reach an altitude of 10,000 feet.

Sketch a graph that represents this situation.

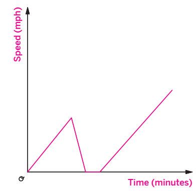
Sample response shown.



5. Tyler got on his rollerblades. He was increasing his speed but then fell down. He got up and continued rollerblading, gradually increasing his speed.

Sketch a graph that shows Tyler's speed, in miles per hour, after several minutes. Be sure to label the axes.

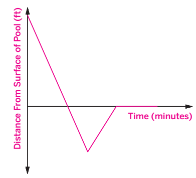
Sample response shown.



6. Kiran was on a water slide. He slid down the slide and dove into the water. He was below the surface water and then swam back to the surface of the water. He swam at the surface of the water for a few minutes.

Sketch a graph that shows Kiran's distance from the surface of the water as a function of time in minutes. Be sure to label the axes.

Sample response shown.



Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.5
2	1	8.F.A.5
3	2	8.F.A.5
4	2	8.F.A.5
5	2	8.F.A.5
6	3	8.F.A.5

Notes:

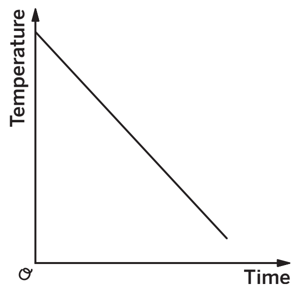
Name: .... Date: .... Period: ....

Additional Practice

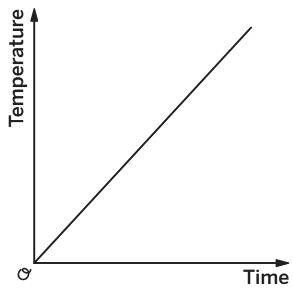
ACC7.5.06

1. Which graph(s) represents an increasing function? Select *all* that apply.

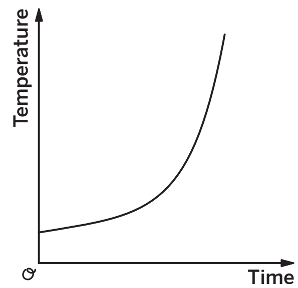
- [ ] A



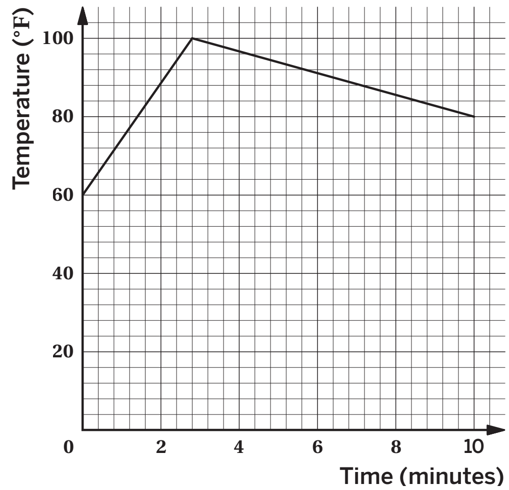
- [x] B



- [x] C



2. The graph shows the temperature of a liquid in a mug over time. Which best describes the graph of the function?



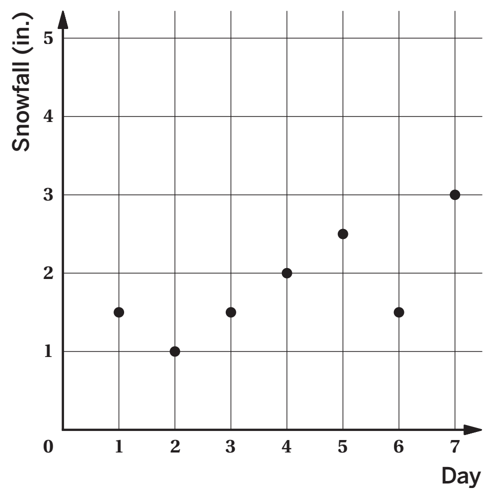
A. The temperature of the liquid slowly increases, and then slowly decreases over time.

B. The temperature of the liquid quickly decreases, and then slowly increases over time.

C. The temperature of the liquid quickly increases, and then decreases even faster over time.

D. The temperature of the liquid quickly increases, and then decreases more slowly over time.

3. The graph shows the total amount of snow that fell each day over a 7-day period.



a What was the amount of snowfall for Day 2?

1 in.

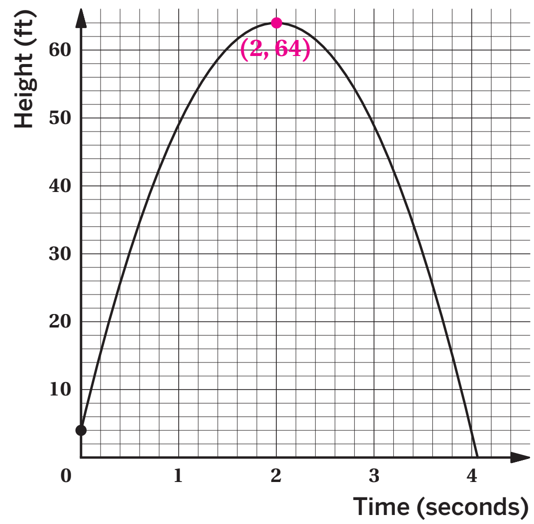
b On which day(s) was the amount of snowfall 1.5 in.?

Days 1, 3, 6

c Is the amount of snowfall a function of the day or is the day a function of the amount of snowfall? Explain your thinking. Then determine the independent and dependent variables.

The amount of snowfall is a function of the day.
***Explanations vary*. For each day, there is one corresponding amount of snowfall. The independent variable is the day and the dependent variable is the amount of snowfall.**

4. Andre's science class is launching model rockets from a picnic table. The graph represents a model rocket that is launched upward, that then falls to the ground.



a What is the height at which the model rocket was launched?

4 ft

b Plot the point that represents the greatest height of the model rocket. How long did it take for the model rocket to reach that height?

2 sec

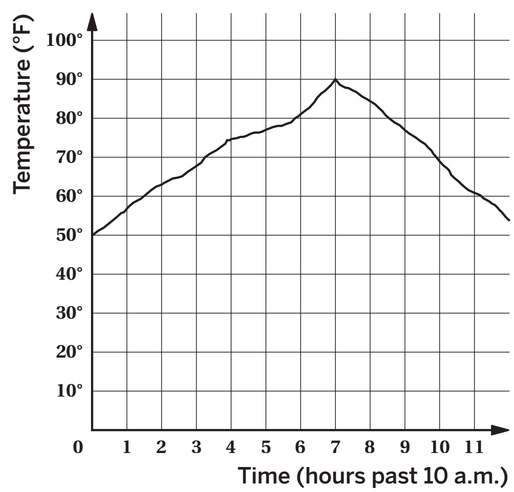
c Determine one time interval when the height of the object was increasing.

***Responses vary.* The model rocket's height was increasing from 0 to 2 seconds.**

d Determine one time interval when the height of the object was decreasing.

***Responses vary.* The model rocket's height was decreasing from 2 to about 4 seconds.**

5. The graph shows the temperature between 10 a.m. and 10 p.m. for one day a city.



a When the input is 7, what is the output? What does that tell you about the time and temperature?

90°. At 5 p.m., the temperature was 90°F.

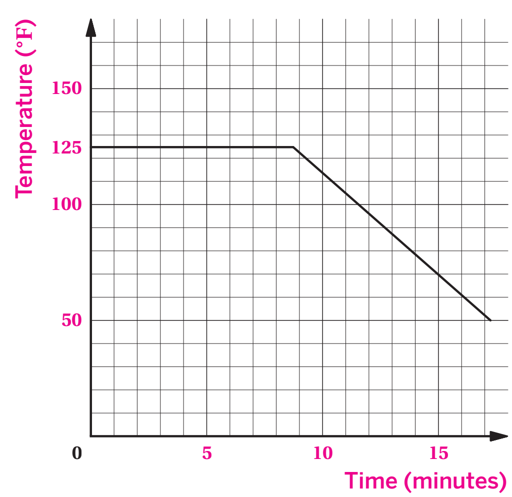
b Determine one time interval when the temperature was increasing.

The temperature was increasing from 10 a.m. to 5 p.m.

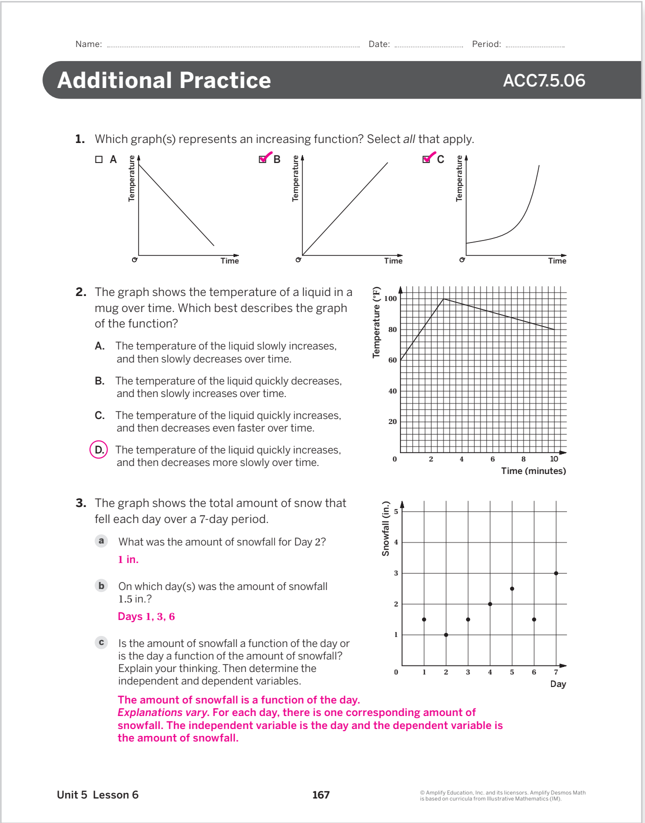
c Determine one time interval when the temperature was decreasing.

The temperature was decreasing from 5 p.m. to 10 p.m.

6. Describe a real-world situation that would represent the graph. Label the axes.



***Responses vary.* A cup of hot tea is kept warm in a kettle at a constant temperature for a time before ice is added to cool it down.**



Name: Date: Period:

Additional Practice ACC7.5.06

1. Which graph(s) represents an increasing function? Select *all* that apply.

☐ A ☑ B ☑ C

2. The graph shows the temperature of a liquid in a mug over time. Which best describes the graph of the function?

A. The temperature of the liquid slowly increases, and then slowly decreases over time.

B. The temperature of the liquid quickly decreases, and then slowly increases over time.

C. The temperature of the liquid quickly increases, and then decreases even faster over time.

D. The temperature of the liquid quickly increases, and then decreases more slowly over time.

3. The graph shows the total amount of snow that fell each day over a 7-day period.

a. What was the amount of snowfall for Day 2?

1 in.

b. On which day(s) was the amount of snowfall 1.5 in.?

Days 1, 3, 6

c. Is the amount of snowfall a function of the day or is the day a function of the amount of snowfall? Explain your thinking. Then determine the independent and dependent variables.

The amount of snowfall is a function of the day.
Explanations vary. **For each day, there is one corresponding amount of snowfall. The independent variable is the day and the dependent variable is the amount of snowfall.**

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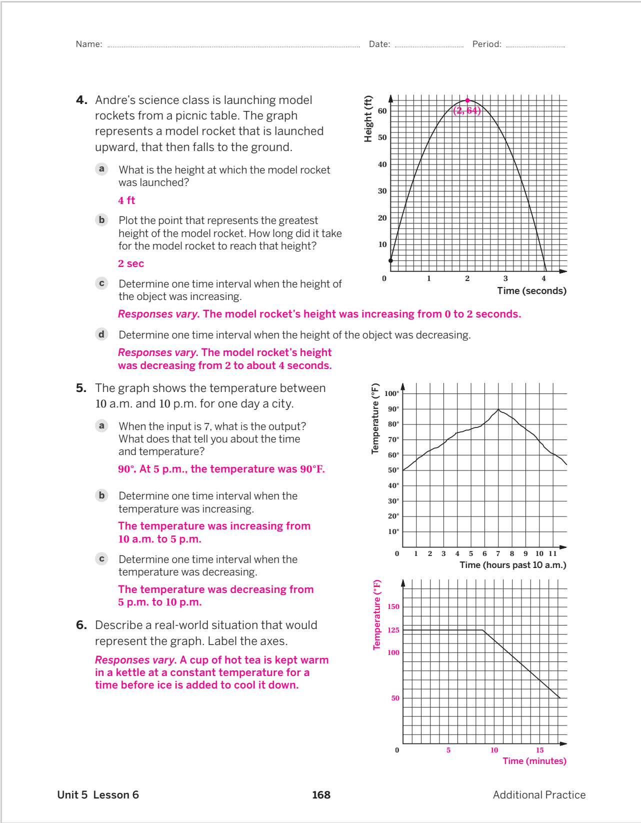


Name: Date: Period:

4. Andre's science class is launching model rockets from a picnic table. The graph represents a model rocket that is launched upward, that then falls to the ground.

a. What is the height at which the model rocket was launched?

4 ft

b. Plot the point that represents the greatest height of the model rocket. How long did it take for the model rocket to reach that height?

2 sec

c. Determine one time interval when the height of the object was increasing.

Responses vary. **The model rocket's height was increasing from 0 to 2 seconds.**

d. Determine one time interval when the height of the object was decreasing.

Responses vary. **The model rocket's height was decreasing from 2 to about 4 seconds.**

5. The graph shows the temperature between 10 a.m. and 10 p.m. for one day a city.

a. When the input is 7, what is the output? What does that tell you about the time and temperature?

90°. At 5 p.m., the temperature was 90°F.

b. Determine one time interval when the temperature was increasing.

The temperature was increasing from 10 a.m. to 5 p.m.

c. Determine one time interval when the temperature was decreasing.

The temperature was decreasing from 5 p.m. to 10 p.m.

6. Describe a real-world situation that would represent the graph. Label the axes.

Responses vary. **A cup of hot tea is kept warm in a kettle at a constant temperature for a time before ice is added to cool it down.**

Unit 5 Lesson 6 168 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.5
2	1	8.F.A.5
3	2	8.F.A.5
4	2	8.F.A.5
5	2	8.F.A.5
6	3	8.F.A.5

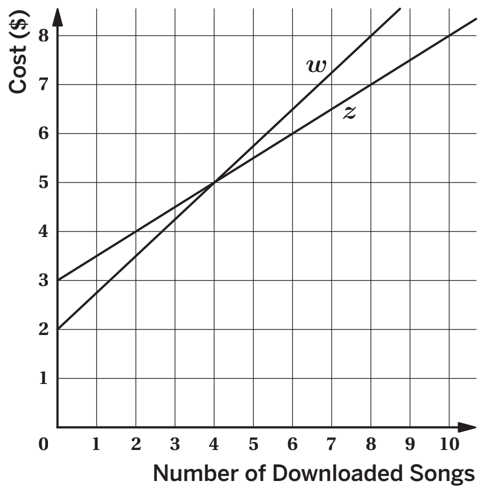
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.5.07

1. Andre is considering two online music subscriptions. Company A charges a flat fee of \$3 plus \$0.50 per downloaded song. Company B charges a flat fee of \$2 plus \$0.80 per downloaded song. Match the companies to the lines w and z shown on the graph.

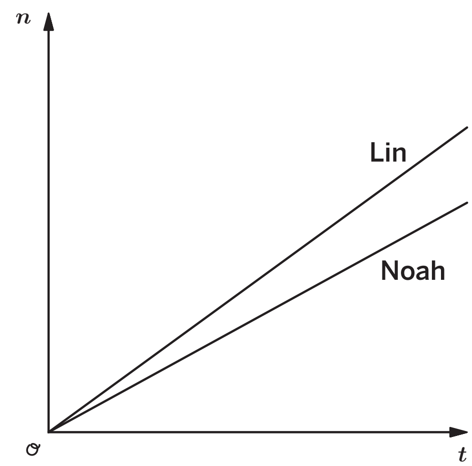


Company A is represented by line z

Company B is represented by line w

2. Noah and Lin are having a typing contest. The graph shows the number of words typed, n, for each student from the start of the contest as a function of time, t. Who is typing faster? Explain your thinking.

Lin is typing faster because Lin's line is steeper than Noah's line.



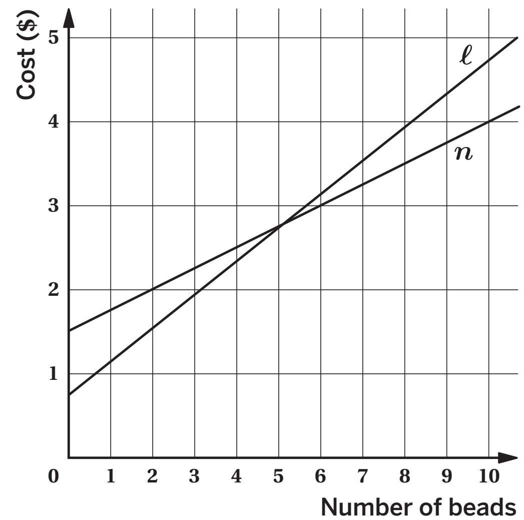
3. Mai and her sibling each want to buy a new scooter. They plan to borrow money from their parents. They each plan to repay their parents the same amount of money every week, but Mai's scooter costs less than her sibling's scooter. Suppose there is a graph that shows the amount they owe their parents, in dollars, from when they begin paying their parents back.

a As you read the graphs from left to right, would you expect the lines to increase or decrease?
Decrease.

b How would you expect the lines representing Mai and her sibling to be different? Explain your thinking.
I expect the y-intercepts to be different. Mai's initial amount owed will be less than her sibling's because her scooter costs less.

c How would you expect the lines representing Mai and her sibling to be alike? Explain your thinking.
I expect the slopes to be the same because they are paying the same amount of money to their parents each week.

4. Shawn and Elena each sell customized bracelets. Shawn charges $0.75 plus $0.40 per bead added to the bracelet. Elena charges $1.50 plus d dollars per bead added to the bracelet.



a Match each person to the lines ℓ and n shown on the graph.

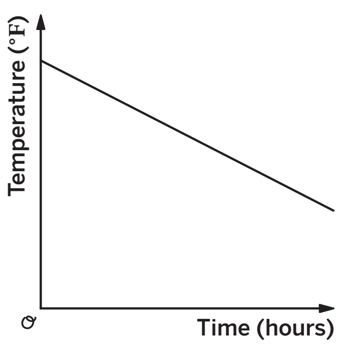
Line ℓ represents Shawn. Line n represents Elena.

b For Elena, is the additional charge per bead greater than or less than $0.40? Explain your thinking.

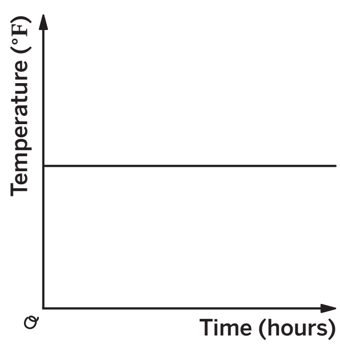
The charge per bead is less than $0.40 because line n is not as steep as line ℓ.

5. One day, a certain city's temperature steadily increased from noon to 6 p.m. Then from 6 p.m. until midnight its temperature steadily decreased.

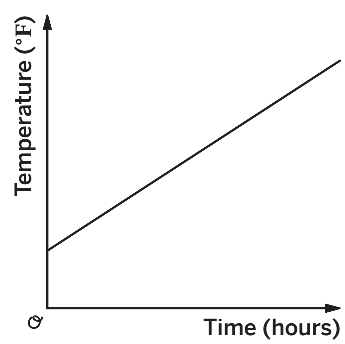
A.



B.



C.



a Which of the graphs is most likely to represent the temperature from noon to 6 p.m.? Explain your thinking.

Graph C. *Explanations vary.* The city's temperature increases at a steady rate, so the line must increase.

b Which of the graphs is most likely to represent the temperature from 6 p.m. to midnight? Explain your thinking.

Graph A. *Explanations vary.* It is the only graph with a decreasing slope, and the temperature decreases during that time span.

c Why does the other graph not likely represent the temperature during either time span? Explain your thinking.

***Responses vary.* Graph B would not make sense because Graph B shows the temperature remaining constant, but the temperature did not remain constant.**

6. Fitness Center A charges a flat fee of $25 plus $2 per visit. The cost of Fitness Center B is represented by $y = 2x + 20$ where x is the number of visits. Which fitness center would you choose? Explain your thinking.

***Responses vary.* I would choose Fitness Center B. The cost will always be greater for Fitness Center A because the cost per visit is the same.**

Name: .... Date: .... Period: ....

Additional Practice ACC7.5.07

1. Andre is considering two online music subscriptions. Company A charges a flat fee of \$3 plus \$0.50 per downloaded song. Company B charges a flat fee of \$2 plus \$0.80 per downloaded song. Match the companies to the lines w and z shown on the graph.

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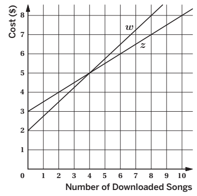

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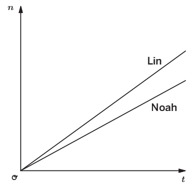


3. Mai and her sibling each want to buy a new scooter. They plan to borrow money from their parents. They each plan to repay their parents the same amount of money every week, but Mai's scooter costs less than her sibling's scooter. Suppose there is a graph that shows the amount they owe their parents, in dollars, from when they begin paying their parents back.

a. As you read the graphs from left to right, would you expect the lines to increase or decrease?
Decrease.

b. How would you expect the lines representing Mai and her sibling to be different? Explain your thinking.
I expect the y-intercepts to be different. Mai's initial amount owed will be less than her sibling's because her scooter costs less.

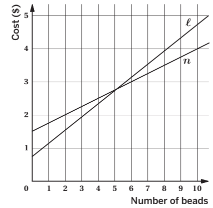
c. How would you expect the lines representing Mai and her sibling to be alike? Explain your thinking.
I expect the slopes to be the same because they are paying the same amount of money to their parents each week.

Name: .... Date: .... Period: ....

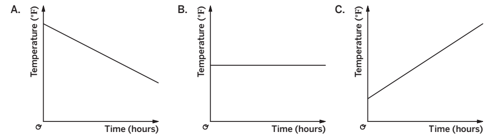
4. Shawn and Elena each sell customized bracelets. Shawn charges \$0.75 plus \$0.40 per bead added to the bracelet. Elena charges \$1.50 plus d dollars per bead added to the bracelet.

a. Match each person to the lines ℓ and n shown on the graph.
Line ℓ represents Shawn. Line n represents Elena.

b. For Elena, is the additional charge per bead greater than or less than \$0.40? Explain your thinking.
The charge per bead is less than \$0.40 because line n is not as steep as line ℓ.



5. One day, a certain city's temperature steadily increased from noon to 6 p.m. Then from 6 p.m. until midnight its temperature steadily decreased.



a. Which of the graphs is most likely to represent the temperature from noon to 6 p.m.? Explain your thinking.
Graph C. *Explanations vary.* The city's temperature increases at a steady rate, so the line must increase.

b. Which of the graphs is most likely to represent the temperature from 6 p.m. to midnight? Explain your thinking.
Graph A. *Explanations vary.* It is the only graph with a decreasing slope, and the temperature decreases during that time span.

c. Why does the other graph not likely represent the temperature during either time span? Explain your thinking.
***Responses vary.* Graph B would not make sense because Graph B shows the temperature remaining constant, but the temperature did not remain constant.**

6. Fitness Center A charges a flat fee of \$25 plus \$2 per visit. The cost of Fitness Center B is represented by $y = 2x + 20$ where x is the number of visits. Which fitness center would you choose? Explain your thinking.

***Responses vary.* I would choose Fitness Center B. The cost will always be greater for Fitness Center A because the cost per visit is the same.**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.A.2
2	1	8.F.A.2
3	2	8.F.A.2
4	2	8.F.A.2
5	2	8.F.A.2
6	3	8.F.A.2

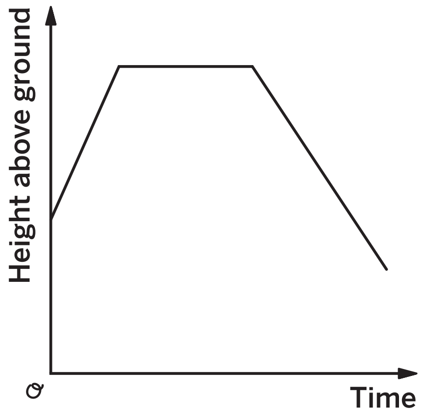
Notes:

Name: .... Date: .... Period: ....

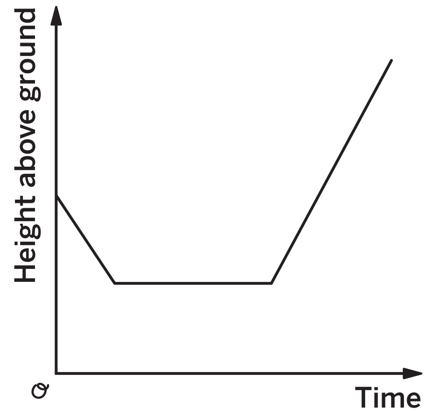
Additional Practice 5.09

1. The graph shows the height above ground of two elevators over time. Match each graph to the situation it represents.

Graph 1



Graph 2



a The elevator starts at the fourth floor, then stops at the second floor, and then goes up to the seventh floor.

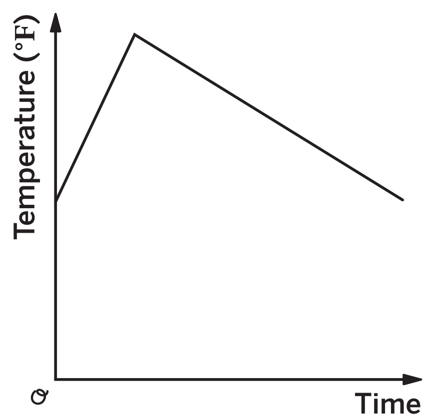
Graph 2

b The elevator starts at the third floor, then stops at the sixth floor, and then goes down to the second floor.

Graph 1

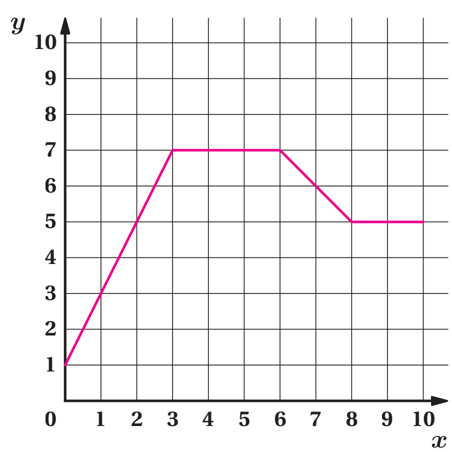
2. Mai had a teacup of water. The graph shows the temperature of water in a teacup. Describe a possible situation that could represent the graph.

Sample response: The water was heated quickly, possibly in a microwave. Then the water cools to room temperature.

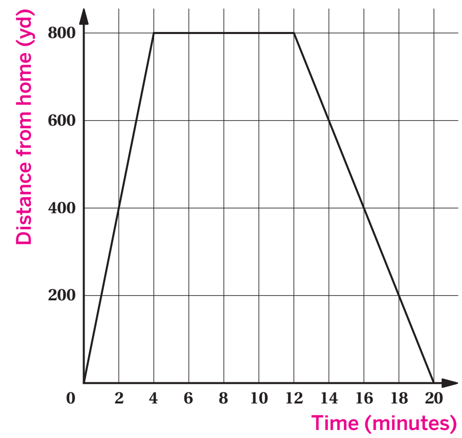


3. The following statements are descriptions of the linear segments that make up a piecewise function. Draw a graph of the piecewise function described.

- Starts at the y-intercept 1.
- Increasing at its greatest constant rate from $x = 0$ to $x = 3$.
- Has a slope of 0 from $x = 3$ to $x = 6$.
- Decreasing at a constant rate from $x = 6$ to $x = 8$.
- Has a slope of 0 from $x = 8$ to $x = 10$.



4. Clare walked her dog to the dog park and let the dog play at the park. Then she walked home.



a The graph shows her distance walked in yards, as a function of time, in minutes. Add the axes labels to the graph to show this.

b When did she reach the dog park?
After 4 minutes

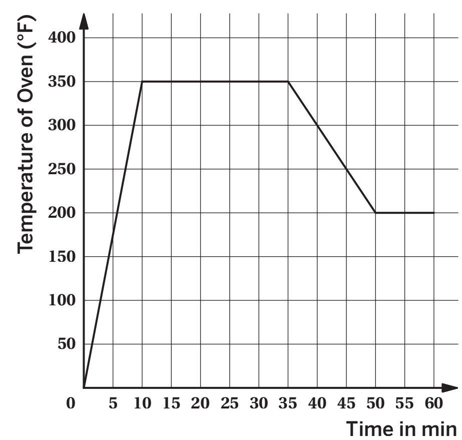
c How far did Clare walk to get to the dog park?
800 yd

d How long were Clare and her dog at the dog park?
8 minutes

e At what rate did she walk to the dog park?
200 yd per minute

f At what rate did she walk home from the dog park?
100 yd per minute

5. Andre's dad is baking a casserole. He preheated the oven. The graph shows the temperature of the oven from when he turned the oven on.



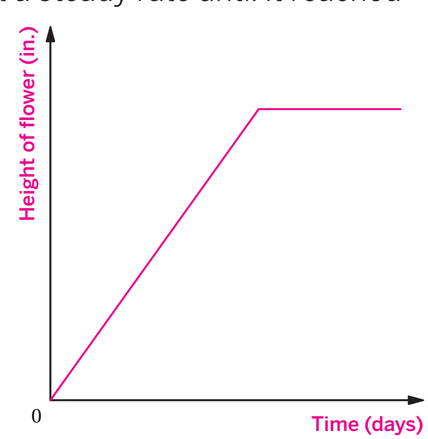
a When was the temperature of the oven increasing?
From 0 to 10 minutes

b What was happening from 10 minutes to 35 minutes after the stove was turned on?
Sample response: The temperature remained constant at 350° F while the casserole was baking.

c What rate did the temperature of the oven increase?
35° F per minute because $\frac{350}{10} = 35$.

6. Noah graphed the growth of his flower. The flower grew at a steady rate until it reached its full height and then stopped growing.

a What does Noah's graph look like? **Sample response:**



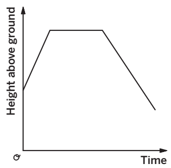
b Han suggested that Noah's flower had a constant growth rate during this time because all parts of the graph can be represented by straight lines. Is he correct? Explain your thinking.
No. Sample response: While sections of Noah's graph represent constant rates of change, the graph as a whole does not have one constant rate of change. The rate of change changes from positive to zero, so it is not one number the entire time.

Name: .... Date: .... Period: ....

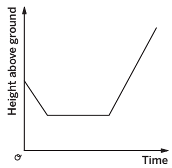
Additional Practice 5.09

1. The graph shows the height above ground of two elevators over time. Match each graph to the situation it represents.

Graph 1



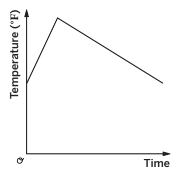
Graph 2



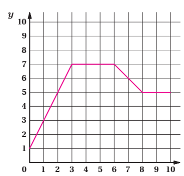
a. The elevator starts at the fourth floor, then stops at the second floor, and then goes up to the seventh floor.
Graph 2

b. The elevator starts at the third floor, then stops at the sixth floor, and then goes down to the second floor.
Graph 1

2. Mai had a teacup of water. The graph shows the temperature of water in a teacup. Describe a possible situation that could represent the graph.
Sample response: The water was heated quickly, possibly in a microwave. Then the water cools to room temperature.



3. The following statements are descriptions of the linear segments that make up a piecewise function. Draw a graph of the piecewise function described.
   - Starts at the y-intercept 1.
   - Increasing at its greatest constant rate from $x = 0$ to $x = 3$.
   - Has a slope of 0 from $x = 3$ to $x = 6$.
   - Decreasing at a constant rate from $x = 6$ to $x = 8$.
   - Has a slope of 0 from $x = 8$ to $x = 10$.



Unit 5 Lesson 9 117

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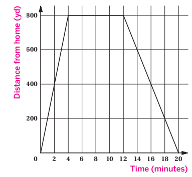
Name: .... Date: .... Period: ....

4. Clare walked her dog to the dog park and let the dog play at the park. Then she walked home.



a. The graph shows her distance walked in yards, as a function of time, in minutes. Add the axes labels to the graph to show this.

b. When did she reach the dog park?
After 4 minutes

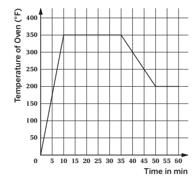
c. How far did Clare walk to get to the dog park?
800 yd

d. How long were Clare and her dog at the dog park?
8 minutes

e. At what rate did she walk to the dog park?
200 yd per minute

f. At what rate did she walk home from the dog park?
100 yd per minute

5. Andre's dad is baking a casserole. He preheated the oven. The graph shows the temperature of the oven from when he turned the oven on.



a. When was the temperature of the oven increasing?
From 0 to 10 minutes

b. What was happening from 10 minutes to 35 minutes after the stove was turned on?
Sample response: The temperature remained constant at 350° F while the casserole was baking.

c. What rate did the temperature of the oven increase?
35° F per minute because $\frac{350}{10} = 35$.

6. Noah graphed the growth of his flower. The flower grew at a steady rate until it reached its full height and then stopped growing.

a. What does Noah's graph look like? **Sample response:**

b. Han suggested that Noah's flower had a constant growth rate during this time because all parts of the graph can be represented by straight lines. Is he correct? Explain your thinking.
No. Sample response: While sections of Noah's graph represent constant rates of change, the graph as a whole does not have one constant rate of change. The rate of change changes from positive to zero, so it is not one number the entire time.

Unit 5 Lesson 9 118 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.B.5
2	1	8.F.B.5
3	2	8.F.B.5
4	2	8.F.B.5
5	2	8.F.B.5
6	3	8.F.B.5

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.6.10

1. The table shows the results of a survey about news-reading habits among different age groups. What does the number 132 represent in the table?

	Internet Media	Print Media	Total
18–25 Year Olds	151	28	179
26–45 Year Olds	132	72	204
Total	283	100	383

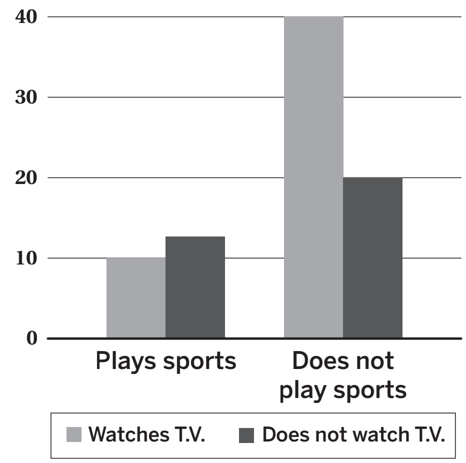
A. 132 people, who are 18–25 years old and read print media

B. 132 people, who are 26–45 years old and read print media

C. 132 people, who are 18–25 years old and read internet media

D. 132 people, who are 26–45 years old and read internet media

2. The bar graph shows data about watching T.V. and playing sports for several students in a middle school. How many students do not play sports and do not watch T.V.?



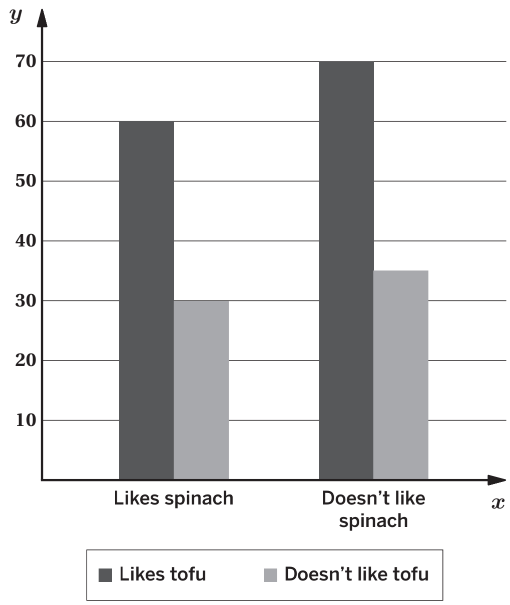
A. 10

B. 12

C. 20

D. 40

3. Create table based on the information in the bar graph.

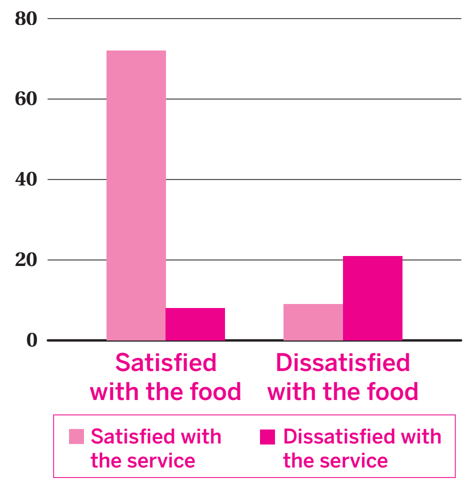


	Likes spinach	Doesn't like spinach	Total
Likes tofu	60	70	130
Doesn't like tofu	30	35	65
Total	90	105	195

Name: .... Date: .... Period: ....

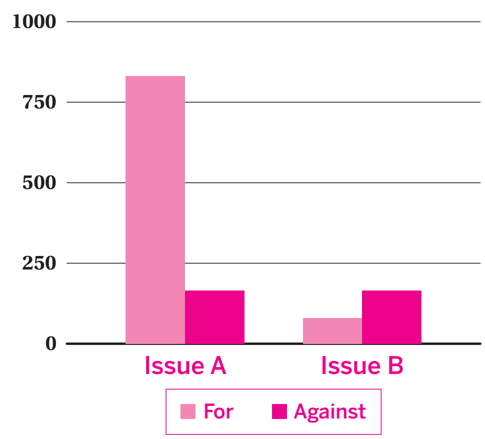
4. A restaurant manager records whether people were satisfied with their food and service. Complete the two-way table. Then create a bar graph based on the information in the table.

	Satisfied With the Service	Dissatisfied With the Service	Total
Satisfied With the Food	**72**	8	**80**
Dissatisfied With the Food	9	**21**	30
Total	**81**	29	110

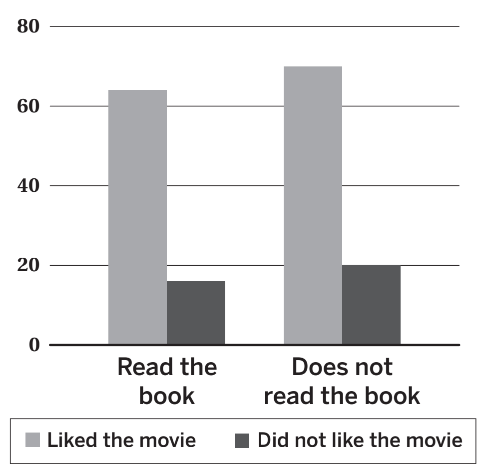


5. A journalist surveys several people about whether they will vote for or against two issues. Complete the two-way table. Then create a bar graph to represent the data.

	For	Against	Total
Issue A	832	**165**	997
Issue B	**80**	160	**240**
Total	912	**325**	1,237



6. A teacher wants to know whether reading a certain book affects liking the movie adaptation of the book for middle school students. Several students' responses were recorded and then graphed using the bar graph shown. What claim might the teacher make based on this data? Explain your thinking.



	Liked the Movie	Disliked the Movie	Total
Read the Book	64	16	80
Did Not Read the Book	70	20	90
Total	134	36	170

***Responses vary.* While more students liked the movie than did not like the movie, the students had similar preferences for the movie whether they read the book or not.**

Name: .... Date: .... Period: ....

Additional Practice ACC7.6.10

1. The table shows the results of a survey about news-reading habits among different age groups. What does the number 132 represent in the table?

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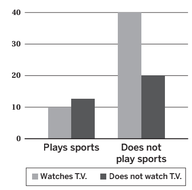
A. 10

B. 12

C. 20

D. 40



3. Create table based on the information in the bar graph.

	Likes spinach	Doesn't like spinach	Total
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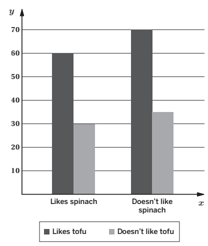

Name: .... Date: .... Period: ....

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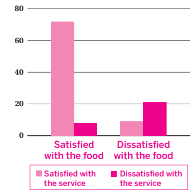

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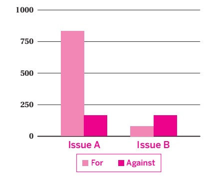

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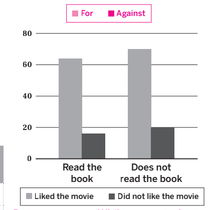


Responses vary. While more students liked the movie than did not like the movie, the students had similar preferences for the movie whether they read the book or not.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.SP.A.4
2	1	8.SP.A.4
3	2	8.SP.A.4
4	2	8.SP.A.4
5	2	8.SP.A.4
6	3	8.SP.A.4

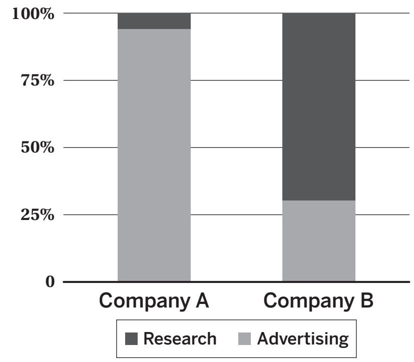
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.6.11

1. The segmented bar graph displays two companies' yearly budgets, in millions of U.S. dollars. Which of the following describes the data?



A. There is no association; the relative frequency for each company is roughly the same.

B. There is a greater percentage of budget spent on research by Company A.

C. There is a greater percentage of budget spent on advertising by Company A.

D. There is a greater percentage of budget spent on advertising by Company B.

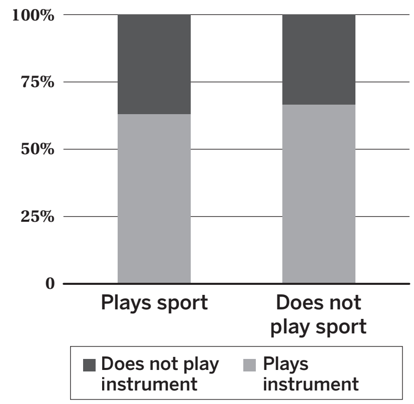
2. Complete the table to show the relative frequencies for each column.

	Class A	Class B
Prefers Math	6	8
Prefers Science	2	10
Total	8	18

	Class A	Class B
Prefers Math	75%	**44%**
Prefers Science	**25%**	**56%**
Total	**100%**	**100%**

3. Students were asked if they play a sport or play a musical instrument. The results are shown in the table and on the segmented bar graph. Is there evidence of an association between playing a sport and playing an instrument? Explain your thinking.

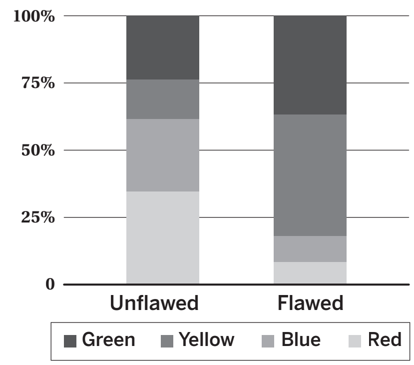
	Plays Instrument	Does not Play Instrument	Total
Plays Sport	12	7	19
Does Not Play Sport	10	5	15
Total	22	12	34



No. *Explanations vary.* The relative frequency for playing or not playing a sport is roughly the same whether they play an instrument or not.

4. The manager of an eraser factory notices some flaws in certain erasers. Each color eraser is made through a different machine. The manager collects data on the number of flawed and unflawed erasers of each color. The results are recorded in the table and are shown on the segmented bar graph. Is there evidence that the flawed erasers are associated with certain colors? Explain your thinking.

	Unflawed	Flawed
Red	285	15
Blue	223	17
Yellow	120	80
Green	195	65



Yes. *Explanations vary*. The segmented bar graph shows a greater frequency of yellow and green flawed erasers.

5. A scientist is interested in whether certain plants attract more bees or butterflies. Do these data show an association between bees and butterflies and flower types? Explain your thinking.

	Daisies	Lavender
Bees	17	23
Butterflies	22	28

No. *Explanations vary*. Bees and butterflies appear to prefer each flower with similar relative frequencies. 42.5% of bees prefer daisies while 44% of butterflies prefer daisies and 57.5% of bees prefer lavender while 56% of butterflies prefer lavender flowers.

6. Several students at a middle school were asked whether they preferred drama or band classes. The results are shown in the table.

Write a question that could be answered by the data in the table.

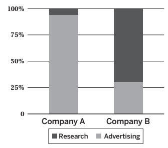
***Responses vary*. Are students in 8th grade more likely to take drama than students in 7th grade?**

	Drama	Band	Total
7th Grade	72	51	123
8th Grade	85	32	117
Total	157	83	240

Name: Date: Period:

Additional Practice ACC7.6.11

1. The segmented bar graph displays two companies' yearly budgets, in millions of U.S. dollars. Which of the following describes the data?



A. There is no association; the relative frequency for each company is roughly the same.

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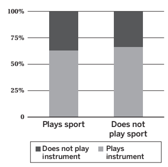

No. *Explanations vary.* The relative frequency for playing or not playing a sport is roughly the same whether they play an instrument or not.

Name: Date: Period:

4. The manager of an eraser factory notices some flaws in certain erasers. Each color eraser is made through a different machine. The manager collects data on the number of flawed and unflawed erasers of each color. The results are recorded in the table and are shown on the segmented bar graph. Is there evidence that the flawed erasers are associated with certain colors? Explain your thinking.

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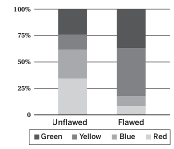

Yes. *Explanations vary.* The segmented bar graph shows a greater frequency of yellow and green flawed erasers.

5. A scientist is interested in whether certain plants attract more bees or butterflies. Do these data show an association between bees and butterflies and flower types? Explain your thinking.

	Daisies	Lavender
Bees	17	23
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6. Several students at a middle school were asked whether they preferred drama or band classes. The results are shown in the table.

Write a question that could be answered by the data in the table.

	Drama	Band	Total
7th Grade	72	51	123
8th Grade	85	32	117
Total	157	83	240

Responses vary. Are students in 8th grade more likely to take drama than students in 7th grade?

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.SP.A.4
2	1	8.SP.A.4
3	2	8.SP.A.4
4	2	8.SP.A.4
5	2	8.SP.A.4
6	3	8.SP.A.4

Notes:

Name: .... Date: .... Period: ....

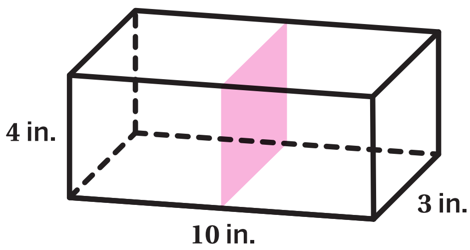
Additional Practice

ACC7.7.01

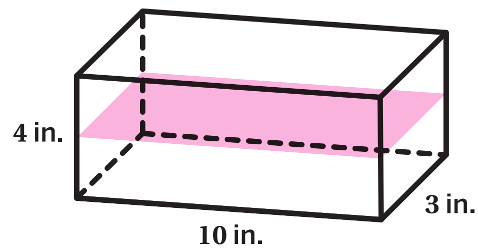
Problems 1–4: Show how to cut a rectangular prism to make each cross section.

Sample shown on figures.

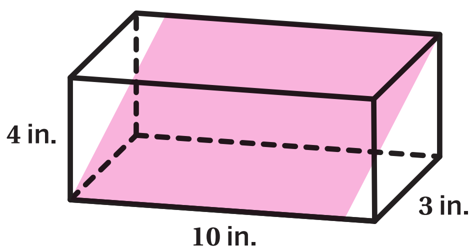
1. A 3-by-4-inch rectangle



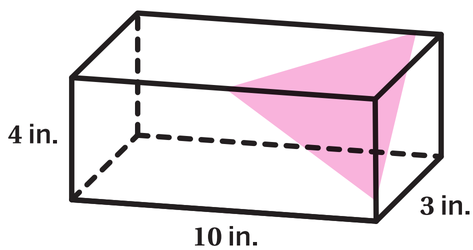
2. A 4-by-10-inch rectangle



3. A different rectangle than Problem 1 or 2

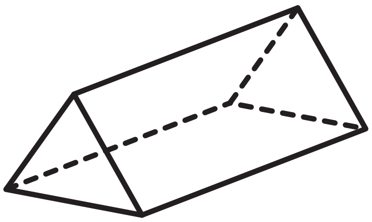


4. A triangle



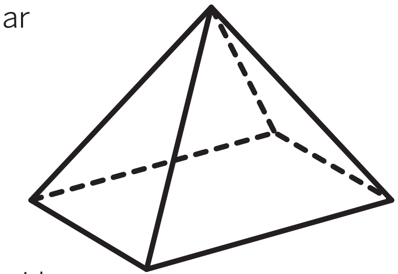
5. Select *all* of the cross sections you can make from the triangular prism shown.

- [x] **A.** Triangle
- [ ] **B.** Square
- [x] **C.** Rectangle
- [ ] **D.** Trapezoid
- [ ] **E.** Hexagon



Name: .... Date: .... Period: ....

6. Which of the following descriptions could result in a triangular cross section for the rectangular pyramid shown?



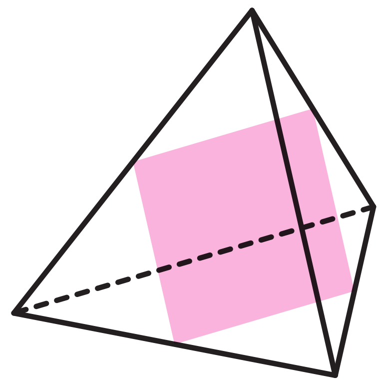
A. Cut the pyramid parallel to its base halfway up.

B. Cut the pyramid parallel to its base near the top.

C. Cut the pyramid parallel to its base near the bottom.

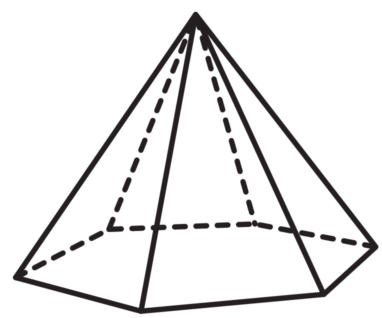
D. Cut the pyramid vertical to its base through the top of the pyramid.

7. Ben says: *No matter which way you cut this triangular pyramid figure, the cross section will be a triangle.* Harper says: *I'm not so sure.* Describe or show a cut that Harper might be thinking of.



***Responses vary.* Sample shown on figure. Harper might be thinking of cutting the pyramid diagonally from the side of the pyramid to the base.**

Problems 8–10: The pyramid shown has a hexagonal base. The side lengths of the hexagon are equal. Describe the cross section that will result if the pyramid is cut:



8. Parallel to the base

A hexagon smaller than the base

9. Vertical to the base through the top point of the pyramid

A triangle

10. Describe another way you could cut the pyramid that would result in a different cross section.

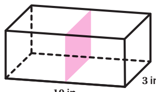
***Responses vary.* A rectangular cross-section by cutting diagonally from one side of the pyramid to the edge of the opposite side of the base**

Name: ______ Date: ______ Period: ______

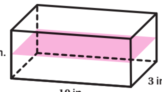
Additional Practice ACC7.7.01

Problems 1–4: Show how to cut a rectangular prism to make each cross section.
Sample shown on figures.

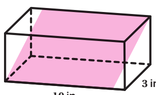
1. A 3-by-4-inch rectangle



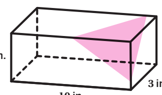
2. A 4-by-10-inch rectangle



3. A different rectangle than Problem 1 or 2

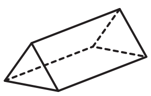


4. A triangle



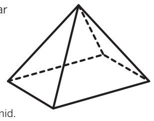
5. Select *all* of the cross sections you can make from the triangular prism shown.

- [x] **A.** Triangle
- [ ] **B.** Square
- [x] **C.** Rectangle
- [ ] **D.** Trapezoid
- [ ] **E.** Hexagon



Name: ______ Date: ______ Period: ______

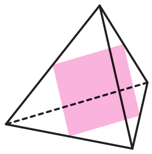
6. Which of the following descriptions could result in a triangular cross section for the rectangular pyramid shown?



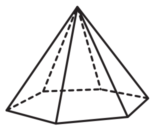
A. Cut the pyramid parallel to its base halfway up.
B. Cut the pyramid parallel to its base near the top.
C. Cut the pyramid parallel to its base near the bottom.
D. Cut the pyramid vertical to its base through the top of the pyramid.

7. Ben says: *No matter which way you cut this triangular pyramid figure, the cross section will be a triangle.* Harper says: *I'm not so sure.* Describe or show a cut that Harper might be thinking of.

Responses vary. Sample shown on figure. Harper might be thinking of cutting the pyramid diagonally from the side of the pyramid to the base.



Problems 8–10: The pyramid shown has a hexagonal base. The side lengths of the hexagon are equal. Describe the cross section that will result if the pyramid is cut:



8. Parallel to the base

A hexagon smaller than the base

9. Vertical to the base through the top point of the pyramid

A triangle

10. Describe another way you could cut the pyramid that would result in a different cross section.

Responses vary. A rectangular cross-section by cutting diagonally from one side of the pyramid to the edge of the opposite side of the base

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	7.G.A.3
2	2	7.G.A.3
3	2	7.G.A.3
4	2	7.G.A.3
5	2	7.G.A.3
6	2	7.G.A.3
7	2	7.G.A.3
8	2	7.G.A.3
9	2	7.G.A.3
10	2	7.G.A.3

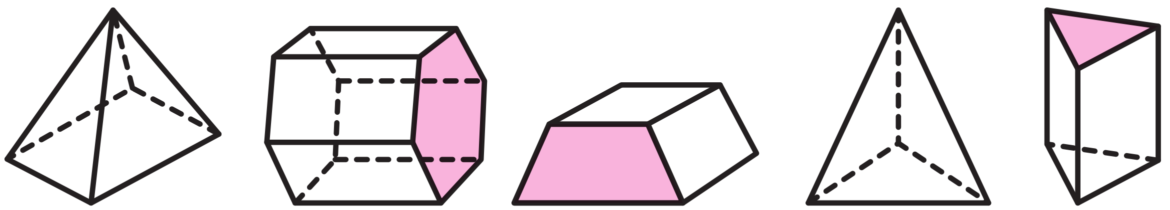
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.7.02

Problems 1–2: Here is a set of 3-D objects.



Object A   Object B   Object C   Object D   Object E

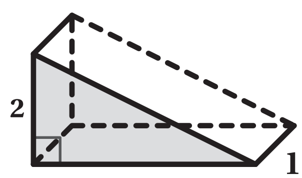
1. Circle all of the prisms.

2. For each prism, shade one of the bases. **Sample shown on figures.**

Problems 3–5: Here are three prisms with the same base.

3. Determine the volume of prism A.

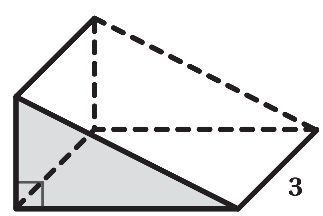
   4 cubic units



A

4. Determine the volume of prism B.

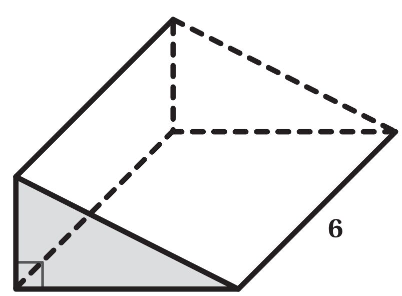
   12 cubic units



B

5. Determine the volume of prism C.

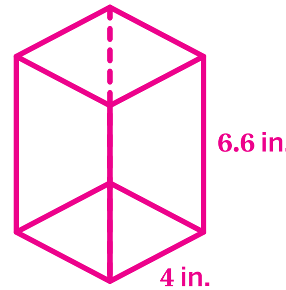
   24 cubic units



C

Name: .... Date: .... Period: ....

Problems 6–7: The base of a rectangular prism is a square with edges of 4 inches. The height of the prism is 6.6 inches.



6. Draw the prism and label the measurements.

Drawings vary. Sample shown.

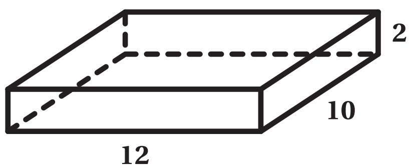
7. Calculate the volume of the prism. Show or explain your thinking.

105.6 cubic inches. *Explanations vary.*

$V = (4 \cdot 4) \cdot 6.6 = 105.6$ cubic inches

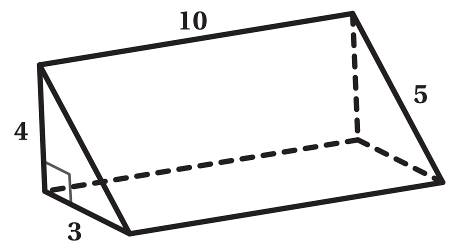
Problems 8–9: Determine the volume of each prism. Show or explain your thinking.

8.



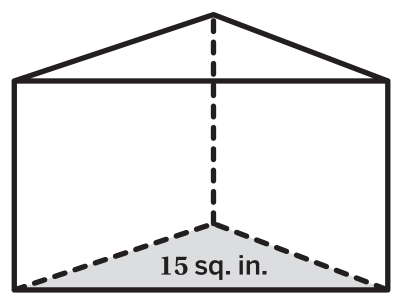
240 cubic units. *Explanations vary.* This shape is a prism, so the volume is the area of the base multiplied by the height. $120 \cdot 2 = 240$ cubic units

9.



60 cubic units. *Explanations vary.* This shape is a prism, so the volume is the area of the base multiplied by the height. $\frac{1}{2} \cdot 4 \cdot 3 \cdot 10 = 60$ cubic units

10. The volume of this prism is between 60 cubic inches and 80 cubic inches. The area of the base is 15 square inches. What is a possible height of the prism? Show or explain your thinking.



Between 4 and $5\frac{1}{3}$ inches. *Work varies.*

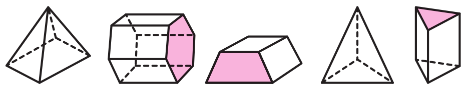
If $V = 60$ cubic inches:	If $V = 80$ cubic inches:
$V = B \cdot h$	$V = B \cdot h$
$60 = 15 \cdot h$	$80 = 15 \cdot h$
4 inches $= h$	$5\frac{1}{3}$ inches $= h$

Additional Practice | Answer Key

Name: .... Date: .... Period: ....

Additional Practice ACC7.7.02

Problems 1–2: Here is a set of 3-D objects.

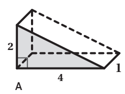


Object A | Object B | Object C | Object D | Object E

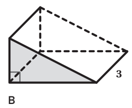
1. Circle all of the prisms.
2. For each prism, shade one of the bases. **Sample shown on figures.**

Problems 3–5: Here are three prisms with the same base.

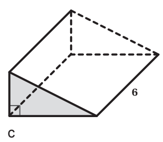
3. Determine the volume of prism A.
4 cubic units



4. Determine the volume of prism B.
12 cubic units

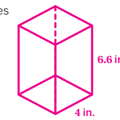


5. Determine the volume of prism C.
24 cubic units



Name: .... Date: .... Period: ....

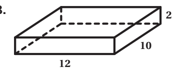
Problems 6–7: The base of a rectangular prism is a square with edges of 4 inches. The height of the prism is 6.6 inches.



6. Draw the prism and label the measurements.
***Drawings vary.* Sample shown.**
7. Calculate the volume of the prism. Show or explain your thinking.
105.6 cubic inches. *Explanations vary.*
$V = (4 \cdot 4) \cdot 6.6 = 105.6$ **cubic inches**

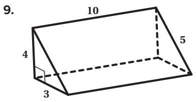
Problems 8–9: Determine the volume of each prism. Show or explain your thinking.

8.



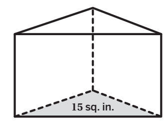
240 cubic units. *Explanations vary.* This shape is a prism, so the volume is the area of the base multiplied by the height. $120 \cdot 2 = 240$ **cubic units**

9.



60 cubic units. *Explanations vary.* This shape is a prism, so the volume is the area of the base multiplied by the height. $\frac{1}{2} \cdot 4 \cdot 3 \cdot 10 = 60$ **cubic units**

10. The volume of this prism is between 60 cubic inches and 80 cubic inches. The area of the base is 15 square inches. What is a possible height of the prism? Show or explain your thinking.



Between 4 and $5\frac{1}{3}$ inches. *Work varies.*

If $V = 60$ cubic inches:	If $V = 80$ cubic inches:
$V = B \cdot h$	$V = B \cdot h$
$60 = 15 \cdot h$	$80 = 15 \cdot h$
4 inches $= h$	$5\frac{1}{3}$ inches $= h$

Practice Problem Analysis

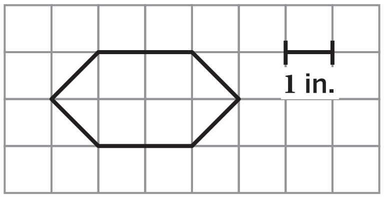
Problem	DOK	Standard(s)
1	2	7.G.B.6
2	2	7.G.B.6
3	1	7.G.B.6
4	1	7.G.B.6
5	1	7.G.B.6
6	2	7.G.B.6
7	2	7.G.B.6
8	2	7.G.B.6
9	2	7.G.B.6
10	3	7.G.B.6

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.7.03

Problems 1–2: The base of a prism is shown.



1. If the height of the prism is 5 inches, what is the volume of the prism? Show or explain your thinking.

30 cubic inches. ***Work varies.***

The area of the base is 6 square inches.

$V = B \cdot h$

$V = 6 \cdot 5$

$V = 30$ **cubic inches**

2. If the volume of the prism is 99 cubic inches, what is the height of the prism? Show or explain your thinking.

16.5 inches. ***Work varies.***

$V = B \cdot h$

$99 = 6 \cdot h$

$\frac{99}{6} = h$

$16.5 = h$

Problems 3–4: Finnias calculated the volume of the prism but made an error.

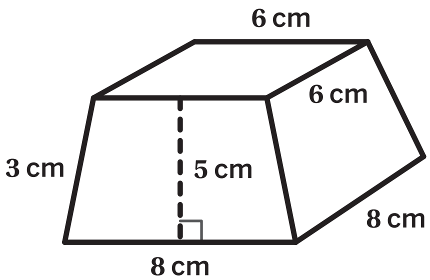
Finnias' Method

$V =$ Base $\cdot$ height

$V = (8 \cdot 8) \cdot 5$

$V = 64 \cdot 5$

$V = 320$ cubic cm



3. Explain why Finnias' method is incorrect.

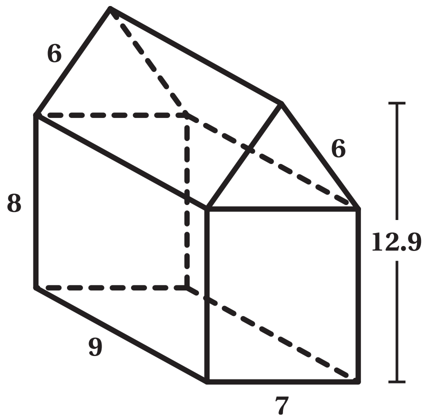
Explanations vary. **Finnias incorrectly identified the base and height of the prism. The base of the prism is a trapezoid and the height of the prism is the distance between the trapezoids.**

4. Calculate the volume of the prism.

280 cubic cm

Problems 5–8: Determine the volume of each prism. Show or explain your thinking.

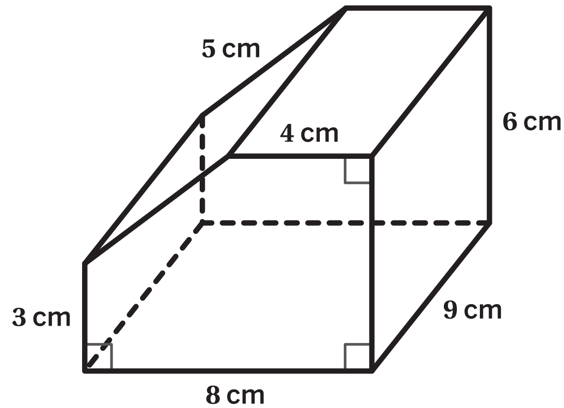
5.



658.35 cubic units. *Explanations vary.*

The base of the prism is a rectangle and a triangle. The area of the rectangle is 56 square units. The triangle has a base of 7 units and a height of $12.9 - 8 = 4.9$ units, so the area of the triangle is $\frac{1}{2} \cdot 7 \cdot 4.9 = 17.15$ square units. The total area of the base is $56 + 17.15 = 73.15$ square units. The volume is $73.15 \cdot 9 = 658.35$.

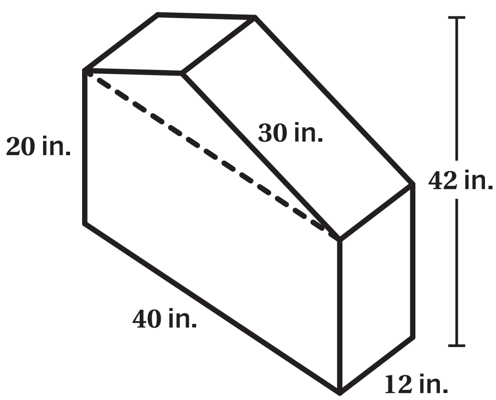
6.



378 cubic cm. *Explanations vary.*

The base of the prism is a 8-by-6 rectangle minus a right triangle with an area of $\frac{1}{2} \cdot 4 \cdot 3 = 6$. So the base has an area of $8 \cdot 6 - 6 = 42$ square centimeters. The volume is Base • height $= 42 \cdot 9 = 378$.

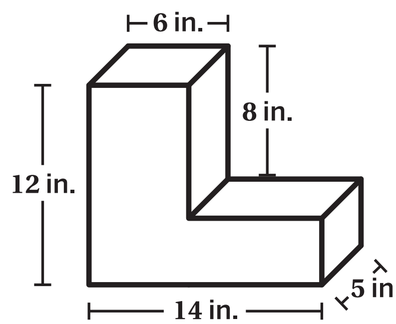
7.



14,880 cubic inches. *Explanations vary.*

The base of the prism is a 40-by-20 rectangle and a triangle with an area of $\frac{1}{2} \cdot 40 \cdot (42 - 20) = 440$. So the base has a total area of $800 + 440 = 1240$ square inches. The volume is Base • height $= 1240 \cdot 12 = 14880$.

8.



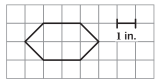
520 cubic inches. *Explanations vary.*

The base of the prism is a 6-by-12 rectangle and an 8-by-4 rectangle. So the base has an area of $(6 \cdot 12) + (8 \cdot 4) = 104$ square inches. The volume is Base • height $= 104 \cdot 5 = 520$.

Name: .... Date: .... Period: ....

Additional Practice ACC7.7.03

Problems 1–2: The base of a prism is shown.

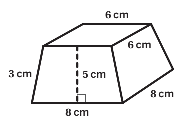


1. If the height of the prism is 5 inches, what is the volume of the prism? Show or explain your thinking.
   30 cubic inches. *Work varies.*
   The area of the base is 6 square inches.
   $V = B \cdot h$
   $V = 6 \cdot 5$
   $V = 30$ cubic inches

2. If the volume of the prism is 99 cubic inches, what is the height of the prism? Show or explain your thinking.
   16.5 inches. *Work varies.*
   $V = B \cdot h$
   $99 = 6 \cdot h$
   $\frac{99}{6} = h$
   $16.5 = h$

Problems 3–4: Finnias calculated the volume of the prism but made an error.

Finnias' Method
$V =$ Base • height
$V = (8 \cdot 8) \cdot 5$
$V = 64 \cdot 5$
$V = 320$ cubic cm



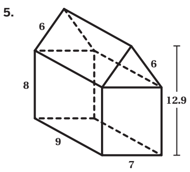
3. Explain why Finnias' method is incorrect.
   Explanations vary. Finnias incorrectly identified the base and height of the prism. The base of the prism is a trapezoid and the height of the prism is the distance between the trapezoids.

4. Calculate the volume of the prism.
   280 cubic cm

Name: .... Date: .... Period: ....

Problems 5–8: Determine the volume of each prism. Show or explain your thinking.

5.



658.35 cubic units. *Explanations vary.*
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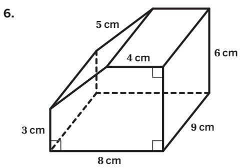
6.



378 cubic cm. *Explanations vary.*
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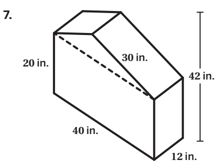

14,880 cubic inches. *Explanations vary.*
The base of the prism is a 40-by-20 rectangle and a triangle with an area of $\frac{1}{2} \cdot 40 \cdot (42 - 20) = 440$. So the base has a total area of $800 + 440 = 1240$ square inches. The volume is Base • height $= 1240 \cdot 12 = 14880$.

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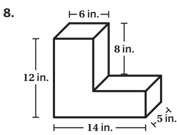

520 cubic inches. *Explanations vary.*
The base of the prism is a 6-by-12 rectangle and an 8-by-4 rectangle. So the base has an area of $(6 \cdot 12) + (8 \cdot 4) = 104$ square inches. The volume is Base • height $= 104 \cdot 5 = 520$.

Practice Problem Analysis

Problem	DOK	Standard(s)
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2	2	7.G.B.6
3	2	7.G.B.6
4	2	7.G.B.6
5	2	7.G.B.6
6	2	7.G.B.6
7	2	7.G.B.6
8	2	7.G.B.6

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.7.04

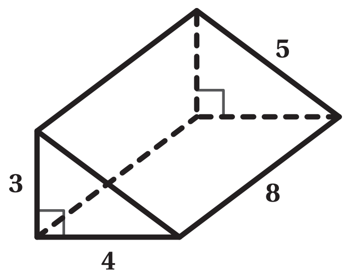
1. Select *all* the situations where knowing the surface area of an object would be useful.

- [x] **A.** The amount of cardboard needed to make a box of cereal
- [x] **B.** How much wood is needed to build cubed-shaped blocks
- [x] **C.** The amount of paint needed to paint a playhouse
- [ ] **D.** The amount of milk remaining in a jug of milk
- [x] **E.** How much brown paper is needed to cover a package
- [x] **F.** Advertising space on a highway sign

Problems 2–3: Determine the volume and surface area of each prism. Show your thinking.

Work varies.

2.



Volume: 48 cubic units

$V =$ Base • height

$V = \frac{1}{2}(3 \cdot 4) \cdot 8$

$V =$ 48 cubic units

Surface Area: 108 square units

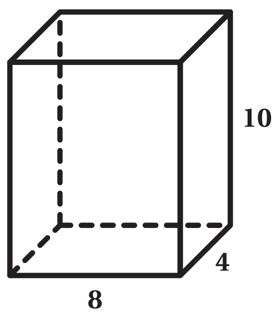
$S.A. =$ 2(Bases) + Face + Face + Face

$S.A. = 2(6) + (4 \cdot 8) + (3 \cdot 8) + (5 \cdot 8)$

$S.A. = 12 + 32 + 24 + 40$

$S.A. =$ 108 square units

3.



Volume: 320 cubic units

$V =$ Base • height

$V = (4 \cdot 8) \cdot 10$

$V =$ 320 cubic units

Surface Area: 304 square units

$S.A. =$ 2(Bases) + 2(Face) + 2(Face)

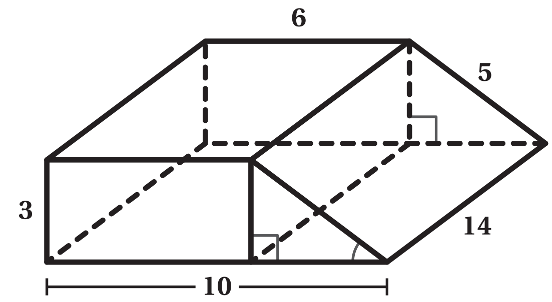
$S.A. = 2(32) + 2(4 \cdot 10) + 2(8 \cdot 10)$

$S.A. = 64 + 80 + 160$

$S.A. =$ 304 square units

Name: .... Date: .... Period: ....

Problems 4–5: Here is a three-dimensional shape.



4. Determine the surface area using two different methods. Show your thinking.

Method #1:

384 square units. ***Work varies.***

$S.A. = 2(\text{Bases}) + \text{Face}_1 + \text{Face}_2 + \text{Face}_3 + \text{Face}_4$

$S.A. = 2(3 \cdot 6 + \frac{1}{2} \cdot 3 \cdot 4) + (3 \cdot 14) + (10 \cdot 14) + (5 \cdot 14) + (6 \cdot 14)$

$S.A. = 2(18 + 6) + 42 + 140 + 70 + 84$

$S.A. = 2(24) + 336$

$S.A. = 384$ square units

Method #2:

384 square units. ***Work varies.***

$S.A. = 2(\text{Bases}) + (\text{Perimeter}_{\text{base}} \cdot \text{height of prism})$

$S.A. = 2(3 \cdot 6 + \frac{1}{2} \cdot 3 \cdot 4) + (10 + 5 + 6 + 3) \cdot 14$

$S.A. = 2(24) + 24 \cdot 14$

$S.A. = 48 + 336$

$S.A. = 384$ square units

5. Determine the volume. Show your thinking.

336 cubic units. ***Work varies.***

$V = \text{Base} \cdot \text{height}$

$V = 24 \cdot 14$

$V = 336$ cubic units

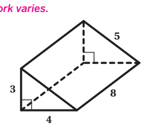
Name: ______ Date: ______ Period: ______

Additional Practice ACC7.7.04

1. Select *all* the situations where knowing the surface area of an object would be useful.
   - ☑ A. The amount of cardboard needed to make a box of cereal
   - ☑ B. How much wood is needed to build cubed-shaped blocks
   - ☑ C. The amount of paint needed to paint a playhouse
   - ☐ D. The amount of milk remaining in a jug of milk
   - ☑ E. How much brown paper is needed to cover a package
   - ☑ F. Advertising space on a highway sign

Problems 2–3: Determine the volume and surface area of each prism. Show your thinking.
Work varies.

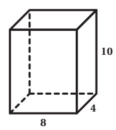
2.



Volume: 48 cubic units
$V =$ Base • height
$V = \frac{1}{2}(3 \cdot 4) \cdot 8$
$V = 48$ cubic units

Surface Area: 108 square units
$S.A. =$ 2(Bases) + Face + Face + Face
$S.A. = 2(6) + (4 \cdot 8) + (3 \cdot 8) + (5 \cdot 8)$
$S.A. = 12 + 32 + 24 + 40$
$S.A. = 108$ square units

3.



Volume: 320 cubic units
$V =$ Base • height
$V = (4 \cdot 8) \cdot 10$
$V = 320$ cubic units

Surface Area: 304 square units
$S.A. =$ 2(Bases) + 2(Face) + 2(Face)
$S.A. = 2(32) + 2(4 \cdot 10) + 2(8 \cdot 10)$
$S.A. = 64 + 80 + 160$
$S.A. = 304$ square units

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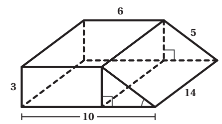
Name: ______ Date: ______ Period: ______

Problems 4–5: Here is a three-dimensional shape.



4. Determine the surface area using two different methods. Show your thinking.

Method #1:

384 square units. *Work varies.*
$S.A. = 2(\text{Bases}) + \text{Face}_1 + \text{Face}_2 + \text{Face}_3 + \text{Face}_4$
$S.A. = 2(3 \cdot 6 + \frac{1}{2} \cdot 3 \cdot 4) + (3 \cdot 14) + (10 \cdot 14) + (5 \cdot 14) + (6 \cdot 14)$
$S.A. = 2(18 + 6) + 42 + 140 + 70 + 84$
$S.A. = 2(24) + 336$
$S.A. = 384$ square units

Method #2:

384 square units. *Work varies.*
$S.A. = 2(\text{Bases}) + (\text{Perimeter}_{\text{base}} \cdot \text{height of prism})$
$S.A. = 2(3 \cdot 6 + \frac{1}{2} \cdot 3 \cdot 4) + (10 + 5 + 6 + 3) \cdot 14$
$S.A. = 2(24) + 24 \cdot 14$
$S.A. = 48 + 336$
$S.A. = 384$ square units

5. Determine the volume. Show your thinking.

336 cubic units. *Work varies.*
$V =$ Base • height
$V = 24 \cdot 14$
$V = 336$ cubic units

Unit 7 Lesson 4 206 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	7.G.B
2	2	7.G.B.6
3	2	7.G.B.6
4	2	7.G.B.6
5	2	7.G.B.6

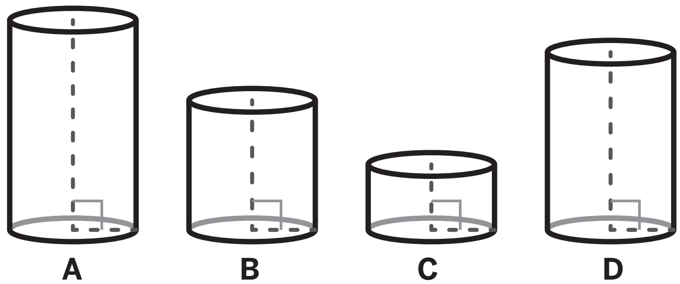
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.7.05

1. Cylinders A, B, C, and D have the same radius but different heights.

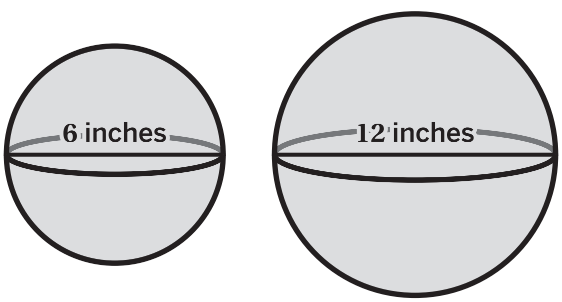


Order the cylinders from *least* volume to *greatest* volume.

C	B	D	A

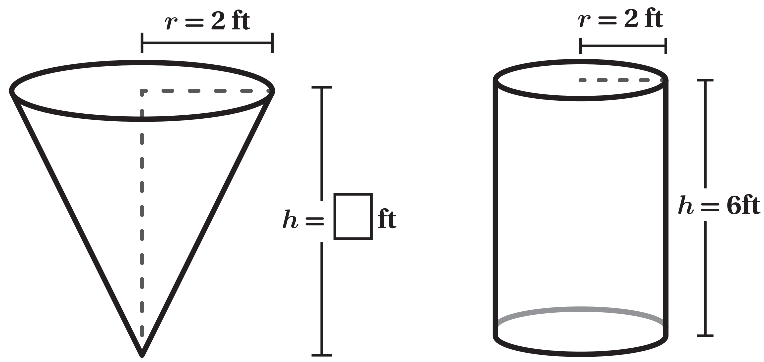
Least **Greatest**

2. The diameter of a balloon measures 6 inches. It has a volume of approximately 113 cubic inches. Kai would like a balloon that has double the diameter. He finds a spherical balloon that measures 12 inches in diameter. What is the approximate volume of this balloon? Explain your thinking.



≈ 904 cubic inches. *Explanations vary.* When the diameter of a sphere is doubled, the original volume of the sphere will be multiplied by 8.

3. Here is a cylinder and a cone. What does the height of the cone need to be so that the cylinder will have a volume that is three times greater than the cone?



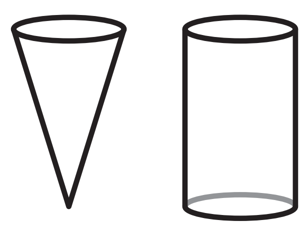
A. 2 feet

B. 3 feet

C. 6 feet

D. 18 feet

4. Maia is filling a water bottle with a paper cup. The paper cup is in the shape of a cone and has a height of 12 centimeters and a diameter of 6 centimeters. The water bottle is in the shape of a cylinder and has a height of 12 centimeters and a diameter of 6 centimeters. How many paper cups would it take to fill the water bottle? Explain your thinking.

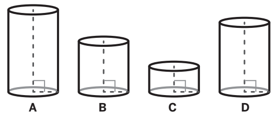


It would take 3 paper cups to fill the water bottle. *Explanations vary*. When the height and diameter are equal for a cylinder and cone, then the cylinder will have a volume that is three times greater than the cone.

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.7.05

1. Cylinders A, B, C, and D have the same radius but different heights.

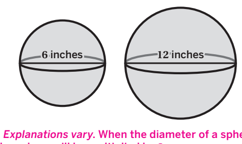


Order the cylinders from *least* volume to *greatest* volume.

C	B	D	A

Least **Greatest**

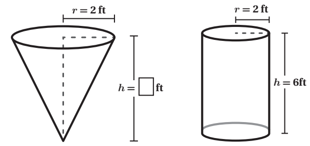
2. The diameter of a balloon measures 6 inches. It has a volume of approximately 113 cubic inches. Kai would like a balloon that has double the diameter. He finds a spherical balloon that measures 12 inches in diameter. What is the approximate volume of this balloon? Explain your thinking.



≈ 904 cubic inches. *Explanations vary.* When the diameter of a sphere is doubled, the original volume of the sphere will be multiplied by 8.

Name: ______ Date: ______ Period: ______

3. Here is a cylinder and a cone. What does the height of the cone need to be so that the cylinder will have a volume that is three times greater than the cone?



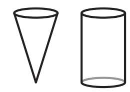
A. 2 feet

B. 3 feet

C. 6 feet

D. 18 feet

4. Maia is filling a water bottle with a paper cup. The paper cup is in the shape of a cone and has a height of 12 centimeters and a diameter of 6 centimeters. The water bottle is in the shape of a cylinder and has a height of 12 centimeters and a diameter of 6 centimeters. How many paper cups would it take to fill the water bottle? Explain your thinking.



It would take 3 paper cups to fill the water bottle. *Explanations vary.* When the height and diameter are equal for a cylinder and cone, then the cylinder will have a volume that is three times greater than the cone.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.C
2	2	8.G.C
3	2	8.G.C
4	2	8.G.C

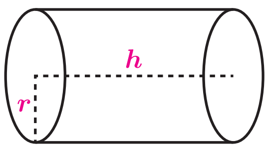
Notes:

Name: .... Date: .... Period: ....

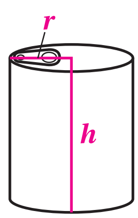
Additional Practice ACC7.7.06

1. For Cylinders A–D, draw the radius and the height. Label each radius with r and each height with h.

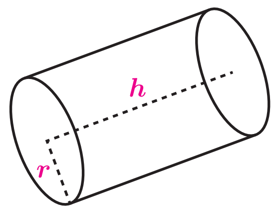
Cylinder A



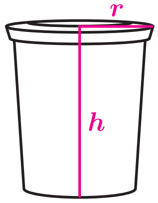
Cylinder B



Cylinder C



Cylinder D



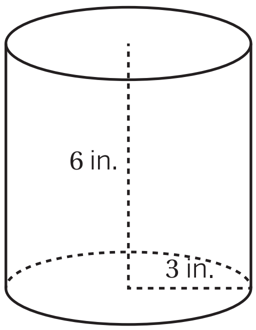
2. What is the volume of the cylinder shown?

A. 9π in³

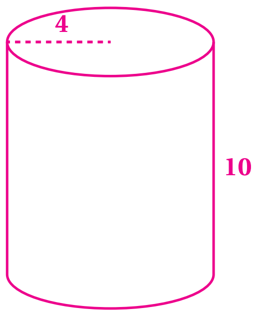
B. 18π in³

C. 54π in³

D. 108π in³



3. **a** Draw a cylinder. Label the radius 4 units and the height 10 units.



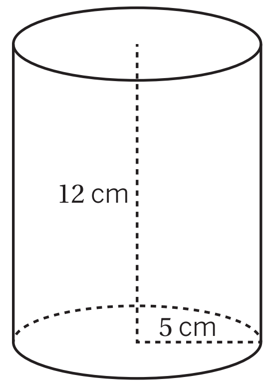
b Determine the area of the base. Write your response in terms of π.

16π square units

c Determine the volume of the cylinder. Write your response in terms of π.

160π cubic units

4. The cylinder shown has a height of 12 cm and a radius of 5 cm.



a How many cubic centimeters of fluid can fill this cylinder? Write your response in terms of π.

300π cm^3

b Give a decimal approximation of your answer to the nearest hundredths place.

942.48 cm^3

5. A preschool teacher stores modeling clay in two types of containers. One container is in the shape of a rectangular prism. The base has side lengths 14 cm and 21 cm. The second container is in the shape of a cylinder with a diameter of 20 cm. Both containers have a height of 8 cm.

a Which is greater in area, the rectangular base of the rectangular container or the circular base of the cylinder container? Show your thinking.

The base of the cylinder. *Explanations vary.* The rectangular base area is $14 \cdot 21 = 294$ cm^2, and the cylindrical base area is $\pi \cdot 10^2 = 100\pi$ cm^2, which is approximately 314 cm^2.

b Which is greater in volume, the rectangular container or the cylinder container? Show your thinking.

The cylinder. *Explanations vary.* Both figures have the same height, so the figure with the greater base area has the greater volume.

6. Cylinder A has a radius of 9 inches. Cylinder B has the same height as Cylinder A and a radius of one third as long as Cylinder A. What fraction of the volume of Cylinder A is the volume of Cylinder B? Show or explain your thinking.

$\frac{1}{9}$. *Explanations vary.* For any height h, Cylinder A has a volume of $81\pi h$ in^3 and Cylinder B has a volume of $9\pi h$ in^3.

7. Andre says that if you double the height of a cylinder, it will have the same effect on the volume as doubling the radius of the cylinder. Do you agree with Andre? Show or explain your thinking.

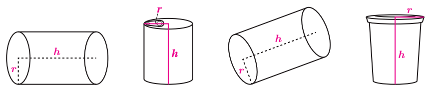
No. *Explanations vary.* A cylinder with a radius of 2 units and a height of 4 units has a volume of 16π cubic units. Doubling the height gives a volume of 32π cubic units. Doubling the radius gives a volume of 64π cubic units.

Name: .... Date: .... Period: ....

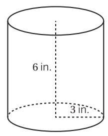
Additional Practice ACC7.7.06

1. For Cylinders A–D, draw the radius and the height. Label each radius with r and each height with h.

Cylinder A Cylinder B Cylinder C Cylinder D

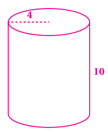


2. What is the volume of the cylinder shown?



A. 9π in³
B. 18π in³
C. 54π in³
D. 108π in³

3. a Draw a cylinder. Label the radius 4 units and the height 10 units.



b Determine the area of the base. Write your response in terms of π.

16π square units

c Determine the volume of the cylinder. Write your response in terms of π.

160π cubic units

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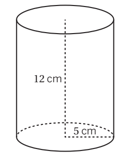
Name: .... Date: .... Period: ....

4. The cylinder shown has a height of 12 cm and a radius of 5 cm.



a How many cubic centimeters of fluid can fill this cylinder? Write your response in terms of π.

300π cm³

b Give a decimal approximation of your answer to the nearest hundredths place.

942.48 cm³

5. A preschool teacher stores modeling clay in two types of containers. One container is in the shape of a rectangular prism. The base has side lengths 14 cm and 21 cm. The second container is in the shape of a cylinder with a diameter of 20 cm. Both containers have a height of 8 cm.

a Which is greater in area, the rectangular base of the rectangular container or the circular base of the cylinder container? Show your thinking.

The base of the cylinder. *Explanations vary.* The rectangular base area is $14 \cdot 21 = 294$ cm², and the cylindrical base area is $\pi \cdot 10^2 = 100\pi$ cm², which is approximately 314 cm².

b Which is greater in volume, the rectangular container or the cylinder container? Show your thinking.

The cylinder. *Explanations vary.* Both figures have the same height, so the figure with the greater base area has the greater volume.

6. Cylinder A has a radius of 9 inches. Cylinder B has the same height as Cylinder A and a radius of one third as long as Cylinder A. What fraction of the volume of Cylinder A is the volume of Cylinder B? Show or explain your thinking.

$\frac{1}{9}$. *Explanations vary.* For any height h, Cylinder A has a volume of $81\pi h$ in³ and Cylinder B has a volume of $9\pi h$ in³.

7. Andre says that if you double the height of a cylinder, it will have the same effect on the volume as doubling the radius of the cylinder. Do you agree with Andre? Show or explain your thinking.

No. *Explanations vary.* A cylinder with a radius of 2 units and a height of 4 units has a volume of 16π cubic units. Doubling the height gives a volume of 32π cubic units. Doubling the radius gives a volume of 64π cubic units.

Unit 7 Lesson 6 210 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.C.9
2	1	8.G.C.9
3	2	8.G.C.9
4	2	8.G.C.9
5	2	8.G.C.9
6	2	8.G.C.9
7	3	8.G.C.9

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.7.07

1. Each row of this table lists information about a specific cylinder. Complete the table.

Diameter (units)	Area of Base (sq. units)	Height (units)	Volume (cu. units)
6	9π	3	**27π**
4	**4π**	5	**20π**
2	**π**	**2**	2π
10	25π	6	**150π**

Problems 2–5: Each soup can is a cylinder with a height of 5 inches.

2. Write an equation that represents the relationship between the volume, V, and the radius, r, for all the soup cans with a height of 5 inches.

$V = 5\pi r^2$

3. Complete this table.

r (in.)	4	5	6
V (cubic in.)	**80π**	**125π**	**180π**

4. The jumbo soup can is also in the shape of a cylinder and has a height of 5 inches, but the radius is 8 inches. What is the volume of this jumbo soup can in cubic inches?

A. 80π

B. 160π

C. 240π

D. 320π

5. If the radius of the soup can is doubled, does the volume double?

Yes

No

Maybe

6. A cylinder has a volume of 216π cubic inches and a height represented by h. Complete this table with the volumes of other cylinders that have the same radius but different heights.

Height (in.)	Volume (cu. in.)
$\frac{h}{2}$	**108π**
$\frac{h}{3}$	**72π**
$2h$	**432π**
$3h$	**648π**
$4h$	**864π**

7. Using the volumes from the table, what do you think the volume of a cylinder with a height of $8h$ will be? Explain your thinking.

1728π cubic inches. *Explanations vary.* Since the volume of the cylinder with a height of $4h$ is 864π, the volume will double for a cylinder with a height of $8h$.

Name: .... Date: .... Period: ....

Additional Practice ACC7.7.07

1. Each row of this table lists information about a specific cylinder. Complete the table.

Diameter (units)	Area of Base (sq. units)	Height (units)	Volume (cu. units)
6	9π	3	27π
4	4π	5	20π
2	π	2	2π
10	25π	6	150π

Problems 2–5: Each soup can is a cylinder with a height of 5 inches.

2. Write an equation that represents the relationship between the volume, V, and the radius, r, for all the soup cans with a height of 5 inches.
$V = 5\pi r^2$

3. Complete this table.

r (in.)	4	5	6
V (cubic in.)	80π	125π	180π

4. The jumbo soup can is also in the shape of a cylinder and has a height of 5 inches, but the radius is 8 inches. What is the volume of this jumbo soup can in cubic inches?
A. 80π
B. 160π
C. 240π
D. 320π

5. If the radius of the soup can is doubled, does the volume double?
Yes
No
Maybe

Name: .... Date: .... Period: ....

6. A cylinder has a volume of 216π cubic inches and a height represented by h. Complete this table with the volumes of other cylinders that have the same radius but different heights.

Height (in.)	Volume (cu. in.)
$\frac{h}{2}$	108π
$\frac{h}{3}$	72π
$2h$	432π
$3h$	648π
$4h$	864π

7. Using the volumes from the table, what do you think the volume of a cylinder with a height of $8h$ will be? Explain your thinking.
1728π cubic inches. *Explanations vary.* Since the volume of the cylinder with a height of $4h$ is 864π, the volume will double for a cylinder with a height of $8h$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.F.B, 8.G.C.9
2	2	8.F.B, 8.G.C.9
3	1	8.F.B, 8.G.C.9
4	2	8.F.B, 8.G.C.9
5	2	8.F.B, 8.G.C.9
6	1	8.F.B, 8.G.C.9
7	2	8.F.B, 8.G.C.9

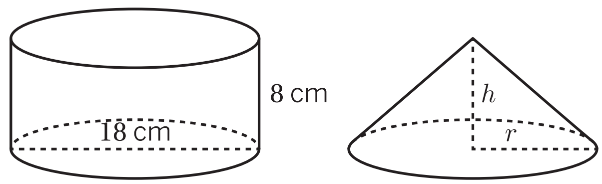
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.7.08

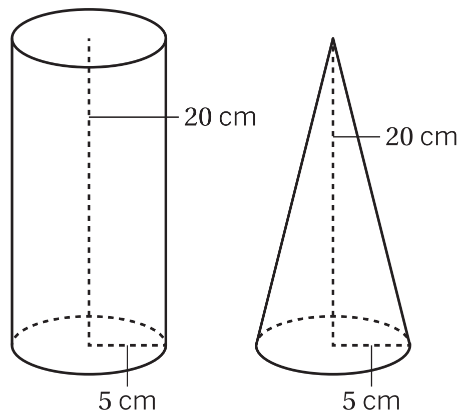
1. The cylinder and cone shown have the same height and the same base area.



a What is the radius r of the cone?
9 cm

b What is the height h of the cone?
8 cm

2. Which is a true statement about the volumes of the cylinder and cone?



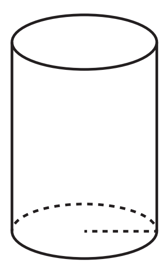
A. The volumes are equal.

B. The volume of the cone is $\frac{1}{3}$ times the volume of the cylinder.

C. The volume of the cylinder is $\frac{1}{3}$ times the volume of the cone.

D. The volume of the cone is 3 times the volume of the cylinder.

3. The volume of this cylinder is 72π mm^3. What is the volume of a cone that has the same base area and the same height?



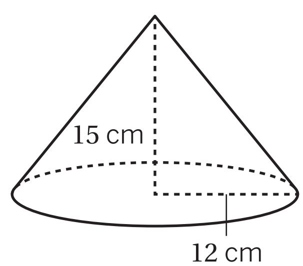
A. 24π mm^3

B. 36π mm^3

C. 72π mm^3

D. 216π mm^3

4. What is the volume of the cone shown?



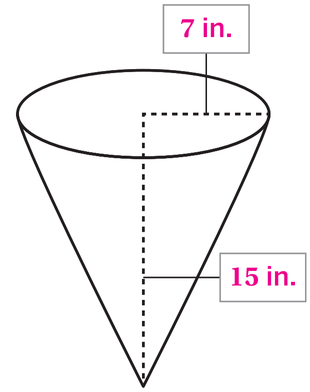
A. 60π cm^3

B. 180π cm^3

C. 720π cm^3

D. 900π cm^3

5. A cone-shaped hanging basket is used to grow flowers. The basket has a diameter of 14 inches and a height of 15 inches.



a Label the height and radius of the hanging basket.

b If the container is filled completely with potting soil, about how many cubic inches can the container hold? Round to the nearest tenth. Show your thinking.

769.7 in^3. *Explanations vary.*

$V = \frac{1}{3}\pi r^2 h$; **If** $r = 7$ **and** $h = 15$, **then**

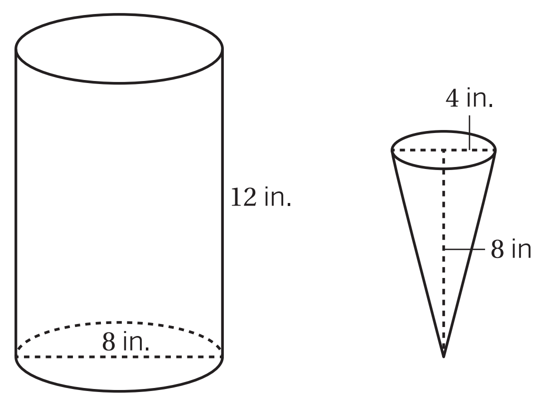
$V = \frac{1}{3} \cdot \pi \cdot 7^2 \cdot 15$

$V = 245\pi$ **or about 769.7 in^3**

6. The volume of a cylinder is 96π cubic units. What is the volume of a cone that has the same base area and the same height as the cylinder? Explain your thinking.

32π cubic units. *Explanations vary.* The volume of a cone is one third times the volume of a cylinder with the same height and base area.

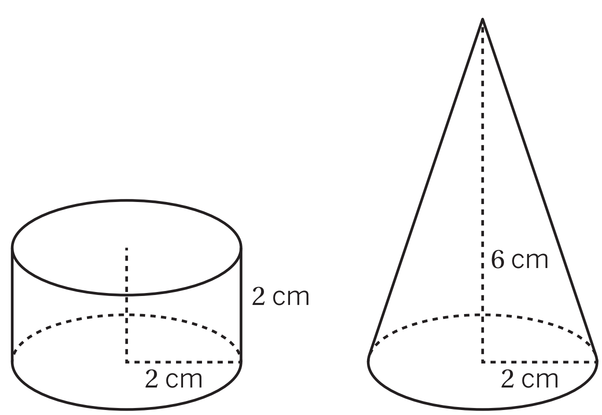
7. At a frozen yogurt shop, a cylinder-shaped container of frozen yogurt has the base diameter and height shown. The shop serves the frozen yogurt in waffle cones that have a base diameter of 4 inches and a height of 8 inches. If each waffle cone is filled completely with frozen yogurt, how many cones can be filled from one container of frozen yogurt? Explain your thinking.



18 cones. *Explanations vary.* The container has a volume of 192π in^3 and each cone has a volume of $\frac{32\pi}{3}$ in^3. $192\pi \div \frac{32\pi}{3} = 18$, so 18 cones can be filled.

8. Bard claims that the cone has a greater volume than the cylinder. Han argues that the cylinder has the greater volume. Who is correct? Circle one.

Bard Han Both Neither



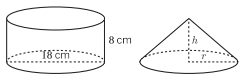
Explain your thinking.

Neither. *Explanations vary.* A cone with the same base area and the same height as a cylinder has a volume that is one third the volume of the cylinder. Here, the cone has the same base area as the cylinder, but its height is 3 times the height of the cylinder. So, its volume is the same as the volume of the cylinder. Both figures have a volume of 8π cm^3.

Name: ________ Date: ________ Period: ________

Additional Practice ACC7.7.08

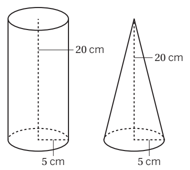
1. The cylinder and cone shown have the same height and the same base area.



a. What is the radius r of the cone?
9 cm

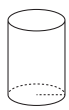
b. What is the height h of the cone?
8 cm

2. Which is a true statement about the volumes of the cylinder and cone?



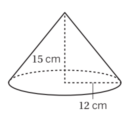
A. The volumes are equal.
B. The volume of the cone is $\frac{1}{3}$ times the volume of the cylinder.
C. The volume of the cylinder is $\frac{1}{3}$ times the volume of the cone.
D. The volume of the cone is 3 times the volume of the cylinder.

3. The volume of this cylinder is 72π mm³. What is the volume of a cone that has the same base area and the same height?



A. 24π mm³
B. 36π mm³
C. 72π mm³
D. 216π mm³

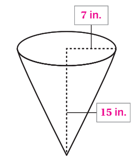
4. What is the volume of the cone shown?



A. 60π cm³
B. 180π cm³
C. 720π cm³
D. 900π cm³

Name: ________ Date: ________ Period: ________

5. A cone-shaped hanging basket is used to grow flowers. The basket has a diameter of 14 inches and a height of 15 inches.



a. Label the height and radius of the hanging basket.

b. If the container is filled completely with potting soil, about how many cubic inches can the container hold? Round to the nearest tenth. Show your thinking.

769.7 in³. *Explanations vary.*

$V = \frac{1}{3}\pi r^2 h$; If $r = 7$ and $h = 15$, then

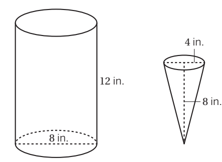
$V = \frac{1}{3} \cdot \pi \cdot 7^2 \cdot 15$

$V = 245\pi$ or about 769.7 in³

6. The volume of a cylinder is 96π cubic units. What is the volume of a cone that has the same base area and the same height as the cylinder? Explain your thinking.

32π cubic units. *Explanations vary.* The volume of a cone is one third times the volume of a cylinder with the same height and base area.

7. At a frozen yogurt shop, a cylinder-shaped container of frozen yogurt has the base diameter and height shown. The shop serves the frozen yogurt in waffle cones that have a base diameter of 4 inches and a height of 8 inches. If each waffle cone is filled completely with frozen yogurt, how many cones can be filled from one container of frozen yogurt? Explain your thinking.

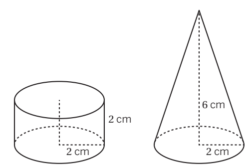


18 cones. *Explanations vary.* The container has a volume of 192π in³ and each cone has a volume of $\frac{32\pi}{3}$ in³. $192\pi \div \frac{32\pi}{3} = 18$, so 18 cones can be filled.

8. Bard claims that the cone has a greater volume than the cylinder. Han argues that the cylinder has the greater volume. Who is correct? Circle one.

Bard Han Both Neither

Explain your thinking.



Neither. *Explanations vary.* A cone with the same base area and the same height as a cylinder has a volume that is one third the volume of the cylinder. Here, the cone has the same base area as the cylinder, but its height is 3 times the height of the cylinder. So, its volume is the same as the volume of the cylinder. Both figures have a volume of 8π cm³.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.C.9
2	1	8.G.C.9
3	2	8.G.C.9
4	2	8.G.C.9
5	2	8.G.C.9
6	2	8.G.C.9
7	3	8.G.C.9
8	3	8.G.C.9

Notes:

Name: .... Date: .... Period: ....

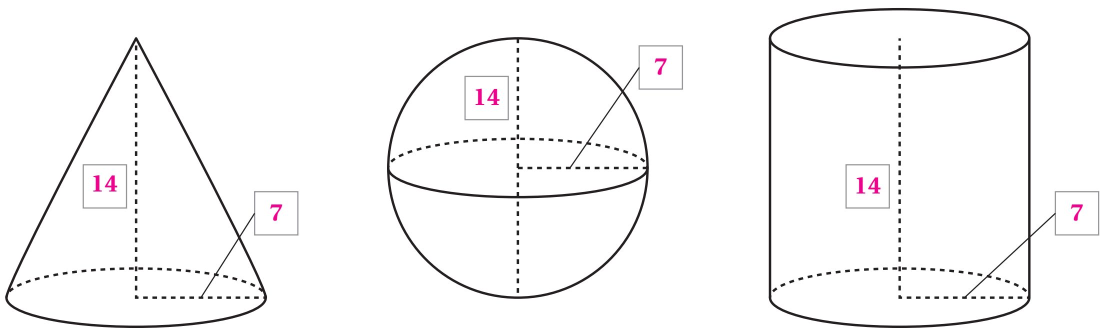
Additional Practice

ACC7.7.10

1. Complete the following table for the dimensions of different spheres.

Radius	14 in.	**8 cm**	$\frac{500}{3}$ ft	**$\frac{350}{3}$ yd**	4.09 m	**2.045 m**
Diameter	**28 in.**	16 cm	**$\frac{1000}{3}$ ft**	$\frac{700}{3}$ yd	**8.18 m**	4.09 m

2. The cylinder, cone, and sphere all have the same radius and height. The radius of the cone is 7 units. Label the radius and height on each figure, in units.



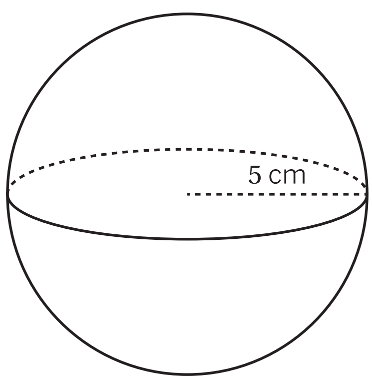
3. Use the formula $V = \frac{4}{3}\pi r^3$ to determine the volume of the sphere with a radius of 5 cm.

A. $\frac{125}{3}\pi$ cm³

B. $\frac{500}{3}\pi$ cm³

C. 125π cm³

D. 500π cm³



4. A sphere has a diameter of 12 inches. What is the *exact* volume of the sphere? Write your response in terms of π.

288π in^3

Name: .... Date: .... Period: ....

5. Which of the following are true statements about the volumes of a sphere, a cone, and a cylinder with the same dimensions? Select *all* that apply.

- ☐ **A.** The sphere has the greatest volume.
- ☑ **B.** The cone's volume is half the sphere's volume.
- ☐ **C.** The cone's volume is half the cylinder's volume.
- ☑ **D.** The sphere's volume is double the cone's volume.
- ☑ **E.** The cylinder's volume is $\frac{3}{2}$ the sphere's volume.
- ☑ **F.** The sphere's volume is $\frac{2}{3}$ the cylinder's volume.

6. Match the description of each sphere to its volume.

a. Sphere A: radius of 7 cm — b — $\frac{32}{3}\pi$ cm³

b. Sphere B: radius of 2 cm — c — $\frac{256}{3}\pi$ cm³

c. Sphere C: diameter of 8 cm — a — $\frac{1372}{3}\pi$ cm³

d. Sphere D: radius of 9 cm — d — 972π cm³

7. A cube's volume is 216 in³.

a What is the length of its edge?

6 in.

b If a sphere fits snugly inside this cube, what is the volume of the sphere? Show or explain your thinking.

36π in³. *Explanations vary.* The diameter is 6 in., so the radius is 3 in. If $r = 3$, then $V = \frac{4}{3}\pi \cdot 3^3$ or 36π.

c What percent of the cube is taken up by the sphere? Round to the nearest whole percent. Show or explain your thinking.

52%. *Explanations vary.* $\frac{36\pi}{216} \approx 0.52$, so about 52% of the cube is taken up by the sphere.

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.7.10

1. Complete the following table for the dimensions of different spheres.

Radius	14 in.	**8 cm**	$\frac{500}{3}$ ft	**$\frac{350}{3}$ yd**	4.09 m	**2.045 m**
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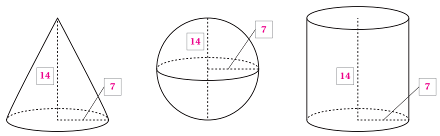

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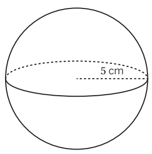
B. $\frac{500}{3}\pi$ cm³

C. 125π cm³

D. 500π cm³



4. A sphere has a diameter of 12 inches. What is the *exact* volume of the sphere? Write your response in terms of π.

288π in³

Name: ______ Date: ______ Period: ______

5. Which of the following are true statements about the volumes of a sphere, a cone, and a cylinder with the same dimensions? Select *all* that apply.

- [ ] A. The sphere has the greatest volume.
- [x] B. The cone's volume is half the sphere's volume.
- [ ] C. The cone's volume is half the cylinder's volume.
- [x] D. The sphere's volume is double the cone's volume.
- [x] E. The cylinder's volume is $\frac{3}{2}$ the sphere's volume.
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c. What percent of the cube is taken up by the sphere? Round to the nearest whole percent. Show or explain your thinking.

52%. *Explanations vary.* $\frac{36\pi}{216} \approx 0.52$, so about 52% of the cube is taken up by the sphere.

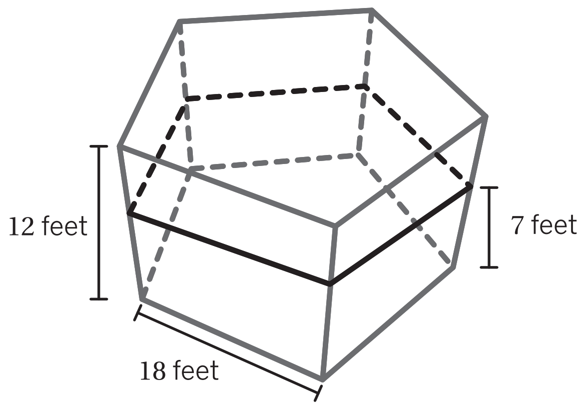
Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.C.9
2	1	8.G.C.9
3	2	8.G.C.9
4	2	8.G.C.9
5	2	8.G.C.9
6	2	8.G.C.9
7	3	8.G.C.9

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.7.11



Problems 1–3: Here is a shed design.

1. 3,899 cubic feet of hay is needed to fill the shed up to 7 feet. What is the area of the shed's base?

557 square feet

2. If the owner wants to fill the shed up to 12 feet, how much more hay is needed?

2,785 cubic feet

3. The owner wants to paint the five sides of the shed (not including the bottom). How many square feet will she need to cover with paint? Explain your thinking.

1,080 square feet. *Explanations vary.* She will need $12 \cdot 18 = 216$ square feet of paint for each side and $216 \cdot 5 = 1080$ square feet in total.

Problems 4–5: There are two boxes of granola in the shape of rectangular prisms. The first box has a height of 12 inches, a length of 7 inches, and width of 4 inches. The second box has a height of 12 inches, a length of 6.5 inches, and a width of 2.5 inches.

4. What is the difference in volume, in cubic inches, between the two boxes of granola?

141 cubic inches

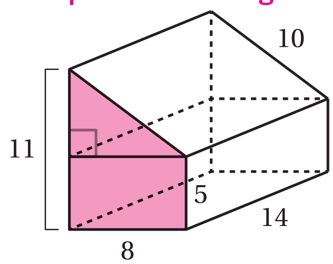
5. What is the difference in surface area, in square inches, between the two boxes of granola?

71.5 square inches

6. Here is a prism

a Shade a base of the prism.

Sample shown on figure.



b Determine the area of the base.

64 square units

c Determine the volume of the prism.

896 cubic units

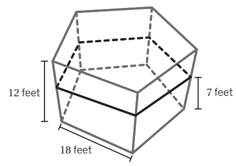
d Determine the surface area of the prism.

604 square units

Name: ________ Date: ________ Period: ________

Additional Practice ACC7.7.11

Problems 1–3: Here is a shed design.



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557 square feet

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1,080 square feet. ***Explanations vary.*** **She will need $12 \cdot 18 = 216$ square feet of paint for each side and $216 \cdot 5 = 1080$ square feet in total.**

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141 cubic inches

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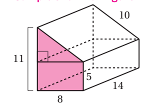
71.5 square inches

Name: ________ Date: ________ Period: ________

6. Here is a prism

a. Shade a base of the prism.

Sample shown on figure.



b. Determine the area of the base.

64 square units

c. Determine the volume of the prism.

896 cubic units

d. Determine the surface area of the prism.

604 square units

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	7.G.B.6
2	2	7.G.B.6
3	2	7.G.B.6
4	2	7.G.B.6
5	2	7.G.B.6
6	2	7.G.B.6

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.01

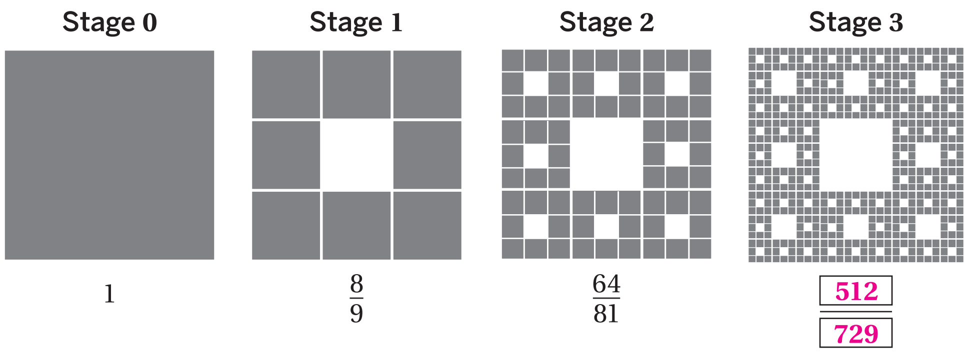
1. Match each exponent expression with its expanded expression.

Exponent Expression	**Expanded Expression**
a. 2^6	d $3 \cdot 3 \cdot 3 \cdot 3$
b. 6^2	c $4 \cdot 4 \cdot 4$
c. 4^3	b $6 \cdot 6$
d. 3^4	a $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

2. Complete the table.

Exponent Expression	Expanded Expression
10^2	$10 \cdot 10$
4^5	$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$
$\left(\frac{1}{2}\right)^6$	$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$
$\left(\frac{7}{9}\right)^1$	$\frac{7}{9}$
$\left(1\frac{2}{3}\right)^4$	$1\frac{2}{3} \cdot 1\frac{2}{3} \cdot 1\frac{2}{3} \cdot 1\frac{2}{3}$

3. The pattern shown is called Sierpiński's carpet. Study the pattern. The shaded area of each square in stages 0–2 is written as a fraction of the total area, in square units. Write the fraction that represents the shaded area for stage 3.



Problems 4–6: Write an equivalent expression that uses exponents.

4. $5 \cdot 7 \cdot 5 \cdot 5 \cdot 7 \cdot 5$ **$5^4 \cdot 7^2$ (or equivalent)**

5. $4 \cdot 4 + 4 \cdot 4 \cdot 4$ **$4^2 + 4^3$ (or equivalent)**

6. 6^4 **$6^3 \cdot 6^1$ (or equivalent)**

7. The first three terms of a pattern are shown.

$\frac{3}{5}, \frac{9}{25}, \frac{27}{125}, \ldots$

a What is the 10th term of this pattern? Show or explain your thinking.

$\frac{59049}{9765625}$. *Explanations vary*. The pattern can also be expressed as $\left(\frac{3}{5}\right)^1, \left(\frac{3}{5}\right)^2, \left(\frac{3}{5}\right)^3$. The 10th term would be $\left(\frac{3}{5}\right)^{10}$.

b Write an expression for the nth term of this pattern. Show or explain your thinking.

$\left(\frac{3}{5}\right)^n$. *Explanations vary*. Because the pattern can be expressed as $\left(\frac{3}{5}\right)$ raised to the exponent of the term number, n, any term will be $\left(\frac{3}{5}\right)^n$.

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.01

1. Match each exponent expression with its expanded expression.

Exponent Expression

a. 2^6

b. 6^2

c. 4^3

d. 3^4

Expanded Expression

d $3 \cdot 3 \cdot 3 \cdot 3$

c $4 \cdot 4 \cdot 4$

b $6 \cdot 6$

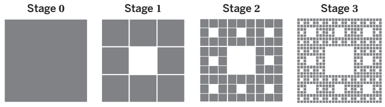
a $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

2. Complete the table.

Exponent Expression	Expanded Expression
10^2	$10 \cdot 10$
4^5	$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$
$\left(\frac{1}{2}\right)^6$	$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$
$\left(\frac{7}{9}\right)^1$	$\frac{7}{9}$
$\left(1\frac{2}{3}\right)^4$	$1\frac{2}{3} \cdot 1\frac{2}{3} \cdot 1\frac{2}{3} \cdot 1\frac{2}{3}$

3. The pattern shown is called Sierpiński's carpet. Study the pattern. The shaded area of each square in stages 0–2 is written as a fraction of the total area, in square units. Write the fraction that represents the shaded area for stage 3.

Stage 0 Stage 1 Stage 2 Stage 3



1 $\quad \frac{8}{9}$ $\quad \frac{64}{81}$ $\quad \frac{\boxed{512}}{\boxed{729}}$

Name: .... Date: .... Period: ....

Problems 4–6: Write an equivalent expression that uses exponents.

4. $5 \cdot 7 \cdot 5 \cdot 5 \cdot 7 \cdot 5$ $\quad 5^4 \cdot 7^2$ (or equivalent)

5. $4 \cdot 4 + 4 \cdot 4 \cdot 4$ $\quad 4^2 + 4^3$ (or equivalent)

6. 6^4 $\quad 6^3 \cdot 6^1$ (or equivalent)

7. The first three terms of a pattern are shown.

$\frac{3}{5}, \frac{9}{25}, \frac{27}{125}, \ldots$

a. What is the 10th term of this pattern? Show or explain your thinking.

$\frac{59049}{9765625}$. *Explanations vary.* The pattern can also be expressed as $\left(\frac{3}{5}\right)^1, \left(\frac{3}{5}\right)^2, \left(\frac{3}{5}\right)^3$. The 10th term would be $\left(\frac{3}{5}\right)^{10}$.

b. Write an expression for the nth term of this pattern. Show or explain your thinking.

$\left(\frac{3}{5}\right)^n$. *Explanations vary.* Because the pattern can be expressed as $\left(\frac{3}{5}\right)$ raised to the exponent of the term number, n, any term will be $\left(\frac{3}{5}\right)^n$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.A.1
2	1	8.EE.A.1
3	2	8.EE.A.1
4	1	8.EE.A.1
5	1	8.EE.A.1
6	1	8.EE.A.1
7	2	8.EE.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.02

1. For each expression, write an equivalent expression with a single power.

a. $7^4 \cdot 7^7$

7^{11}

b. $25^8 \cdot 25^3$

25^{11}

c. $\left(\frac{2}{3}\right)^9 \cdot \left(\frac{2}{3}\right)^5$

$\left(\frac{2}{3}\right)^{14}$

d. $(5.4)^{13} \cdot (5.4)^6$

$(5.4)^{19}$

e. $(-3) \cdot (-3)^6$

$(-3)^7$

f. $6^{15} \cdot 16^7 \cdot 16^3$

6^{25}

2. Select *all* the expressions that are equivalent to 10^{16}.

- [ ] A. $10^4 \cdot 10^4$
- [ ] B. $10^2 \cdot 10^8$
- [x] C. $10^8 \cdot 10^8$
- [ ] D. $10^{16} \cdot 10$
- [x] E. $10^{15} \cdot 10$

3. Two of the following expressions are equivalent. Circle the expression that is *not* equivalent. Show or explain your thinking.

10^6 $\qquad$ $10 + 10 + 10 + 10 + 10 + 10$ $\qquad$ $10^3 \cdot 10 \cdot 10 \cdot 10$

***Explanations vary*. The other expressions are equivalent to 10^6. This expression equals 60.**

4. A large rectangular aquarium is 81 inches long, 81 inches wide, and 9 inches deep. The aquarium is filled to the top with water.

a. Write each measurement of the aquarium as a single power of 9.

Length: 9^2 in. **Width:** 9^2 in. **Depth:** 9^1 in.

b. Write an expression with exponents to represent the volume, in cubic inches, of the aquarium.

$9^2 \cdot 9^2 \cdot 9^1$

c. How much water does the aquarium hold? Write your response as a single power of 9.

9^5 cubic inches

5. Fill in the empty box with a single power of 4 to make each equation true.

a. $4^7 \cdot \boxed{4^9} = 4^{16}$

b. $\boxed{4} \cdot 4^6 = 4^7$

c. $4^{10} \cdot \boxed{4^7} \cdot 4 = 4^{18}$

6. Fill in the empty box with a single power of a to make each equation true.

a. $a^9 \cdot \boxed{a} = a^{10}$

b. $\boxed{a^{10}} \cdot a^{10} = a^{20}$

c. $a^3 \cdot \boxed{a^2} \cdot a = a^6$

7. Lin wants to write a multiplication expression that is equivalent to 2^8. She writes the expression $2^4 \cdot 2^2$. Is her expression correct? Explain your thinking and correct Lin's expression, if necessary.

No. *Explanations vary.* Her expression could be $2^4 \cdot 2^4$. Lin needed to add the exponents, not multiply them.

8. If $a^b \cdot a^b = a^c$ is true, is $\frac{c}{2}$ greater than, less than, or equal to b? Show or explain your thinking.

$\frac{c}{2} = b$. *Explanations vary.* If $a^b \cdot a^b = a^c$, then $b + b = c$, and $2b = c$ or $\frac{c}{2} = b$.

Name: ________ Date: ________ Period: ________

Additional Practice ACC7.8.02

1. For each expression, write an equivalent expression with a single power.

a. $7^4 \cdot 7^7$
7^{11}

b. $25^8 \cdot 25^3$
25^{11}

c. $\left(\frac{2}{3}\right)^9 \cdot \left(\frac{2}{3}\right)^5$
$\left(\frac{2}{3}\right)^{14}$

d. $(5.4)^{13} \cdot (5.4)^6$
$(5.4)^{19}$

e. $(-3) \cdot (-3)^6$
$(-3)^7$

f. $6^{15} \cdot 16^7 \cdot 16^3$
6^{25}

2. Select *all* the expressions that are equivalent to 10^{16}.

- [ ] A. $10^4 \cdot 10^4$
- [ ] B. $10^2 \cdot 10^8$
- [x] C. $10^8 \cdot 10^8$
- [ ] D. $10^{16} \cdot 10$
- [x] E. $10^{15} \cdot 10$

3. Two of the following expressions are equivalent. Circle the expression that is *not* equivalent. Show or explain your thinking.

10^6 $\qquad$ $10 + 10 + 10 + 10 + 10 + 10$ $\qquad$ $10^3 \cdot 10 \cdot 10 \cdot 10$

Explanations vary. The other expressions are equivalent to 10^6. This expression equals 60.

4. A large rectangular aquarium is 81 inches long, 81 inches wide, and 9 inches deep. The aquarium is filled to the top with water.

a. Write each measurement of the aquarium as a single power of 9.

Length: 9^2 in. **Width:** 9^2 in. **Depth:** 9^1 in.

b. Write an expression with exponents to represent the volume, in cubic inches, of the aquarium.
$9^2 \cdot 9^2 \cdot 9^1$

c. How much water does the aquarium hold? Write your response as a single power of 9.
9^5 cubic inches

Name: ________ Date: ________ Period: ________

5. Fill in the empty box with a single power of 4 to make each equation true.

a. $4^7 \cdot \boxed{4^9} = 4^{16}$

b. $\boxed{4} \cdot 4^6 = 4^7$

c. $4^{10} \cdot \boxed{4^7} \cdot 4 = 4^{18}$

6. Fill in the empty box with a single power of a to make each equation true.

a. $a^9 \cdot \boxed{a} = a^{10}$

b. $\boxed{a^{10}} \cdot a^{10} = a^{20}$

c. $a^3 \cdot \boxed{a^2} \cdot a = a^6$

7. Lin wants to write a multiplication expression that is equivalent to 2^8. She writes the expression $2^4 \cdot 2^2$. Is her expression correct? Explain your thinking and correct Lin's expression, if necessary.

No. *Explanations vary.* Her expression could be $2^4 \cdot 2^4$. Lin needed to add the exponents, not multiply them.

8. If $a^b \cdot a^b = a^c$ is true, is $\frac{c}{2}$ greater than, less than, or equal to b? Show or explain your thinking.

$\frac{c}{2} = b$. *Explanations vary.* If $a^b \cdot a^b = a^c$, then $b + b = c$, and $2b = c$ or $\frac{c}{2} = b$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.A.1
2	1	8.EE.A.1
3	2	8.EE.A.1
4	2	8.EE.A.1
5	2	8.EE.A.1
6	2	8.EE.A.1
7	2	8.EE.A.1
8	3	8.EE.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.04

1. Select *all* the expressions that are equivalent to 12^4.

- [x] A. $\frac{12^7}{12^3}$
- [ ] B. $2 \cdot 6^4$
- [ ] C. $12^{10} - 12^6$
- [x] D. $12 \cdot 12^3$
- [x] E. $(12^2)^2$

Problems 2–9: Rewrite each expression as a single power.

2. $3^2 \cdot 4^2 = 12^2$

3. $\frac{7^{12}}{7^8} = 7^4$

4. $(16^2)^3 = 16^6$

5. $\frac{5^8 \cdot 5^4}{5^9} = 5^3$

6. $31^2 \cdot 31^{10} = 31^{12}$

7. $\frac{9^5}{9} = 9^4$

8. $(11^4)^5 \cdot 11 = 11^{21}$

9. $\frac{8^{11}}{4^{11}} = 2^{11}$

Name: .... Date: .... Period: ....

10. Michael and Amelia were asked to rewrite the expression $\frac{4^{10}}{4^4} \cdot 4$ as a single power.

Michael says the answer is 4^7. Amelia says the answer is 2^{14}.

Who is correct? Circle one.

Michael Amelia Neither Both

Show or explain your thinking.

Explanations vary. **Michael is correct because he simplified the expression $\frac{4^{10}}{4^4} \cdot 4$ to 4^7 by subtracting the exponents 10 and 4 to get $4^6 \cdot 4$ and then adding the exponents to get 4^7. Amelia is also correct because $4^7 = (2^2)^7 = 2^{14}$.**

11. What is the value of m in the equation $5^m = 5^8 \cdot 5^{10}$.

18

12. Order the expressions from *least* to *greatest*.

$3^7 \cdot 8^7$	$4^5 \cdot 6^5$	$(24^2)^4$	$24^2 \cdot 24^2$	$24^2 \cdot (24^4)^3$

$24^2 \cdot 24^2$	$4^5 \cdot 6^5$	$3^7 \cdot 8^7$	$(24^2)^4$	$24^2 \cdot (24^4)^3$

Least **Greatest**

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.04

1. Select *all* the expressions that are equivalent to 12^4.
   - ☑ A. $\frac{12^7}{12^3}$
   - ☐ B. $2 \cdot 6^4$
   - ☐ C. $12^{10} - 12^6$
   - ☑ D. $12 \cdot 12^3$
   - ☑ E. $(12^2)^2$

Problems 2–9: Rewrite each expression as a single power.

2. $3^2 \cdot 4^2 = 12^2$
3. $\frac{7^{12}}{7^8} = 7^4$
4. $(16^2)^3 = 16^6$
5. $\frac{5^8 \cdot 5^4}{5^9} = 5^3$
6. $31^2 \cdot 31^{10} = 31^{12}$
7. $\frac{9^5}{9} = 9^4$
8. $(11^4)^5 \cdot 11 = 11^{21}$
9. $\frac{8^{11}}{4^{11}} = 2^{11}$

Name: .... Date: .... Period: ....

10. Michael and Amelia were asked to rewrite the expression $\frac{4^{10}}{4^4} \cdot 4$ as a single power.

    Michael says the answer is 4^7. Amelia says the answer is 2^{14}.

    Who is correct? Circle one.

    Michael Amelia Neither Both

    Show or explain your thinking.

    ***Explanations vary.* Michael is correct because he simplified the expression $\frac{4^{10}}{4^4} \cdot 4$ to 4^7 by subtracting the exponents 10 and 4 to get $4^6 \cdot 4$ and then adding the exponents to get 4^7. Amelia is also correct because $4^7 = (2^2)^7 = 2^{14}$.**

11. What is the value of m in the equation $5^m = 5^8 \cdot 5^{10}$.

    18

12. Order the expressions from *least* to *greatest*.

$3^7 \cdot 8^7$	$4^5 \cdot 6^5$	$(24^2)^4$	$24^2 \cdot 24^2$	$24^2 \cdot (24^4)^3$

$24^2 \cdot 24^2$	$4^5 \cdot 6^5$	$3^7 \cdot 8^7$	$(24^2)^4$	$24^2 \cdot (24^4)^3$

Least **Greatest**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.1
2	1	8.EE.A.1
3	1	8.EE.A.1
4	1	8.EE.A.1
5	1	8.EE.A.1
6	1	8.EE.A.1
7	1	8.EE.A.1
8	1	8.EE.A.1
9	1	8.EE.A.1
10	2	8.EE.A.1
11	2	8.EE.A.1
12	2	8.EE.A.1

Notes:

Science Mom Lesson 103

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.05

1. Rewrite each expression as a single power with a *negative* exponent.

a $\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10}$

10^{-5}

b $\frac{1}{6 \cdot 6 \cdot 6 \cdot 6}$

6^{-4}

c $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$

2^{-6}

d $\frac{1}{y \cdot y \cdot y}$

y^{-3}

2. Rewrite each expression as a single power with a *positive* exponent.

a 10^{-2}

$\frac{1}{10^2}$

b 5^{-5}

$\frac{1}{5^5}$

c 8^{-7}

$\frac{1}{8^7}$

d 2^{-8}

$\frac{1}{2^8}$

3. Rewrite each expression as a single power with a *positive* exponent.

a $10^7 \cdot 10^{-2}$

10^5

b $9^{-3} \cdot 9^{-5}$

$\frac{1}{9^8}$

c $\frac{10^6}{10^8}$

$\frac{1}{10^2}$

d $\frac{4^2}{4^7}$

$\frac{1}{4^5}$

4. Select *all* the expressions that are equivalent to $\frac{1}{1000}$.

- ☐ A. $(-10)^3$
- ☑ B. 10^{-3}
- ☑ C. 1000^{-1}
- ☐ D. 100^{-2}
- ☐ E. -1000

5. Select *all* the expressions that are equivalent to 10^{-5}.

- ☑ A. $\frac{10^5}{10^{10}}$
- ☐ B. -50
- ☐ C. $10^{-2} + 10^{-3}$
- ☐ D. $10^{-5} \bullet 10$
- ☑ E. $\frac{1}{10} \bullet \frac{1}{10} \bullet \frac{1}{10} \bullet \frac{1}{10} \bullet \frac{1}{10}$

6. Fill in the empty box with a single power of 10 to make each equation true.

a. $\frac{10^3}{\boxed{10^7}} = \frac{1}{10^4}$

b. $\frac{\boxed{10^3}}{10^4} = \frac{1}{10}$

c. $10^{-4} \bullet \boxed{10^2} \bullet 10 = \frac{1}{10}$

7. Without evaluating, order the expressions 12^{-3}, 12^2, and 12^0 from *least* to *greatest*. Explain your thinking.

$12^{-3}, 12^0, 12^2$. *Explanations vary.* The expression 12^{-3} written with a positive exponent is $\frac{1}{12^3}$ and is less than 1, 12^0 equals 1, and 12^2 is greater than 1.

8. Mai states the missing single power of b in the expression $\frac{b^{-12}}{\square} = \frac{1}{b^{10}}$ is b^2. Do you agree? Show or explain your thinking.

No. *Explanations vary.* The missing power is b^{-2} because $b^{-12-(-2)} = b^{-10}$ or $\frac{1}{b^{10}}$.

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.05

1. Rewrite each expression as a single power with a *negative* exponent.

a. $\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10}$

10^{-5}

b. $\frac{1}{6 \cdot 6 \cdot 6 \cdot 6}$

6^{-4}

c. $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$

2^{-6}

d. $\frac{1}{y \cdot y \cdot y}$

y^{-3}

2. Rewrite each expression as a single power with a *positive* exponent.

a. 10^{-2}

$\frac{1}{10^2}$

b. 5^{-5}

$\frac{1}{5^5}$

c. 8^{-7}

$\frac{1}{8^7}$

d. 2^{-8}

$\frac{1}{2^8}$

3. Rewrite each expression as a single power with a *positive* exponent.

a. $10^7 \cdot 10^{-2}$

10^5

b. $9^{-3} \cdot 9^{-5}$

$\frac{1}{9^8}$

c. $\frac{10^6}{10^8}$

$\frac{1}{10^2}$

d. $\frac{4^2}{4^7}$

$\frac{1}{4^5}$

4. Select *all* the expressions that are equivalent to $\frac{1}{1000}$.

- [ ] A. $(-10)^3$
- [x] B. 10^{-3}
- [x] C. 1000^{-1}
- [ ] D. 100^{-2}
- [ ] E. -1000

Name: .... Date: .... Period: ....

5. Select *all* the expressions that are equivalent to 10^{-5}.

- [x] A. $\frac{10^5}{10^{10}}$
- [ ] B. -50
- [ ] C. $10^{-2} + 10^{-3}$
- [ ] D. $10^{-5} \cdot 10$
- [x] E. $\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10}$

6. Fill in the empty box with a single power of 10 to make each equation true.

a. $\frac{10^3}{\boxed{10^7}} = \frac{1}{10^4}$

b. $\frac{\boxed{10^3}}{10^4} = \frac{1}{10}$

c. $10^{-4} \cdot \boxed{10^2} \cdot 10 = \frac{1}{10}$

7. Without evaluating, order the expressions 12^{-3}, 12^2, and 12^0 from *least* to *greatest*. Explain your thinking.

$12^{-3}, 12^0, 12^2$. *Explanations vary.* The expression 12^{-3} written with a positive exponent is $\frac{1}{12^3}$ and is less than 1, 12^0 equals 1, and 12^2 is greater than 1.

8. Mai states the missing single power of b in the expression $\frac{b^{-12}}{\square} = \frac{1}{b^{10}}$ is b^2. Do you agree? Show or explain your thinking.

No. *Explanations vary.* The missing power is b^{-2} because $b^{-12-(-2)} = b^{-10}$ or $\frac{1}{b^{10}}$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.A.1
2	1	8.EE.A.1
3	1	8.EE.A.1
4	2	8.EE.A.1
5	2	8.EE.A.1
6	2	8.EE.A.1
7	2	8.EE.A.1
8	2	8.EE.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.07

1. Rewrite each expression three different ways using powers of 10.

a 576,000

Responses vary.

$576 \cdot 10^3$ $57.6 \cdot 10^4$ $5.76 \cdot 10^5$

b 9,510,000

Responses vary.

$9510 \cdot 10^3$ $951 \cdot 10^4$ $95.1 \cdot 10^5$

2. Rewrite each value as a combination of powers of 10.

a One light-year is about 6 trillion miles. *Responses vary.*

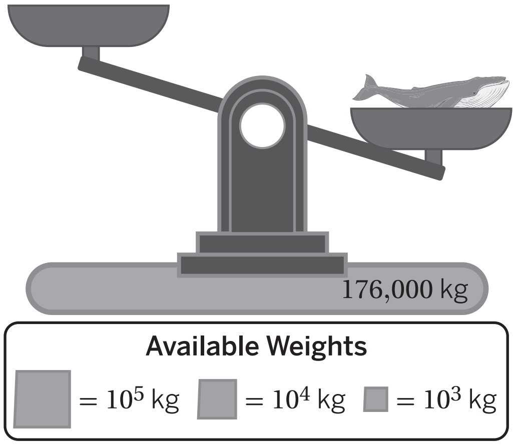
$6 \cdot 10^{12}$

b On average, the human brain has 86 billion neurons. *Responses vary.*

$86 \cdot 10^9$

Problems 3–4: Three students tried to balance the scale using weights measuring 10^5 kilograms, 10^4 kilograms, and 10^3 kilograms. They each wrote the weight of the whale as:

- **Expression A:** $176 \cdot 10^2$ kilograms
- **Expression B:** $17.6 \cdot 10^4$ kilograms
- **Expression C:** $1 \cdot 10^5 + 7 \cdot 10^4 + 6 \cdot 10^3$ kilograms



3. Which of these expressions are accurate?

Expressions B and C

4. Write an expression to represent a different combination of available weights that will balance the scale.

Responses vary. $1.76 \cdot 10^5$ kilograms

5. Which value is less: $2 \cdot 10^4$ or $2 \cdot 10^5$? Show or explain your thinking.

$2 \cdot 10^4$. *Explanations vary.* $2 \cdot 10^4$ is smaller because the exponent 4 is less than the exponent 5.

6. Which value is greater: $500{,}000$ or $50 \cdot 10^4$? Show or explain your thinking.

They are equal. *Explanations vary.* When written as multiples of a power of 10, the exponent on the power on 10 for $500{,}000$ is the same as the exponent on the power of 10 for $50 \cdot 10^4$.

7. Which of the following numbers has the greatest value?

A. $571.3 \cdot 10^2$

B. $5.713 \cdot 10^3$

C. $571.3 \cdot 10^3$

D. $571.3 \cdot 10^4$

8. Order the expressions from *least* to *greatest*.

$2.612 \cdot 10^5$	$26.12 \cdot 10^2$	$2612 \cdot 10^3$	$2612 \cdot 10^4$

$26.12 \cdot 10^2$	**$2.612 \cdot 10^5$**	**$2612 \cdot 10^3$**	**$2612 \cdot 10^4$**

Least **Greatest**

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.07

1. Rewrite each expression three different ways using powers of 10.

a. 576,000

Responses vary.

$576 \cdot 10^3$ | $57.6 \cdot 10^4$ | $5.76 \cdot 10^5$

b. 9,510,000

Responses vary.

$9510 \cdot 10^3$ | $951 \cdot 10^4$ | $95.1 \cdot 10^5$

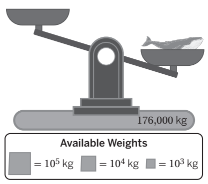
2. Rewrite each value as a combination of powers of 10.

a. One light-year is about 6 trillion miles. *Responses vary.*
$6 \cdot 10^{12}$

b. On average, the human brain has 86 billion neurons. *Responses vary.*
$86 \cdot 10^9$

Problems 3–4: Three students tried to balance the scale using weights measuring 10^5 kilograms, 10^4 kilograms, and 10^3 kilograms. They each wrote the weight of the whale as:

- **Expression A:** $176 \cdot 10^2$ kilograms
- **Expression B:** $17.6 \cdot 10^4$ kilograms
- **Expression C:** $1 \cdot 10^5 + 7 \cdot 10^4 + 6 \cdot 10^3$ kilograms



3. Which of these expressions are accurate?
Expressions B and C

4. Write an expression to represent a different combination of available weights that will balance the scale.
Responses vary. $1.76 \cdot 10^5$ kilograms

Name: .... Date: .... Period: ....

5. Which value is less: $2 \cdot 10^4$ or $2 \cdot 10^5$? Show or explain your thinking.
$2 \cdot 10^4$. *Explanations vary.* $2 \cdot 10^4$ is smaller because the exponent 4 is less than the exponent 5.

6. Which value is greater: 500,000 or $50 \cdot 10^4$? Show or explain your thinking.
They are equal. *Explanations vary.* When written as multiples of a power of 10, the exponent on the power on 10 for 500,000 is the same as the exponent on the power of 10 for $50 \cdot 10^4$.

7. Which of the following numbers has the greatest value?

A. $571.3 \cdot 10^2$
B. $5.713 \cdot 10^3$
C. $571.3 \cdot 10^3$
D. $571.3 \cdot 10^4$

8. Order the expressions from *least* to *greatest*.

$2.612 \cdot 10^5$	$26.12 \cdot 10^2$	$2612 \cdot 10^3$	$2612 \cdot 10^4$
$26.12 \cdot 10^2$	$2.612 \cdot 10^5$	$2612 \cdot 10^3$	$2612 \cdot 10^4$
Least			**Greatest**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.3
2	2	8.EE.A.3
3	2	8.EE.A.3
4	2	8.EE.A.3
5	2	8.EE.A.3
6	2	8.EE.A.3
7	2	8.EE.A.3
8	2	8.EE.A.3

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.08

1. Select *all* expressions that are equivalent to $\frac{1}{100}$.

- ☐ **A.** $-1 \cdot 10^2$
- ☑ **B.** $1 \cdot 10^{-2}$
- ☐ **C.** $-1 \cdot 10^2$
- ☑ **D.** $10 \cdot 10^{-3}$
- ☐ **E.** $-1 \cdot 10^{-3}$

2. Order the expressions from *least* to *greatest*.

$-4 \cdot 10^0$	$-5 \cdot 10^{-2}$	$4 \cdot 10^2$	$5 \cdot 10^{-2}$	$-4 \cdot 10^2$

$-4 \cdot 10^2$	$-4 \cdot 10^0$	$-5 \cdot 10^{-2}$	$5 \cdot 10^{-2}$	$4 \cdot 10^2$

Least **Greatest**

Problems 3–4: Write each sum as a decimal.

3. $7 \cdot 10^{-3} + 3 \cdot 10^{-4} + 2 \cdot 10^{-5}$

0.00732

4. $5 \cdot 10^{-6} + 1 \cdot 10^{-3} + 4 \cdot 10^{-1}$

0.401005

Problems 5–6: Write each value as a number times a single power of 10.

5. $\frac{8}{100000} = 8 \cdot 10^{-5}$

6. $0.012 = 12 \cdot 10^{-3}$ **(or equivalent)**

7. Write $-625{,}000$ three different ways, using a single power of 10.

Responses vary.

$-6.25 \cdot 10^5$ $-62.5 \cdot 10^4$ $-625 \cdot 10^3$

8. Write 0.000986 three different ways, using a single power of 10.

Responses vary.

$9.86 \cdot 10^{-4}$ $98.6 \cdot 10^{-5}$ $986 \cdot 10^{-6}$

9. Which value is greater: 0.091 or $9 \cdot 10^{-3}$. Explain your thinking.

0.091. *Explanations vary.* $9 \cdot 10^{-3}$ is equal to 0.009, and $0.091 > 0.0090$.

10. For what values of x and y will this statement be true: $5 \cdot 10^{-x} > 5 \cdot 10^{-y}$? Show or explain your thinking.

This statement will be true when $x < y$. *Explanations vary.* $5 \cdot 10^{-x} = \frac{5}{10^x}$ and $5 \cdot 10^{-y} = \frac{5}{10^y} \cdot \frac{5}{10^x} > \frac{5}{10^y}$ when $x < y$ because $10^x < 10^y$, and a smaller divisor results in a larger quotient.

Name: ______ Date: ______ Period: ______

Additional Practice

ACC7.8.08

1. Select *all* expressions that are equivalent to $\frac{1}{100}$.

- ☐ A. $-1 \cdot 10^2$
- ☑ B. $1 \cdot 10^{-2}$
- ☐ C. $-1 \cdot 10^2$
- ☑ D. $10 \cdot 10^{-3}$
- ☐ E. $-1 \cdot 10^{-3}$

2. Order the expressions from *least* to *greatest*.

$-4 \cdot 10^0$	$-5 \cdot 10^{-2}$	$4 \cdot 10^2$	$5 \cdot 10^{-2}$	$-4 \cdot 10^2$

$-4 \cdot 10^2$	$-4 \cdot 10^0$	$-5 \cdot 10^{-2}$	$5 \cdot 10^{-2}$	$4 \cdot 10^2$

Least **Greatest**

Problems 3–4: Write each sum as a decimal.

3. $7 \cdot 10^{-3} + 3 \cdot 10^{-4} + 2 \cdot 10^{-5}$

0.00732

4. $5 \cdot 10^{-6} + 1 \cdot 10^{-3} + 4 \cdot 10^{-1}$

0.401005

Problems 5–6: Write each value as a number times a single power of 10.

5. $\frac{8}{100000} = 8 \cdot 10^{-5}$

6. $0.012 = 12 \cdot 10^{-3}$ **(or equivalent)**

Name: ______ Date: ______ Period: ______

7. Write $-625{,}000$ three different ways, using a single power of 10.

Responses vary.

$-6.25 \cdot 10^5$ $-62.5 \cdot 10^4$ $-625 \cdot 10^3$

8. Write 0.000986 three different ways, using a single power of 10.

Responses vary.

$9.86 \cdot 10^{-4}$ $98.6 \cdot 10^{-5}$ $986 \cdot 10^{-6}$

9. Which value is greater: 0.091 or $9 \cdot 10^{-3}$. Explain your thinking.

0.091. *Explanations vary.* $9 \cdot 10^{-3}$ is equal to 0.009, and $0.091 > 0.0090$.

10. For what values of x and y will this statement be true: $5 \cdot 10^{-x} > 5 \cdot 10^{-y}$? Show or explain your thinking.

This statement will be true when $x < y$. *Explanations vary.* $5 \cdot 10^{-x} = \frac{5}{10^x}$ and $5 \cdot 10^{-y} = \frac{5}{10^y} \cdot \frac{5}{10^x} > \frac{5}{10^y}$ when $x < y$ because $10^x < 10^y$, and a smaller divisor results in a larger quotient.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.3
2	2	8.EE.A.3
3	1	8.EE.A.3
4	1	8.EE.A.3
5	1	8.EE.A.3
6	1	8.EE.A.3
7	2	8.EE.A.3
8	2	8.EE.A.3
9	1	8.EE.A.3
10	2	8.EE.A.3

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.09

1. Decide whether each statement is true or false. If the statement is false, explain your thinking.

a) 18.2 billion written in scientific notation is 18.2×10^9.

False. *Explanations vary.* The first part of the number is 18.2 but in scientific notation the first part should be less than 10, 1.82×10^{10}.

b) 95 million written in scientific notation is 9.5×10^7.

True

c) 0.00083 written in scientific notation is 83×10^{-4}.

False. *Explanations vary.* The first part of the number is 83 but in scientific notation the first part should be less than 10, 8.3×10^{-4}.

d) 0.0005 written in scientific notation is 5×10^{-4}.

True

2. Complete the table by filling in the scientific notation or equivalent number.

Number	Scientific Notation
23,400	**2.34×10^4**
0.0035	**3.5×10^{-3}**
64,000,000	6.4×10^7
0.00097	9.7×10^{-4}
3,500	**3.5×10^3**
0.06	6×10^{-2}
0.35	**3.5×10^{-1}**
3.5	**3.5×10^0**
350,000,000,000	**3.5×10^{11}**
0.000000006	6×10^{-9}

3. Han and Clare were determining the correct way to write the approximate radius of a planet, 71 million meters, in scientific notation. Han wrote the value as 71×10^6 meters. Clare wrote the value as 7.1×10^7 meters. Who is correct? Explain your thinking.

Clare is correct. *Explanations vary*. The first factor must be greater than or equal to 1, but less than 10.

4. Bard and Elena were determining the correct way to write the approximate diameter of a red blood cell, 0.000007 meters, in scientific notation. Bard wrote the value as 7×10^{-6} meters. Elena wrote the value as 0.7×10^{-5} meters. Who is correct? Explain your thinking.

Bard is correct. *Explanations vary*. The first factor must be greater than or equal to 1, not less than 1.

5. The radius of a certain fish egg is 0.0028 meters. Andre claims that the number written in scientific notation is 2.8×10^3 meters. Andre is incorrect. Without evaluating, how do you know he is incorrect? Explain your thinking.

***Explanations vary*. Numbers greater than 0, but less than 1, are written using negative exponents on the power of 10.**

6. Shawn analyzed a microorganism in science class. Using a microscope, Shawn measured the diameter to be 0.0000035 centimeters. Shawn was asked to write this value in meters and rewrote the value as 3.5×10^{-6} meters. Did Shawn correctly rewrite the value in scientific notation? Explain your thinking.

No. *Explanations vary*. The value in scientific notation is 3.5×10^{-8}. Shawn's solution shows the value in centimeters.

7. The volume of a drop of a liquid is 0.05 milliliters. Priya was asked to write this value in liters, so she rewrote the value as 5×10^{-6} liters. Did she correctly rewrite the value in scientific notation? Explain your thinking.

No. *Explanations vary*. The value in scientific notation is 5×10^{-5} liters.

Name: ____________ Date: ________ Period: ________

Additional Practice ACC7.8.09

1. Decide whether each statement is true or false. If the statement is false, explain your thinking.
   - a. 18.2 billion written in scientific notation is 18.2×10^9.
     False. ***Explanations vary.*** **The first part of the number is 18.2 but in scientific notation the first part should be less than 10, 1.82×10^{10}.**
   - b. 95 million written in scientific notation is 9.5×10^7.
     True
   - c. 0.00083 written in scientific notation is 83×10^{-4}.
     False. ***Explanations vary.*** **The first part of the number is 83 but in scientific notation the first part should be less than 10, 8.3×10^{-4}.**
   - d. 0.0005 written in scientific notation is 5×10^{-4}.
     True

2. Complete the table by filling in the scientific notation or equivalent number.

Number	Scientific Notation
23,400	**2.34×10^4**
0.0035	**3.5×10^{-3}**
64,000,000	6.4×10^7
0.00097	9.7×10^{-4}
3,500	**3.5×10^3**
0.06	6×10^{-2}
0.35	**3.5×10^{-1}**
3.5	**3.5×10^0**
350,000,000,000	**3.5×10^{11}**
0.000000006	6×10^{-9}

Name: ____________ Date: ________ Period: ________

3. Han and Clare were determining the correct way to write the approximate radius of a planet, 71 million meters, in scientific notation. Han wrote the value as 71×10^6 meters. Clare wrote the value as 7.1×10^7 meters. Who is correct? Explain your thinking.
   Clare is correct. ***Explanations vary.*** **The first factor must be greater than or equal to 1, but less than 10.**

4. Bard and Elena were determining the correct way to write the approximate diameter of a red blood cell, 0.000007 meters, in scientific notation. Bard wrote the value as 7×10^{-6} meters. Elena wrote the value as 0.7×10^{-5} meters. Who is correct? Explain your thinking.
   Bard is correct. ***Explanations vary.*** **The first factor must be greater than or equal to 1, not less than 1.**

5. The radius of a certain fish egg is 0.0028 meters. Andre claims that the number written in scientific notation is 2.8×10^3 meters. Andre is incorrect. Without evaluating, how do you know he is incorrect? Explain your thinking.
   Explanations vary. **Numbers greater than 0, but less than 1, are written using negative exponents on the power of 10.**

6. Shawn analyzed a microorganism in science class. Using a microscope, Shawn measured the diameter to be 0.0000035 centimeters. Shawn was asked to write this value in meters and rewrote the value as 3.5×10^{-6} meters. Did Shawn correctly rewrite the value in scientific notation? Explain your thinking.
   No. ***Explanations vary.*** **The value in scientific notation is 3.5×10^{-8}. Shawn's solution shows the value in centimeters.**

7. The volume of a drop of a liquid is 0.05 milliliters. Priya was asked to write this value in liters, so she rewrote the value as 5×10^{-6} liters. Did she correctly rewrite the value in scientific notation? Explain your thinking.
   No. ***Explanations vary.*** **The value in scientific notation is 5×10^{-5} liters.**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.4
2	2	8.EE.A.4
3	2	8.EE.A.4
4	2	8.EE.A.4
5	2	8.EE.A.4
6	2	8.EE.A.4
7	2	8.EE.A.4

Notes:

Science Mom Lesson 107

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.11

1. Which number is greater? Circle one.

$24 \cdot 10^7$ $5 \cdot 10^7$

About how many times greater is it than the other number?

$24 \cdot 10^7$ is about 5 times greater than $5 \cdot 10^7$.

2. Which number is greater? Circle one.

$3 \cdot 10^8$ $6.214 \cdot 10^8$

About how many times greater is it than the other number?

$6.214 \cdot 10^8$ is about 2 times greater than $3 \cdot 10^8$.

3. Which number is greater? Circle one.

$8.6 \cdot 10^6$ $41 \cdot 10^5$

About how many times greater is it than the other number?

$8.6 \cdot 10^6$ is about 2 times greater than $41 \cdot 10^5$.

Problems 4–5: Complete each sentence by writing a number in scientific notation.

Responses vary.

4. $12.8 \cdot 10^5$ is about **$6 \cdot 10^3$** times as large as $2.2 \cdot 10^2$.

5. $10.1 \cdot 10^{12}$ is about $2 \cdot 10^8$ times as large as **$5 \cdot 10^4$**.

6. On planet Zerg, there are two different types of alien species, zings and zangs. One zing has a mass of $5.6 \cdot 10^6$ kilograms and one zang has a mass of $2.1 \cdot 10^2$ kilograms. About how many times less in mass is one zang than one zing? Write your answer in scientific notation.

The mass of one zang is about $3 \cdot 10^4$ times less than the mass of one zing.

7. The mass of one Brachiosaurus is estimated to have been $7.9 \cdot 10^4$ kilograms. The mass of one antarctic krill is $4.86 \cdot 10^{-4}$ kilograms. About how many times more massive is one Brachiosaurus than one antarctic krill? Write your answer in scientific notation.

About $1.63 \cdot 10^8$ times more massive.

8. The mass of one white-toothed pygmy shrew is $4.86 \cdot 10^{-3}$ kilograms. One of the largest blue whales to ever have lived had a mass of about $1.7 \cdot 10^5$ kilograms. About how many times more massive is the largest blue whale than the mass of one white-toothed pygmy shrew? Write your answer in scientific notation.

$4 \cdot 10^7$ times larger.

Name: ________ Date: ________ Period: ________

Additional Practice ACC7.8.11

1. Which number is greater? Circle one.

   $24 \cdot 10^7$ $5 \cdot 10^7$

   About how many times greater is it than the other number?

   $24 \cdot 10^7$ is about 5 times greater than $5 \cdot 10^7$.

2. Which number is greater? Circle one.

   $3 \cdot 10^8$ $6.214 \cdot 10^8$

   About how many times greater is it than the other number?

   $6.214 \cdot 10^8$ is about 2 times greater than $3 \cdot 10^8$.

3. Which number is greater? Circle one.

   $8.6 \cdot 10^6$ $41 \cdot 10^5$

   About how many times greater is it than the other number?

   $8.6 \cdot 10^6$ is about 2 times greater than $41 \cdot 10^5$.

Problems 4–5: Complete each sentence by writing a number in scientific notation.

Responses vary.

4. $12.8 \cdot 10^5$ is about **$6 \cdot 10^3$** times as large as $2.2 \cdot 10^2$.

5. $10.1 \cdot 10^{12}$ is about $2 \cdot 10^8$ times as large as **$5 \cdot 10^4$**.

Name: ________ Date: ________ Period: ________

6. On planet Zerg, there are two different types of alien species, zings and zangs. One zing has a mass of $5.6 \cdot 10^6$ kilograms and one zang has a mass of $2.1 \cdot 10^2$ kilograms. About how many times less in mass is one zang than one zing? Write your answer in scientific notation.

   The mass of one zang is about $3 \cdot 10^4$ times less than the mass of one zing.

7. The mass of one Brachiosaurus is estimated to have been $7.9 \cdot 10^4$ kilograms. The mass of one antarctic krill is $4.86 \cdot 10^{-4}$ kilograms. About how many times more massive is one Brachiosaurus than one antarctic krill? Write your answer in scientific notation.

   About $1.63 \cdot 10^8$ times more massive.

8. The mass of one white-toothed pygmy shrew is $4.86 \cdot 10^{-3}$ kilograms. One of the largest blue whales to ever have lived had a mass of about $1.7 \cdot 10^5$ kilograms. About how many times more massive is the largest blue whale than the mass of one white-toothed pygmy shrew? Write your answer in scientific notation.

   $4 \cdot 10^7$ times larger.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.3, 8.EE.A.4
2	2	8.EE.A.3, 8.EE.A.4
3	2	8.EE.A.3, 8.EE.A.4
4	2	8.EE.A.3, 8.EE.A.4
5	2	8.EE.A.3, 8.EE.A.4
6	2	8.EE.A.3, 8.EE.A.4
7	2	8.EE.A.3, 8.EE.A.4
8	2	8.EE.A.3, 8.EE.A.4

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.13

1. Decide whether each statement is true or false. Show or explain your thinking.

a $6 \cdot 10^3 + 2 \cdot 10^4 = 8 \cdot 10^7$

False. *Explanations vary.* The digits 6 and 2 do not have the same place value because the powers of 10 are not the same.

b $8 \cdot 10^{-3} + 4 \cdot 10^{-2} = 4.8 \cdot 10^{-2}$

True. *Explanations vary.* $8 \times 10^{-3} = 0.8 \times 10^{-2}$, which will result in a sum of 4.8×10^{-2}.

c $5 \cdot 10^3 - 3.3 \cdot 10^4 = 1.7 \cdot 10^2$

False. *Explanations vary.* 3.3×10^4 is greater than 5×10^3, so the difference must be negative.

2. Evaluate each expression. Write the result in scientific notation.

a $1.9 \cdot 10^{-5} + 8.9 \cdot 10^{-5}$

$\mathbf{1.08 \cdot 10^{-4}}$

b $4.9 \cdot 10^6 - 4.1 \cdot 10^5$

$\mathbf{4.49 \times 10^6}$

c $3.8 \cdot 10^3 + 6.2 \cdot 10^3$

$\mathbf{1 \times 10^4}$

d $5.3 \cdot 10^{-4} - 5.2 \cdot 10^{-4}$

$\mathbf{1 \times 10^{-5}}$

3. Select *all* the expressions that are equal to the expression $1 \cdot 10^5$.

- ☑ **A.** $7.9 \cdot 10^4 + 2.1 \cdot 10^4$
- ☐ **B.** $6.3 \cdot 10^5 - 5.3 \cdot 10^4$
- ☑ **C.** $8 \cdot 10^4 + 2 \cdot 10^4$
- ☐ **D.** $3 \cdot 10^5 + 7 \cdot 10^5$
- ☑ **E.** $2.1 \cdot 10^5 - 1.1 \cdot 10^5$

Name: .... Date: .... Period: ....

4. Select *all* the expressions that are equal to the expression $6 \cdot 10^{-3}$.

- ☐ **A.** $8.8 \cdot 10^{-3} - 8.2 \cdot 10^{-3}$
- ☑ **B.** $7.7 \cdot 10^{-2} - 7.1 \cdot 10^{-2}$
- ☑ **C.** $3 \cdot 10^{-3} + 3 \cdot 10^{-3}$
- ☐ **D.** $4 \cdot 10^{-4} + 2 \cdot 10^{-3}$
- ☑ **E.** $4.7 \cdot 10^{-2} - 4.1 \cdot 10^{-2}$

5. Priya wants to find $4.3 \cdot 10^4 + 3.5 \cdot 10^5$ and writes $4.3 \cdot 10^4 + 3.5 \cdot 10^5 = 7.8 \cdot 10^5$. Explain Priya's mistake and determine the correct sum.

***Explanations vary*. Priya cannot add 4.3 and 3.5 because the powers of 10 are different. The correct sum is $3.93 \cdot 10^5$.**

6. Bard writes $2.8 \cdot 10^3 - 2.7 \cdot 10^3$ in scientific notation as $1 \cdot 10^3$.
Is Bard correct? Show or explain your thinking.

No. *Explanations vary*. The difference is $0.1 \cdot 10^3$. In scientific notation, the first digit should be greater than 1 and less than 10, so $0.1 \cdot 10^3 = 1 \cdot 10^2$.

Additional Practice | Answer Key

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.13

1. Decide whether each statement is true or false. Show or explain your thinking.

a. $6 \cdot 10^3 + 2 \cdot 10^4 = 8 \cdot 10^7$

False. *Explanations vary.* The digits 6 and 2 do not have the same place value because the powers of 10 are not the same.

b. $8 \cdot 10^{-3} + 4 \cdot 10^{-2} = 4.8 \cdot 10^{-2}$

True. *Explanations vary.* $8 \times 10^{-3} = 0.8 \times 10^{-2}$, which will result in a sum of 4.8×10^{-2}.

c. $5 \cdot 10^3 - 3.3 \cdot 10^4 = 1.7 \cdot 10^2$

False. *Explanations vary.* 3.3×10^4 is greater than 5×10^3, so the difference must be negative.

2. Evaluate each expression. Write the result in scientific notation.

a. $1.9 \cdot 10^{-5} + 8.9 \cdot 10^{-5}$

$1.08 \cdot 10^{-4}$

b. $4.9 \cdot 10^6 - 4.1 \cdot 10^5$

4.49×10^6

c. $3.8 \cdot 10^3 + 6.2 \cdot 10^3$

1×10^4

d. $5.3 \cdot 10^{-4} - 5.2 \cdot 10^{-4}$

1×10^{-5}

3. Select *all* the expressions that are equal to the expression $1 \cdot 10^5$.

- ☑ A. $7.9 \cdot 10^4 + 2.1 \cdot 10^4$
- ☐ B. $6.3 \cdot 10^5 - 5.3 \cdot 10^4$
- ☑ C. $8 \cdot 10^4 + 2 \cdot 10^4$
- ☐ D. $3 \cdot 10^5 + 7 \cdot 10^5$
- ☑ E. $2.1 \cdot 10^5 - 1.1 \cdot 10^5$

Name: .... Date: .... Period: ....

4. Select *all* the expressions that are equal to the expression $6 \cdot 10^{-3}$.

- ☐ A. $8.8 \cdot 10^{-3} - 8.2 \cdot 10^{-3}$
- ☑ B. $7.7 \cdot 10^{-2} - 7.1 \cdot 10^{-2}$
- ☑ C. $3 \cdot 10^{-3} + 3 \cdot 10^{-3}$
- ☐ D. $4 \cdot 10^{-4} + 2 \cdot 10^{-3}$
- ☑ E. $4.7 \cdot 10^{-2} - 4.1 \cdot 10^{-2}$

5. Priya wants to find $4.3 \cdot 10^4 + 3.5 \cdot 10^5$ and writes $4.3 \cdot 10^4 + 3.5 \cdot 10^5 = 7.8 \cdot 10^5$. Explain Priya's mistake and determine the correct sum.

***Explanations vary.* Priya cannot add 4.3 and 3.5 because the powers of 10 are different. The correct sum is $3.93 \cdot 10^5$.**

6. Bard writes $2.8 \cdot 10^3 - 2.7 \cdot 10^3$ in scientific notation as $1 \cdot 10^3$. Is Bard correct? Show or explain your thinking.

No. *Explanations vary.* The difference is $0.1 \cdot 10^3$. In scientific notation, the first digit should be greater than 1 and less than 10, so $0.1 \cdot 10^3 = 1 \cdot 10^2$.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.4
2	2	8.EE.A.4
3	2	8.EE.A.4
4	2	8.EE.A.4
5	2	8.EE.A.4
6	2	8.EE.A.4

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.8.14

Problems 1–4: In 2023, a group of countries had an approximate population of 742.3 million people. Russia, Turkey, Germany, and the United Kingdom had the greatest populations out of all the countries.

1. What was the total population of all four countries? Write your answer in scientific notation.

$3.87 \cdot 10^8$ people

Country	Population (people)
Russia	$1.5 \cdot 10^8$
Turkey	$8.5 \cdot 10^7$
Germany	$8.4 \cdot 10^7$
United Kingdom	$6.8 \cdot 10^7$

2. What was the total population of the other countries in the group that are not listed? Write your answer in scientific notation.

$3.553 \cdot 10^8$ people

3. One of the smaller countries in the group is Liechtenstein, which has a population of about 40,000 people. How many times greater is the population of Germany than the population of Liechtenstein?

$\frac{8.4 \cdot 10^7}{4 \cdot 10^4} \approx 2.1 \cdot 10^3$ or 2,100 times greater

Problems 4–6: Scientists measured the weight and length of the following sea creatures:

- Blue whale: $400{,}000$ pounds and 100 feet in length
- Great white shark: $5{,}000$ pounds and 20 feet in length
- Orca whale: $12{,}000$ pounds and 25 feet in length
- Manta ray: $6{,}000$ pounds and 29 feet in length
- Angler fish: 5 pounds and 3.3 feet in length
- Clownfish: 0.15 pounds and 0.33 feet in length
- Sea horse: 1 pound and 0.5 inches in length
- Krill: 0.0044 pounds and 0.2 feet in length

4. What is the sum of the weights of a blue whale, a great white shark, and an orca whale? Write your answer in scientific notation.

$4.17 \cdot 10^5$ pounds

5. How much longer is a manta ray than a clownfish? Write your answer in scientific notation.

2.87×10^1 feet longer

6. A blue whale can consume an astonishing amount of krill each day. On average, a blue whale eats around $8{,}000$ pounds of krill daily during feeding seasons. Approximately how many krill does a blue whale consume in one day? Write your answer in scientific notation.

$2 \cdot 10^6$ krill

Name: .... Date: .... Period: ....

Additional Practice

ACC7.8.14

Problems 1–4: In 2023, a group of countries had an approximate population of 742.3 million people. Russia, Turkey, Germany, and the United Kingdom had the greatest populations out of all the countries.

Country	Population (people)
Russia	$1.5 \cdot 10^8$
Turkey	$8.5 \cdot 10^7$
Germany	$8.4 \cdot 10^7$
United Kingdom	$6.8 \cdot 10^7$

1. What was the total population of all four countries? Write your answer in scientific notation.

$3.87 \cdot 10^8$ people

2. What was the total population of the other countries in the group that are not listed? Write your answer in scientific notation.

$3.553 \cdot 10^8$ people

3. One of the smaller countries in the group is Liechtenstein, which has a population of about 40,000 people. How many times greater is the population of Germany than the population of Liechtenstein?

$\frac{8.4 \cdot 10^7}{4 \cdot 10^4} \approx 2.1 \cdot 10^3$ or 2,100 times greater

Name: .... Date: .... Period: ....

Problems 4–6: Scientists measured the weight and length of the following sea creatures:

- Blue whale: 400,000 pounds and 100 feet in length
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- Sea horse: 1 pound and 0.5 inches in length
- Krill: 0.0044 pounds and 0.2 feet in length

4. What is the sum of the weights of a blue whale, a great white shark, and an orca whale? Write your answer in scientific notation.

$4.17 \cdot 10^5$ pounds

5. How much longer is a manta ray than a clownfish? Write your answer in scientific notation.

2.87×10^1 feet longer

6. A blue whale can consume an astonishing amount of krill each day. On average, a blue whale eats around 8,000 pounds of krill daily during feeding seasons. Approximately how many krill does a blue whale consume in one day? Write your answer in scientific notation.

$2 \cdot 10^6$ krill

Practice Problem Analysis

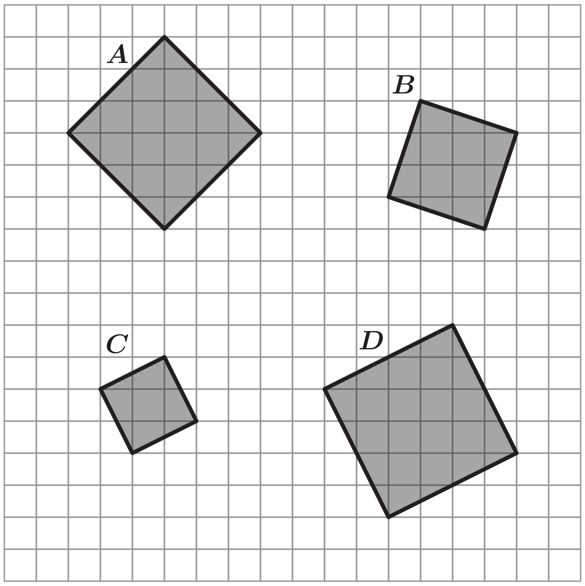
Problem	DOK	Standard(s)
1	2	8.EE.A.4
2	2	8.EE.A.4
3	2	8.EE.A.3
4	2	8.EE.A.3, 8.EE.A.4
5	2	8.EE.A.3, 8.EE.A.4
6	2	8.EE.A.3, 8.EE.A.4

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.01

Problems 1–4: Determine the area of each tilted square. Each square grid represents 1 square unit.



1. Square A

18 square units

2. Square B

10 square units

3. Square C

5 square units

4. Square D

20 square units

Problems 5–6: Determine the area of each square given its side length.

5. Side length: 4 centimeters

16 square centimeters

6. Side length: x units

x^2 square units

Problems 7–9: Here are the areas of three squares. Determine the side length of each square.

7. Area: 36 square meters

6 meters

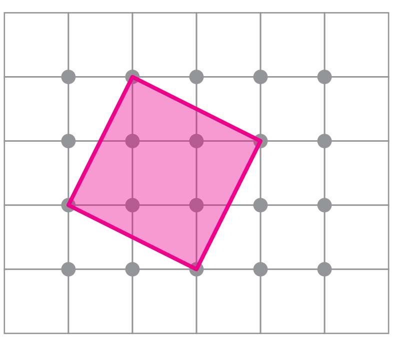
8. Area: $\frac{9}{49}$ square inches

$\frac{3}{7}$ inches

9. Area: w^2 square units

w units

10. Determine the area of the largest tilted square that can be created on the dot grid. Show your thinking. ***Drawings vary.***

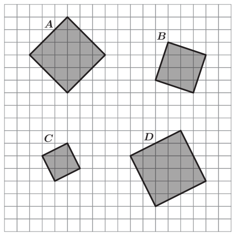


5 square units. *Work varies.* Area $= 9 - 4(1) = 5$ square units

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.01

Problems 1–4: Determine the area of each tilted square. Each square grid represents 1 square unit.



1. Square A
   18 square units

2. Square B
   10 square units

3. Square C
   5 square units

4. Square D
   20 square units

Problems 5–6: Determine the area of each square given its side length.

5. Side length: 4 centimeters
   16 square centimeters

6. Side length: x units
   x^2 square units

Unit 9 Lesson 1 253

Name: .... Date: .... Period: ....

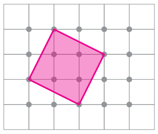
Problems 7–9: Here are the areas of three squares. Determine the side length of each square.

7. Area: 36 square meters
   6 meters

8. Area: $\frac{9}{49}$ square inches
   $\frac{3}{7}$ inches

9. Area: w^2 square units
   w units

10. Determine the area of the largest tilted square that can be created on the dot grid. Show your thinking. *Drawings vary.*



5 square units. *Work varies.* Area $= 9 - 4(1) = 5$ square units

Unit 9 Lesson 1 254 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B*, 8.EE.A.2*
2	1	8.G.B*, 8.EE.A.2*
3	1	8.G.B*, 8.EE.A.2*
4	1	8.G.B*, 8.EE.A.2*
5	1	8.EE.A.2*
6	1	8.EE.A.2*
7	1	8.EE.A.2*
8	1	8.EE.A.2*
9	1	8.EE.A.2*
10	2	8.EE.A.2*

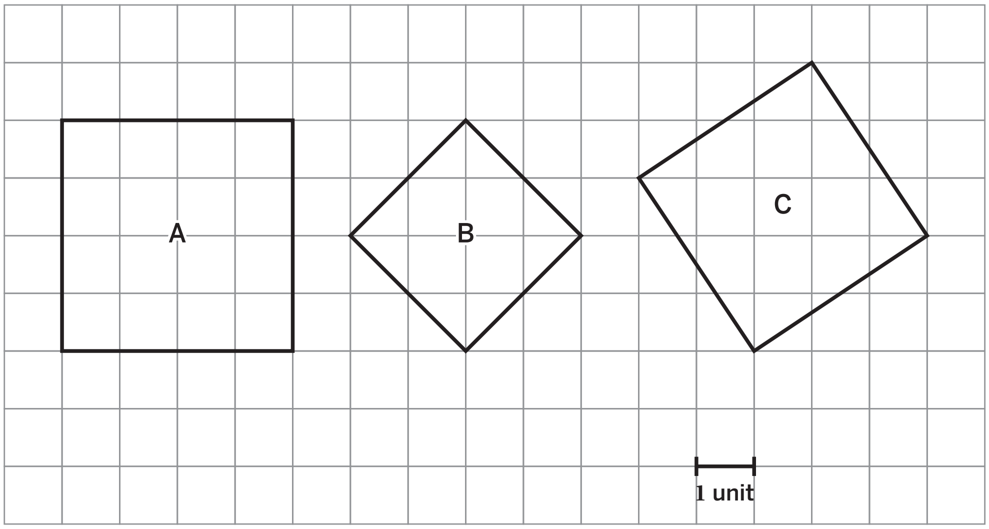
*This problem builds toward the standard shown.

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.02

1. Determine the area and exact side length of each square.



Square A has an area of 16 square units and a side length of 4 units (or $\sqrt{16}$ units).

Square B has an area of 8 square units and a side length of $\sqrt{8}$ units.

Square C has an area of 13 square units and a side length of $\sqrt{13}$ units.

2. Here are the areas of different squares. Determine the side length for each square.

a. Area: 9 square centimeters

3 centimeters

b. Area: $\frac{1}{4}$ square inches

$\frac{1}{2}$ inches

c. Area: 25 square meters

5 meters

d. Area: $\frac{4}{49}$ square feet

$\frac{2}{7}$ feet

3. Here is some information about three squares. Determine which side length matches each square.

- Square A is larger than square B.
- Square B is larger than square C.
- The three squares' side lengths are $\sqrt{10}$, 5.1, and $\sqrt{30}$ units.

Square A: **$\sqrt{30}$** Square B: **5.1** Square C: **$\sqrt{10}$**

4. Square A has an area of 16 square inches. Select *all* the expressions that are equal to the side length of this square (in inches).

- ☐ **A.** 2
- ☑ **B.** 4
- ☐ **C.** 8
- ☐ **D.** $\sqrt{4}$
- ☑ **E.** $\sqrt{16}$

5. Here are the side lengths of different squares. Determine the area of each square.

a Side length: 2 centimeters

4 square centimeters (or equivalent)

b Side length: $\frac{1}{3}$ foot

$\frac{1}{9}$ square foot (or equivalent)

c Side length: $\frac{2}{7}$ inches

$\frac{4}{49}$ square inches (or equivalent)

d Side length: 0.6 units

0.36 square units (or equivalent)

6. Which square has the greatest side length?

A. Square A with an area of 25 square units

B. Square B with an area of 64 square units

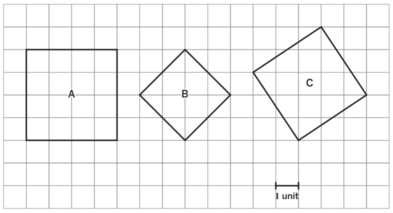
C. Square C with a side length of $\sqrt{49}$ units

D. Square D with a side length of 9 units

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.9.02

1. Determine the area and exact side length of each square.



Square A has an area of 16 square units and a side length of 4 units (or $\sqrt{16}$ units).
Square B has an area of 8 square units and a side length of $\sqrt{8}$ units.
Square C has an area of 13 square units and a side length of $\sqrt{13}$ units.

2. Here are the areas of different squares. Determine the side length for each square.

a. Area: 9 square centimeters
3 centimeters

b. Area: $\frac{1}{4}$ square inches
$\frac{1}{2}$ inches

c. Area: 25 square meters
5 meters

d. Area: $\frac{4}{49}$ square feet
$\frac{2}{7}$ feet

3. Here is some information about three squares. Determine which side length matches each square.

- Square A is larger than square B.
- Square B is larger than square C.
- The three squares' side lengths are $\sqrt{10}$, 5.1, and $\sqrt{30}$ units.

Square A: **$\sqrt{30}$** Square B: **5.1** Square C: **$\sqrt{10}$**

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Name: ______ Date: ______ Period: ______

4. Square A has an area of 16 square inches. Select *all* the expressions that are equal to the side length of this square (in inches).

☐ A. 2 ☑ B. 4 ☐ C. 8 ☐ D. $\sqrt{4}$ ☑ E. $\sqrt{16}$

5. Here are the side lengths of different squares. Determine the area of each square.

a. Side length: 2 centimeters
4 square centimeters (or equivalent)

b. Side length: $\frac{1}{3}$ foot
$\frac{1}{9}$ square foot (or equivalent)

c. Side length: $\frac{2}{7}$ inches
$\frac{4}{49}$ square inches (or equivalent)

d. Side length: 0.6 units
0.36 square units (or equivalent)

6. Which square has the greatest side length?

A. Square A with an area of 25 square units
B. Square B with an area of 64 square units
C. Square C with a side length of $\sqrt{49}$ units
D. Square D with a side length of 9 units

Unit 9 Lesson 2 256 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.A.2*
2	2	8.EE.A.2*
3	2	8.NS.A.2*
4	2	8.EE.A.2*
5	2	8.EE.A.2*
6	3	8.EE.A.2*

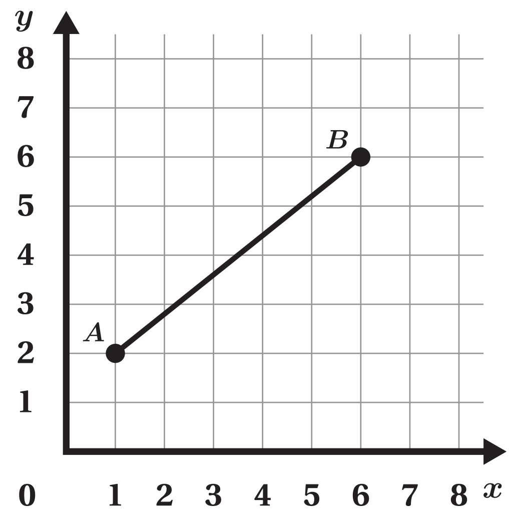
*This problem builds toward the standard shown.

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.03

Problems 1–4: Here is a line segment AB. Each grid square represents 1 square unit. Use a ruler, circle, or square if they help with your thinking.



1. Determine the approximate length of AB.

Responses between 6 and 7 are considered correct.

2. Determine the exact length of AB.

$\sqrt{41}$ units

3. Which method did you choose to help with your thinking? Circle one.

Responses vary.

Ruler Circle Square

4. Why did you choose that particular method?

Responses vary.

Problems 5–6: Determine the value of each square root.

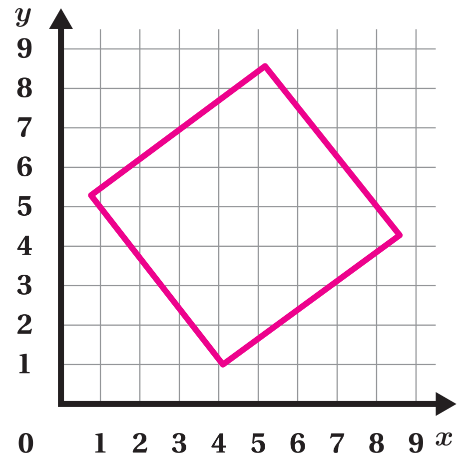
5. $\sqrt{81} = \mathbf{9}$

6. $\sqrt{36} = \mathbf{6}$

Problems 7–8: Estimate each square root. Draw a square if it helps with your thinking.

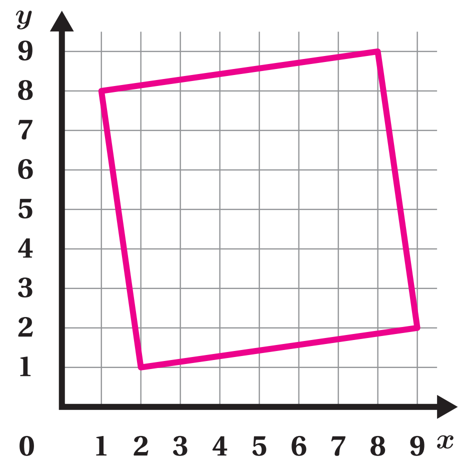
7. $\sqrt{27}$

Responses between 5 and 6 are considered correct.



8. $\sqrt{50}$

Responses between 7 and 8 are considered correct.



Problems 9–10: Determine which two whole numbers each square root is between.

9. $\sqrt{38}$

Between 6 and 7

10. $\sqrt{72}$

Between 8 and 9

11. Here is a list of values ordered from least to greatest. One value is unknown. Which could be the unknown value?

$\frac{9}{2}$, 4.8, ? , $\sqrt{27}$

A. 5.5

B. $(2.5)^2$

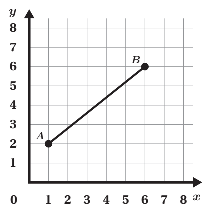
C. 5.1

D. $\frac{5}{2}$

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.9.03

Problems 1–4: Here is a line segment AB. Each grid square represents 1 square unit. Use a ruler, circle, or square if they help with your thinking.



1. Determine the approximate length of AB.
Responses between 6 and 7 are considered correct.

2. Determine the exact length of AB.
$\sqrt{41}$ units

3. Which method did you choose to help with your thinking? Circle one.
Responses vary.
Ruler Circle Square

4. Why did you choose that particular method?
Responses vary.

Problems 5–6: Determine the value of each square root.

5. $\sqrt{81} = 9$

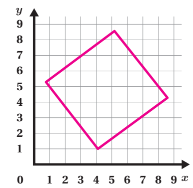
6. $\sqrt{36} = 6$

Unit 9 Lesson 3 257

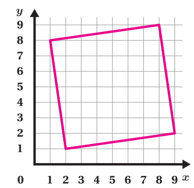
Name: ______ Date: ______ Period: ______

Problems 7–8: Estimate each square root. Draw a square if it helps with your thinking.

7. $\sqrt{27}$
Responses between 5 and 6 are considered correct.



8. $\sqrt{50}$
Responses between 7 and 8 are considered correct.



Problems 9–10: Determine which two whole numbers each square root is between.

9. $\sqrt{38}$
Between 6 and 7

10. $\sqrt{72}$
Between 8 and 9

11. Here is a list of values ordered from least to greatest. One value is unknown. Which could be the unknown value?

$\frac{9}{2}, 4.8, ?, \sqrt{27}$

A. 5.5
B. $(2.5)^2$
C. 5.1
D. $\frac{5}{2}$

Unit 9 Lesson 3 258 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.2
2	2	8.EE.A.2
3	2	8.EE.A.2
4	2	8.EE.A.2
5	2	8.NS.A.2
6	2	8.NS.A.2
7	2	8.NS.A.2
8	2	8.NS.A.2
9	2	8.NS.A.2
10	2	8.NS.A.2
11	2	8.NS.A.2

Notes:

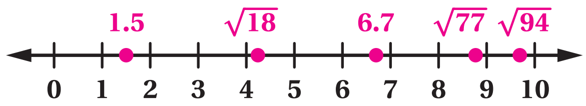
Science Mom Lesson 113

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.04

1. Estimate each number and plot it on the number line.

$\sqrt{77}$, 1.5, 6.7, $\sqrt{94}$, $\sqrt{18}$



2. Estimate each square root to the nearest tenth.

a $\sqrt{13}$

Responses close to, but less than 4 are considered correct.

b $\sqrt{28}$

Responses close to, but greater than 5 are considered correct.

c $\sqrt{78}$

Responses close to, but less than 9 are considered correct.

d $\sqrt{85}$

Responses close to, but greater than 9 are considered correct.

3. Here is a number line with two points plotted between 7 and 8.

a Write a number using square root notation that could represent point A.

Responses vary. $\sqrt{50}$

b Write a number using square root notation that could represent point B.

Responses vary. $\sqrt{63}$

4. Which statement *best* describes the value of $\sqrt{10}$?

A. The value of $\sqrt{10}$ is between 4.5 and 5.

B. The value of $\sqrt{10}$ is between 4 and 4.5.

C. The value of $\sqrt{10}$ is between 3.5 and 4.

D. The value of $\sqrt{10}$ is between 3 and 3.5.

5. Determine whether each statement is *true* or *false*. Circle your response.

a $\sqrt{14}$ is greater than 3.5. True False

b $\sqrt{125}$ is greater than 11.5. True False

c $\sqrt{22}$ is between 5 and 6. True False

d $\sqrt{77}$ is between 8 and 9. True False

6. Explain how you know that $\sqrt{46}$ is a little less than 7.

***Explanations vary.* $\sqrt{49}$ is exactly 7 and $\sqrt{46}$ is a little less than that.**

7. Select *all* the numbers that are greater than 12 and less than 13.

- ☐ **A.** $\sqrt{120}$
- ☑ **B.** $\sqrt{150}$
- ☐ **C.** $\sqrt{140}$
- ☐ **D.** $\sqrt{200}$
- ☑ **E.** $\sqrt{160}$

Name: ________ Date: ________ Period: ________

Additional Practice ACC7.9.04

1. Estimate each number and plot it on the number line.

   $\sqrt{77}$, 1.5, 6.7, $\sqrt{94}$, $\sqrt{18}$

2. Estimate each square root to the nearest tenth.

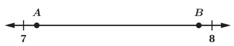
   a. $\sqrt{13}$ Responses close to, but less than 4 are considered correct.

   b. $\sqrt{28}$ Responses close to, but greater than 5 are considered correct.

   c. $\sqrt{78}$ Responses close to, but less than 9 are considered correct.

   d. $\sqrt{85}$ Responses close to, but greater than 9 are considered correct.

3. Here is a number line with two points plotted between 7 and 8.



   a. Write a number using square root notation that could represent point A.
   Responses vary. $\sqrt{50}$

   b. Write a number using square root notation that could represent point B.
   Responses vary. $\sqrt{63}$

4. Which statement *best* describes the value of $\sqrt{10}$?

   A. The value of $\sqrt{10}$ is between 4.5 and 5.

   B. The value of $\sqrt{10}$ is between 4 and 4.5.

   C. The value of $\sqrt{10}$ is between 3.5 and 4.

   D. The value of $\sqrt{10}$ is between 3 and 3.5.

Name: ________ Date: ________ Period: ________

5. Determine whether each statement is *true* or *false*. Circle your response.

   a. $\sqrt{14}$ is greater than 3.5. True False

   b. $\sqrt{125}$ is greater than 11.5. True False

   c. $\sqrt{22}$ is between 5 and 6. True False

   d. $\sqrt{77}$ is between 8 and 9. True False

6. Explain how you know that $\sqrt{46}$ is a little less than 7.
   Explanations vary. $\sqrt{49}$ is exactly 7 and $\sqrt{46}$ is a little less than that.

7. Select *all* the numbers that are greater than 12 and less than 13.

   - [ ] A. $\sqrt{120}$
   - [x] B. $\sqrt{150}$
   - [ ] C. $\sqrt{140}$
   - [ ] D. $\sqrt{200}$
   - [x] E. $\sqrt{160}$

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.EE.A.2, 8.NS.A.2
2	1	8.NS.A.2
3	2	8.NS.A.2
4	1	8.NS.A.2
5	1	8.NS.A.2
6	2	8.EE.A.2, 8.NS.A.2
7	2	8.NS.A.2

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.05

1. Write an equivalent expression that doesn't use a cube root symbol.

a $\sqrt[3]{27}$

$\mathbf{3}$

b $\sqrt[3]{8}$

$\mathbf{2}$

b $\sqrt[3]{125}$

$\mathbf{5}$

2. Complete the table. Write an exact value.

x	x^3
1	$\mathbf{1}$
$\sqrt[3]{10}$	$\mathbf{10}$
$\mathbf{4}$	64
$\sqrt[3]{\mathbf{24}}$	24

3. Order the following from *least* to *greatest*.

$\sqrt[3]{27}$	$\sqrt[3]{120}$	$\sqrt{48}$	$\sqrt{121}$	$\sqrt[3]{\frac{1}{64}}$

$\sqrt[3]{\frac{\mathbf{1}}{\mathbf{64}}}$	$\sqrt[3]{\mathbf{27}}$	$\sqrt[3]{\mathbf{120}}$	$\sqrt{\mathbf{48}}$	$\sqrt{\mathbf{121}}$

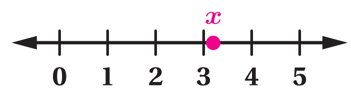
Least **Greatest**

Name: .... Date: .... Period: ....

4. Cube A is smaller than cube B. Cube B is smaller than cube C. The edge lengths of the cubes are $\sqrt[3]{62}$ units, $\sqrt[3]{7}$ units, and 8.5 units. Determine the edge lengths of cube A, cube B, and cube C.

Cube A: $\sqrt[3]{7}$ units
Cube B: $\sqrt[3]{62}$ units
Cube C: 8.5 units

5. Here is the equation $x^3 = 33$. Plot the approximate value of x on the number line.



Response shown on number line.

6. Determine whether each statement is *true* or *false*. Circle your response.

a. $\sqrt[3]{41}$ is less than 3. True **False**

b. $\sqrt{4}$ is equal to $\sqrt[3]{8}$. **True** False

c. $\sqrt[3]{29}$ is between 3 and 4. **True** False

d. $\sqrt[3]{\frac{64}{125}}$ is between 1 and 2. True **False**

7. Cube E has double the edge length of cube F. What could be the volume of each cube?

***Responses vary.* Cube E could have a volume of 8 cubic units, and cube F would have a volume of 1 cubic unit.**

8. Cube G has a volume that is double the volume of cube H. What could be the edge lengths of each cube?

***Responses vary.* Cube G could have an edge length of 2, and cube H would have an edge length of $\sqrt[3]{4}$.**

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.05

1. Write an equivalent expression that doesn't use a cube root symbol.

a $\sqrt[3]{27}$ **3**

b $\sqrt[3]{8}$ **2**

b $\sqrt[3]{125}$ **5**

2. Complete the table. Write an exact value.

x	x^3
1	**1**
$\sqrt[3]{10}$	**10**
4	64
$\sqrt[3]{24}$	24

3. Order the following from *least* to *greatest*.

$\sqrt[3]{27}$	$\sqrt[3]{120}$	$\sqrt{48}$	$\sqrt{121}$	$\sqrt[3]{\frac{1}{64}}$

$\sqrt[3]{\frac{1}{64}}$	$\sqrt[3]{27}$	$\sqrt[3]{120}$	$\sqrt{48}$	$\sqrt{121}$

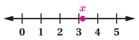
Least **Greatest**

Name: .... Date: .... Period: ....

4. Cube A is smaller than cube B. Cube B is smaller than cube C. The edge lengths of the cubes are $\sqrt[3]{62}$ units, $\sqrt[3]{7}$ units, and 8.5 units. Determine the edge lengths of cube A, cube B, and cube C.

Cube A: $\sqrt[3]{7}$ units
Cube B: $\sqrt[3]{62}$ units
Cube C: 8.5 units

5. Here is the equation $x^3 = 33$. Plot the approximate value of x on the number line.



Response shown on number line.

6. Determine whether each statement is *true* or *false*. Circle your response.

a $\sqrt[3]{41}$ is less than 3. True False

b $\sqrt{4}$ is equal to $\sqrt[3]{8}$. True False

c $\sqrt[3]{29}$ is between 3 and 4. True False

d $\sqrt[3]{\frac{64}{125}}$ is between 1 and 2. True False

7. Cube E has double the edge length of cube F. What could be the volume of each cube?

***Responses vary.* Cube E could have a volume of 8 cubic units, and cube F would have a volume of 1 cubic unit.**

8. Cube G has a volume that is double the volume of cube H. What could be the edge lengths of each cube?

***Responses vary.* Cube G could have an edge length of 2, and cube H would have an edge length of $\sqrt[3]{4}$.**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.EE.A.2
2	1	8.EE.A.2
3	2	8.NS.A.2
4	2	8.NS.A.2
5	2	8.EE.A.2, 8.NS.A.2
6	2	8.EE.A.2, 8.NS.A.2
7	3	8.EE.A.2
8	3	8.EE.A.2

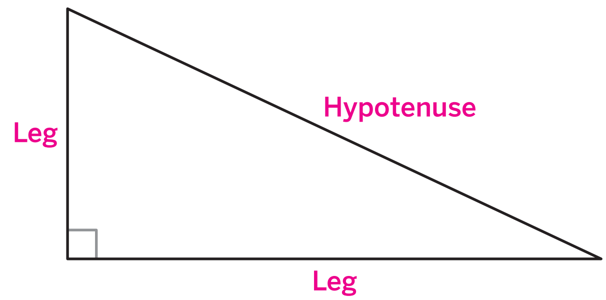
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.9.06

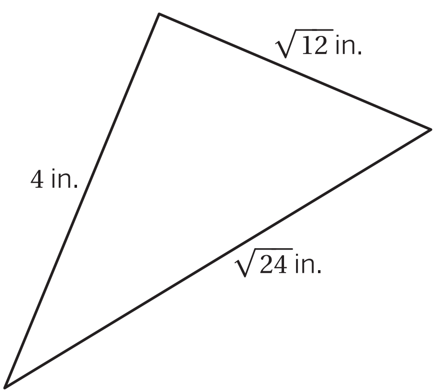
1. Label the legs and hypotenuse of the right triangle.



2. Show that the Pythagorean theorem is true for a right triangle with legs of 3 units and 4 units and a hypotenuse of 5 units.

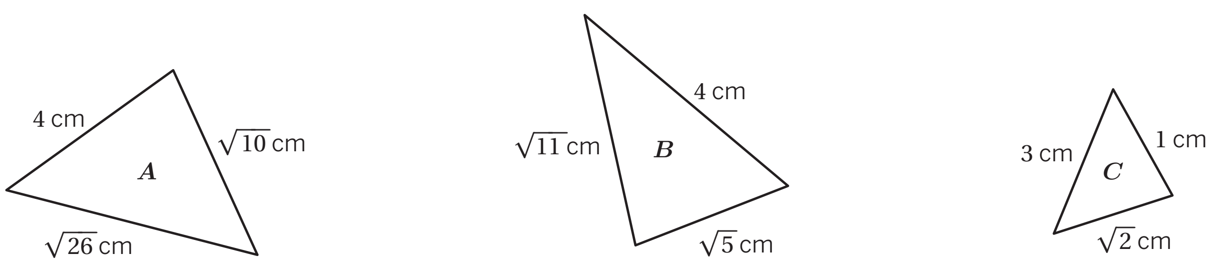
***Work varies.* $3^2 + 4^2 = 25$ and $25 = 5^2$. Therefore, $3^2 + 4^2 = 5^2$, and the Pythagorean theorem is true for this right triangle.**

3. Is the Pythagorean theorem true for the triangle shown? Show or explain your thinking.



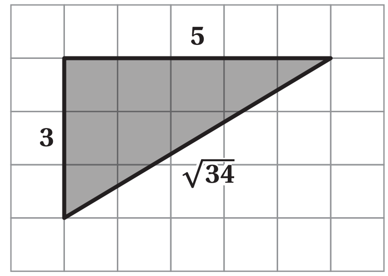
No. *Explanations vary.* The Pythagorean theorem states that $\text{leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$. If leg $= 4$, leg $= \sqrt{12}$, and hypotenuse $= \sqrt{24}$, then $4^2 + \left(\sqrt{12}\right)^2$ should equal $\left(\sqrt{24}\right)^2$ if the Pythagorean theorem is true for this triangle. $4^2 + \left(\sqrt{12}\right)^2$ does not equal $\left(\sqrt{24}\right)^2$, so the Pythagorean theorem is not true for this triangle.

4. For which triangles is $a^2 + b^2 = c^2$ true?



Triangles A and B

5. Select *all* the equations that represent the relationship between the side lengths of the triangle.



- [x] **A.** $(\sqrt{34})^2 = 3^2 + 5^2$
- [x] **B.** $5^2 + 3^2 = (\sqrt{34})^2$
- [ ] **C.** $3^2 + (\sqrt{34})^2 = 5^2$
- [ ] **D.** $(\sqrt{34})^2 + 5^2 = 3^2$
- [x] **E.** $3^2 + 5^2 = (\sqrt{34})^2$

6. Han claims that if $a = \sqrt{5}$, $b = \sqrt{31}$, and $c = 6$, the Pythagorean theorem will be true for this triangle. Is Han correct? Explain your thinking.

Yes. *Explanations vary.* If $a^2 + b^2 = c^2$, and if $a = \sqrt{5}$, $b = \sqrt{31}$, and $c = 6$, then $(\sqrt{5})^2 + (\sqrt{31})^2$ should equal 6^2, or 36. $(\sqrt{5})^2 + (\sqrt{31})^2 = 5 + 31$, or 36, which means those values for the side length measures will make the Pythagorean theorem true for this triangle.

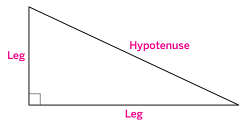
7. Can the Pythagorean theorem ever be true for an isosceles triangle? What about an equilateral triangle? Explain your thinking.
Hint: An isosceles triangle has two congruent sides and an equilateral triangle has three congruent sides.

Yes, for an isosceles triangle, but never for an equilateral triangle. *Explanations vary.* For example, for an isosceles triangle with legs of $\sqrt{2}$ and $\sqrt{2}$, the hypotenuse would be 2 because $(\sqrt{2})^2 + (\sqrt{2})^2 = 4$ and $4 = 2^2$. This would be an example of an isosceles triangle that works with the Pythagorean theorem. For an equilateral triangle, if all sides are equal, then $a = b = c$. If $a = b = c$, then it can never be the case that $a^2 + b^2 = c^2$, which means there can never be a right triangle that is also an equilateral triangle.

Name: ____________ Date: ________ Period: ________

Additional Practice ACC7.9.06

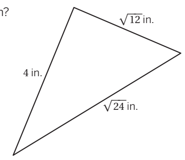
1. Label the legs and hypotenuse of the right triangle.



2. Show that the Pythagorean theorem is true for a right triangle with legs of 3 units and 4 units and a hypotenuse of 5 units.

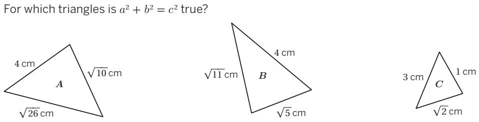
Work varies. **$3^2 + 4^2 = 25$ and $25 = 5^2$. Therefore, $3^2 + 4^2 = 5^2$, and the Pythagorean theorem is true for this right triangle.**

3. Is the Pythagorean theorem true for the triangle shown? Show or explain your thinking.



No. *Explanations vary.* The Pythagorean theorem states that $\text{leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$. If leg $= 4$, leg $= \sqrt{12}$, and hypotenuse $= \sqrt{24}$, then $4^2 + (\sqrt{12})^2$ should equal $(\sqrt{24})^2$ if the Pythagorean theorem is true for this triangle. $4^2 + (\sqrt{12})^2$ does not equal $(\sqrt{24})^2$, so the Pythagorean theorem is not true for this triangle.

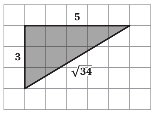
4. For which triangles is $a^2 + b^2 = c^2$ true?



Triangles A and B

Name: ____________ Date: ________ Period: ________

5. Select *all* the equations that represent the relationship between the side lengths of the triangle.



- ☑ A. $(\sqrt{34})^2 = 3^2 + 5^2$
- ☑ B. $5^2 + 3^2 = (\sqrt{34})^2$
- ☐ C. $3^2 + (\sqrt{34})^2 = 5^2$
- ☐ D. $(\sqrt{34})^2 + 5^2 = 3^2$
- ☑ E. $3^2 + 5^2 = (\sqrt{34})^2$

6. Han claims that if $a = \sqrt{5}$, $b = \sqrt{31}$, and $c = 6$, the Pythagorean theorem will be true for this triangle. Is Han correct? Explain your thinking.

Yes. *Explanations vary.* If $a^2 + b^2 = c^2$, and if $a = \sqrt{5}$, $b = \sqrt{31}$, and $c = 6$, then $(\sqrt{5})^2 + (\sqrt{31})^2$ should equal 6^2, or 36. $(\sqrt{5})^2 + (\sqrt{31})^2 = 5 + 31$, or 36, which means those values for the side length measures will make the Pythagorean theorem true for this triangle.

7. Can the Pythagorean theorem ever be true for an isosceles triangle? What about an equilateral triangle? Explain your thinking.
Hint: An isosceles triangle has two congruent sides and an equilateral triangle has three congruent sides.

Yes, for an isosceles triangle, but never for an equilateral triangle. *Explanations vary.* For example, for an isosceles triangle with legs of $\sqrt{2}$ and $\sqrt{2}$, the hypotenuse would be 2 because $(\sqrt{2})^2 + (\sqrt{2})^2 = 4$ and $4 = 2^2$. This would be an example of an isosceles triangle that works with the Pythagorean theorem. For an equilateral triangle, if all sides are equal, then $a = b = c$. If $a = b = c$, then it can never be the case that $a^2 + b^2 = c^2$, which means there can never be a right triangle that is also an equilateral triangle.

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B
2	2	8.G.B
3	2	8.G.B
4	2	8.G.B
5	2	8.G.B
6	2	8.G.B
7	3	8.G.B

Notes:

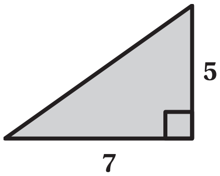
Name: .... Date: .... Period: ....

Additional Practice

ACC7.9.07

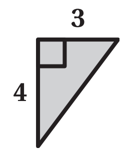
1. Determine the length of each unlabeled side.

a



$\sqrt{74}$ units (or equivalent)

b



5 units (or equivalent)

2. Write an equation that expresses the relationship between the lengths of the three sides of each right triangle.

a $\sqrt{5}$, $\sqrt{3}$, $\sqrt{8}$

$(\sqrt{5})^2 + (\sqrt{3})^2 = (\sqrt{8})^2$ (or equivalent)

b 1, $\sqrt{37}$, 6

$1^2 + 6^2 = (\sqrt{37})^2$ (or equivalent)

c 5, $\sqrt{5}$, $\sqrt{30}$

$5^2 + (\sqrt{5})^2 = (\sqrt{30})^2$ (or equivalent)

d 3, $\sqrt{2}$, $\sqrt{7}$

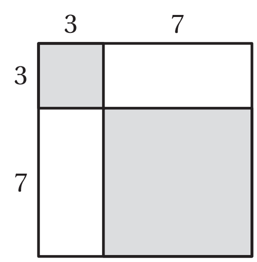
$(\sqrt{2})^2 + (\sqrt{7})^2 = 3^2$ (or equivalent)

3. Here are squares E and F.

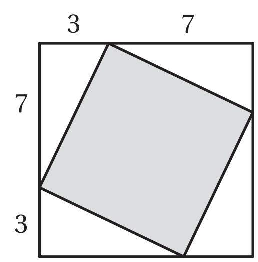
a Determine the total area of the shaded region for each square in square units.

Square E: 58 square units,
Square F: 58 square units

Square E



Square F

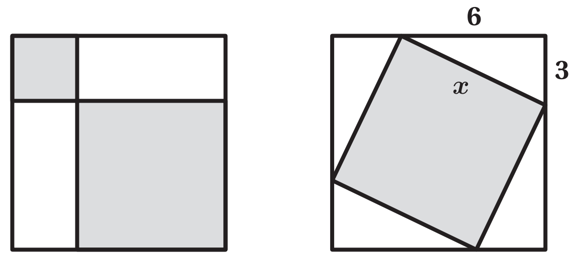


b Determine the exact side length of the shaded square inside square F.

$\sqrt{58}$ units

4. Determine the exact value of x.

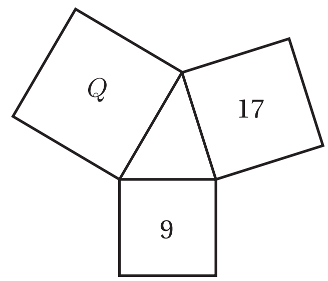
$\sqrt{45}$ units



5. The diagram shows an acute triangle and three squares.

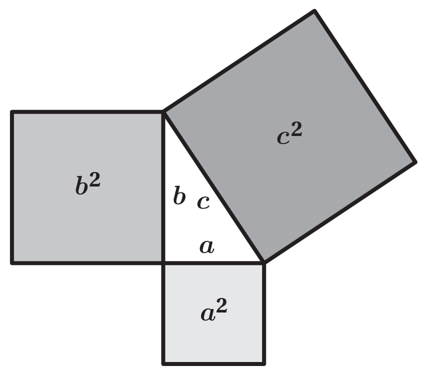
Tyler says the area of square Q is 26 square units because $9 + 17 = 26$. Is Tyler correct? Explain your thinking.

No. *Explanations vary.* The Pythagorean theorem only works for right triangles, and this is an acute triangle.



6. Determine a set of values for a, b, and c that make $a^2 + b^2 = c^2$ true.

Responses vary. $a = 3$, $b = 4$, and $c = 5$



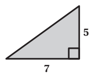
Additional Practice | Answer Key

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.07

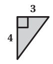
1. Determine the length of each unlabeled side.

a



$\sqrt{74}$ units (or equivalent)

b



5 units (or equivalent)

2. Write an equation that expresses the relationship between the lengths of the three sides of each right triangle.

a $\sqrt{5}, \sqrt{3}, \sqrt{8}$

$(\sqrt{5})^2 + (\sqrt{3})^2 = (\sqrt{8})^2$ (or equivalent)

b $1, \sqrt{37}, 6$

$1^2 + 6^2 = (\sqrt{37})^2$ (or equivalent)

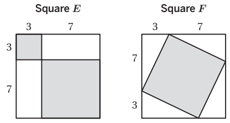
c $5, \sqrt{5}, \sqrt{30}$

$5^2 + (\sqrt{5})^2 = (\sqrt{30})^2$ (or equivalent)

d $3, \sqrt{2}, \sqrt{7}$

$(\sqrt{2})^2 + (\sqrt{7})^2 = 3^2$ (or equivalent)

3. Here are squares E and F.



a Determine the total area of the shaded region for each square in square units.

Square E: 58 square units,
Square F: 58 square units

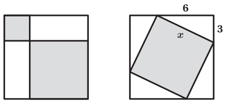
b Determine the exact side length of the shaded square inside square F.

$\sqrt{58}$ units

Name: .... Date: .... Period: ....

4. Determine the exact value of x.

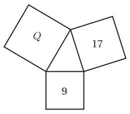
$\sqrt{45}$ units



5. The diagram shows an acute triangle and three squares.

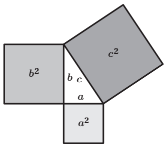
Tyler says the area of square Q is 26 square units because $9 + 17 = 26$. Is Tyler correct? Explain your thinking.

No. *Explanations vary.* The Pythagorean theorem only works for right triangles, and this is an acute triangle.



6. Determine a set of values for a, b, and c that make $a^2 + b^2 = c^2$ true.

Responses vary. $a = 3$, $b = 4$, and $c = 5$



Practice Problem Analysis

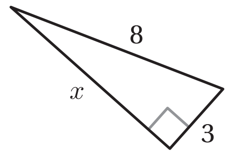
Problem	DOK	Standard(s)
1	1	8.G.B.7
2	1	8.G.B.7
3	2	8.G.B.6
4	2	8.G.B.6
5	2	8.G.B.6
6	2	8.G.B.6

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.08

1. Which equation could be used to solve for the unknown side length, x?



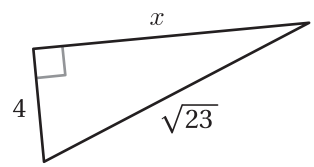
A. $3^2 - 8^2 = x^2$

B. $8^2 + x^2 = 3^2$

C. $8^2 + 3^2 = x^2$

D. $x^2 + 3^2 = 8^2$

2. Which equation could be used to solve for the unknown side length, x?



A. $4 + x = \sqrt{23}$

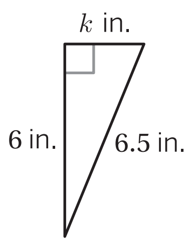
B. $4^2 + x^2 = (\sqrt{23})^2$

C. $4^2 + (\sqrt{23})^2 = x^2$

D. $(\sqrt{23})^2 + x^2 = 4^2$

3. Calculate the exact value of each variable that represents a side length in each right triangle. Show your thinking.

a

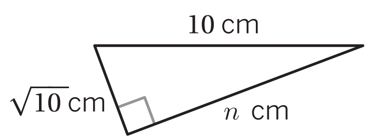


$k = 2.5$ inches (or equivalent)

Work varies.

$$6^2 + k^2 = 6.5^2$$
$$36 + k^2 = 42.25$$
$$k^2 = 6.25$$
$$k = 2.5$$

b



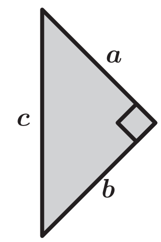
$n = \sqrt{90}$ centimeters (or equivalent)

Work varies.

$$(\sqrt{10})^2 + n^2 = 10^2$$
$$10 + n^2 = 100$$
$$n^2 = 90$$
$$n = \sqrt{90}$$

Name: .... Date: .... Period: ....

4. Here is a right triangle. Complete each equation to show three relationships among a, b, and c.

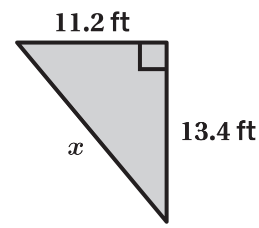


a $c^2 =$ $a^2 + b^2$ or $b^2 + a^2$

b $a^2 =$ $c^2 - b^2$

c $b^2 =$ $c^2 - a^2$

5. This diagram shows a right triangle. Which measurement is closest to the value of x, in feet?



A. 1.5 feet

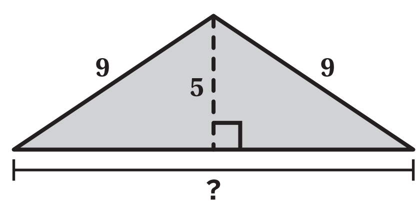
B. 2.2 feet

C. 17.5 feet

D. 30.5 feet

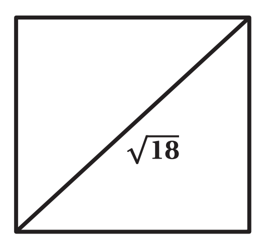
6. Calculate the value of the unknown side length.

15 units



7. What is the side length of this square?

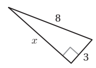
3 units



Name: .... Date: .... Period: ....

Additional Practice ACC7.9.08

1. Which equation could be used to solve for the unknown side length, x?



A. $3^2 - 8^2 = x^2$

B. $8^2 + x^2 = 3^2$

C. $8^2 + 3^2 = x^2$

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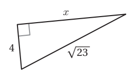
2. Which equation could be used to solve for the unknown side length, x?



A. $4 + x = \sqrt{23}$

B. $4^2 + x^2 = (\sqrt{23})^2$

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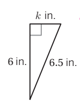
D. $(\sqrt{23})^2 + x^2 = 4^2$

3. Calculate the exact value of each variable that represents a side length in each right triangle. Show your thinking.

a



$k = 2.5$ inches (or equivalent)

Work varies.

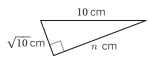
$6^2 + k^2 = 6.5^2$

$36 + k^2 = 42.25$

$k^2 = 6.25$

$k = 2.5$

b



$n = \sqrt{90}$ centimeters (or equivalent)

Work varies.

$(\sqrt{10})^2 + n^2 = 10^2$

$10 + n^2 = 100$

$n^2 = 90$

$n = \sqrt{90}$

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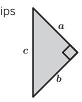
Name: .... Date: .... Period: ....

4. Here is a right triangle. Complete each equation to show three relationships among a, b, and c.



a $c^2 =$ $a^2 + b^2$ or $b^2 + a^2$

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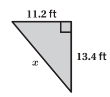


A. 1.5 feet

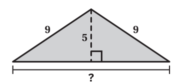
B. 2.2 feet

C. 17.5 feet

D. 30.5 feet

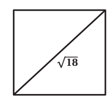
6. Calculate the value of the unknown side length.

15 units



7. What is the side length of this square?

3 units



Unit 9 Lesson 8 268 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B.7
2	1	8.G.B.7
3	1	8.G.B.7
4	2	8.G.B.7
5	1	8.G.B.7
6	2	8.G.B.7
7	2	8.G.B.7

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.9.09

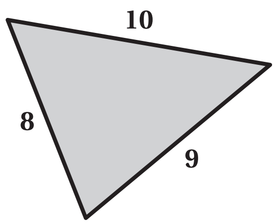
1. A triangle has side lengths $\sqrt{11}$, 2, and $\sqrt{15}$ such that $(\sqrt{11})^2 + 2^2 = (\sqrt{15})^2$. Is it a right triangle?

Yes

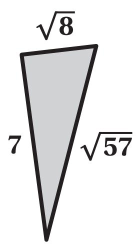
2. The lengths of two sides of a triangle are $\sqrt{7}$ and $\sqrt{32}$. Which side length would make the triangle a right triangle?

A. $\sqrt{40}$ **B.** $\sqrt{50}$ **C.** 4 **D.** 5

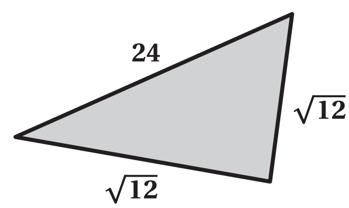
3. Determine whether each triangle is a right triangle.

a


No

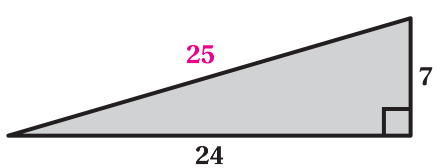
b


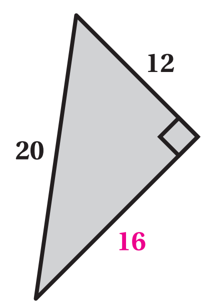
Yes

c


No

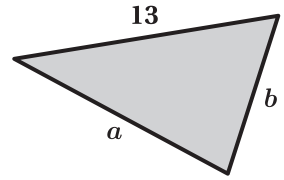
4. Calculate the value of the unknown side length so that each triangle is a right triangle.

a


b


Name: .... Date: .... Period: ....

5. The longest side of this triangle has a length of 13 millimeters. What could the lengths of the other two sides be so that this is a right triangle?



Responses vary.

- **Side a: 5 millimeters**
  Side b: 12 millimeters
- **Side a: 4 millimeters**
  Side b: $\sqrt{153}$ millimeters

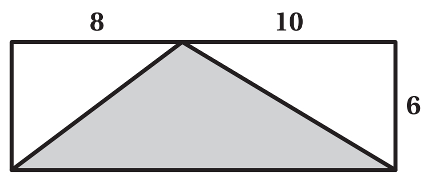
6. The side lengths of a triangle are 3, 6, and 10 units. Does the length of the longest side need to *increase*, *decrease*, or *stay the same* to make the triangle a right triangle? Circle one.

Increase   Decrease   Stay the same

Explain your thinking.

***Explanations vary.* $3^2 + 6^2 < 10^2$. The length of the longest side needs to decrease so that the square of its side length is equal to $3^2 + 6^2$, or 45.**

7. Here is a 18-by-6 rectangle divided into triangles. Is the shaded triangle a right triangle? Circle one.



Yes   No

Explain your thinking.

***Explanations vary.* I used the Pythagorean theorem to calculate the lengths of the legs of the shaded triangle, which are 10 and $\sqrt{136}$. The length of the longest side of the shaded triangle is 18 (the same as the length of the rectangle). For the triangle to be a right triangle, the side lengths must make the equation $a^2 + b^2 = c^2$ true. Because $136 + 100 = 236$, not 324, I know that the shaded triangle is not a right triangle.**

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.09

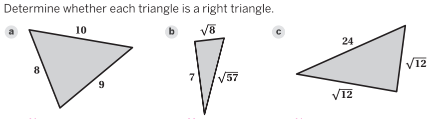
1. A triangle has side lengths $\sqrt{11}$, 2, and $\sqrt{15}$ such that $(\sqrt{11})^2 + 2^2 = (\sqrt{15})^2$. Is it a right triangle?

Yes

2. The lengths of two sides of a triangle are $\sqrt{7}$ and $\sqrt{32}$. Which side length would make the triangle a right triangle?

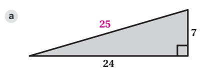
A. $\sqrt{40}$ B. $\sqrt{50}$ C. 4 D. 5

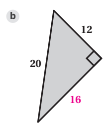
3. Determine whether each triangle is a right triangle.



a. **No** b. **Yes** c. **No**

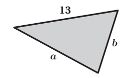
4. Calculate the value of the unknown side length so that each triangle is a right triangle.





Name: .... Date: .... Period: ....

5. The longest side of this triangle has a length of 13 millimeters. What could the lengths of the other two sides be so that this is a right triangle?



Responses vary.

- **Side a: 5 millimeters**
  Side b: 12 millimeters
- **Side a: 4 millimeters**
  Side b: $\sqrt{153}$ millimeters

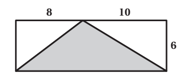
6. The side lengths of a triangle are 3, 6, and 10 units. Does the length of the longest side need to *increase*, *decrease*, or *stay the same* to make the triangle a right triangle? Circle one.

Increase Decrease Stay the same

Explain your thinking.

Explanations vary. **$3^2 + 6^2 < 10^2$. The length of the longest side needs to decrease so that the square of its side length is equal to $3^2 + 6^2$, or 45.**

7. Here is a 18-by-6 rectangle divided into triangles. Is the shaded triangle a right triangle? Circle one.



Yes No

Explain your thinking.

Explanations vary. **I used the Pythagorean theorem to calculate the lengths of the legs of the shaded triangle, which are 10 and $\sqrt{136}$. The length of the longest side of the shaded triangle is 18 (the same as the length of the rectangle). For the triangle to be a right triangle, the side lengths must make the equation $a^2 + b^2 = c^2$ true. Because $136 + 100 = 236$, not 324, I know that the shaded triangle is not a right triangle.**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B, 8.G.B.6
2	1	8.G.B, 8.G.B.6
3	2	8.G.B, 8.G.B.6
4	2	8.G.B, 8.G.B.6
5	2	8.G.B, 8.G.B.6
6	2	8.G.B, 8.G.B.6
7	3	8.G.A.3, 8.G.B.6

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.10

1. Television screens are classified by the length of their diagonal. If a television screen is 22.5 inches tall and 40 inches wide, what is the approximate length of its diagonal in inches? Show your thinking.

46 inches. ***Work varies.***

$$22.5^2 + 40^2 = x^2$$
$$506.25 + 1600 = x^2$$
$$2106.25 = x^2$$
$$\sqrt{2106.25} = x$$
$$45.9 \approx x$$

Rounded to the nearest inch, $x \approx 46$.

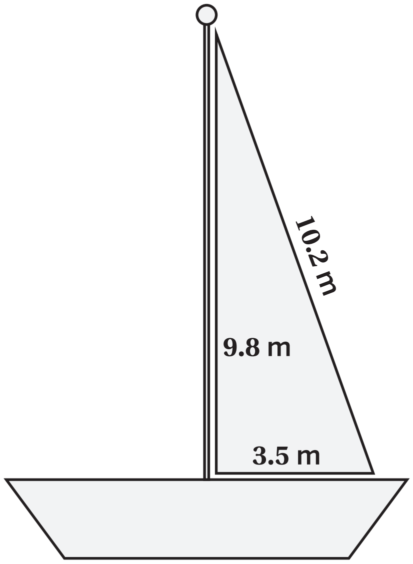
2. Lin leaves her house for a jog. She jogs 4 miles directly north, and then 3 miles directly west. If Lin wants to return home, what is the shortest distance she can travel directly back to her house? Show your thinking.

5 miles. ***Work varies.***

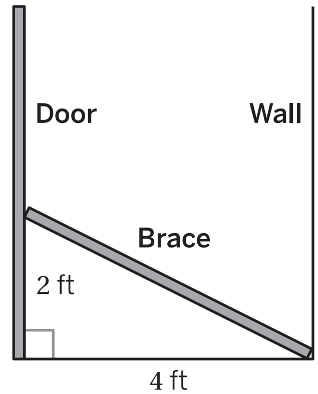
$$4^2 + 3^2 = x^2$$
$$16 + 9 = x^2$$
$$25 = x^2$$
$$5 = x$$

3. Sails come in many shapes and sizes. Is the sail shown a right triangle? Show or explain your thinking.

No. ***Explanations vary.*** **10.2 is the longest side.**
$3.5^2 + 9.8^2 = 108.29$, and 10.2^2 is equal to 104.04.
It is not a right triangle.



4. A carpenter cuts a length of wood that will brace a door against a wall. The wall is 4 feet away from the door, and she wants the brace to rest on the door, 2 feet above the floor. About how long should she cut the brace, to the nearest tenth of a foot? Show or explain your thinking.



About 4.5 feet. *Work varies.*

$2^2 + 4^2 = x^2$

$4 + 16 = x^2$

$20 = x^2$

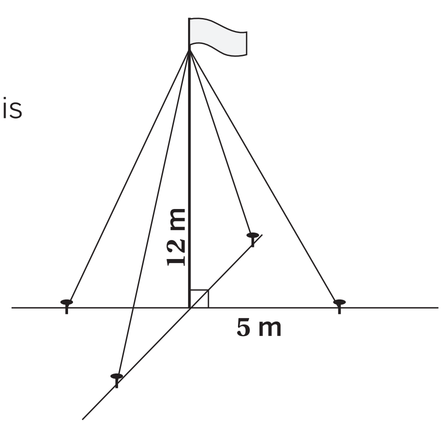
$\sqrt{20} = x$

$4.5 \approx x$

5. Jada is building a marble run track. She wants to create a straight path from the top of one section to the top of a second section. The height of the first section is 8 inches, and the height of the second section is 4 inches. If the distance between the bottom of the two sections is 10 inches, about how long should the connected path be, to the nearest tenth of an inch? Show or explain your thinking.

About 10.8 inches. *Explanations vary*. The difference in heights of the sections is $8 - 4 = 4$ inches, so a right triangle with legs that are 4 inches and 10 inches is created at the top of the track. I used the Pythagorean theorem to get $4^2 + 10^2 = x^2$, and determined the path to be about 10.8 inches long.

6. Four cables are used to mount a 12-meter post. Each cable is mounted at the top of the post and extends to the ground, 5 meters from the bottom of the post. What is the total amount of cable needed to mount the post? Show or explain your thinking.



52 meters. *Explanations vary*. First, I determined the length of one cable. Each cable represents the hypotenuse of a right triangle with legs that are 5 meters and 12 meters. Then, I used the Pythagorean theorem, $5^2 + 12^2 = x^2$, to determine one cable length of 13 meters. Finally, to determine the length for the four cables, I multiplied the length of one cable by four: $4 \cdot 13 = 52$.

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.10

1. Television screens are classified by the length of their diagonal. If a television screen is 22.5 inches tall and 40 inches wide, what is the approximate length of its diagonal in inches? Show your thinking.

46 inches. ***Work varies.***

$22.5^2 + 40^2 = x^2$

$506.25 + 1600 = x^2$

$2106.25 = x^2$

$\sqrt{2106.25} = x$

$45.9 \approx x$

Rounded to the nearest inch, $x \approx 46$.

2. Lin leaves her house for a jog. She jogs 4 miles directly north, and then 3 miles directly west. If Lin wants to return home, what is the shortest distance she can travel directly back to her house? Show your thinking.

5 miles. ***Work varies.***

$4^2 + 3^2 = x^2$

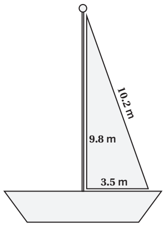
$16 + 9 = x^2$

$25 = x^2$

$5 = x$

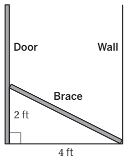
3. Sails come in many shapes and sizes. Is the sail shown a right triangle? Show or explain your thinking.

No. ***Explanations vary.*** **10.2 is the longest side. $3.5^2 + 9.8^2 = 108.29$, and 10.2^2 is equal to 104.04. It is not a right triangle.**



Name: .... Date: .... Period: ....

4. A carpenter cuts a length of wood that will brace a door against a wall. The wall is 4 feet away from the door, and she wants the brace to rest on the door, 2 feet above the floor. About how long should she cut the brace, to the nearest tenth of a foot? Show or explain your thinking.



About 4.5 feet. ***Work varies.***

$2^2 + 4^2 = x^2$

$4 + 16 = x^2$

$20 = x^2$

$\sqrt{20} = x$

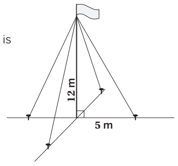
$4.5 \approx x$

5. Jada is building a marble run track. She wants to create a straight path from the top of one section to the top of a second section. The height of the first section is 8 inches, and the height of the second section is 4 inches. If the distance between the bottom of the two sections is 10 inches, about how long should the connected path be, to the nearest tenth of an inch? Show or explain your thinking.

About 10.8 inches. ***Explanations vary.*** **The difference in heights of the sections is $8 - 4 = 4$ inches, so a right triangle with legs that are 4 inches and 10 inches is created at the top of the track. I used the Pythagorean theorem to get $4^2 + 10^2 = x^2$, and determined the path to be about 10.8 inches long.**

6. Four cables are used to mount a 12-meter post. Each cable is mounted at the top of the post and extends to the ground, 5 meters from the bottom of the post. What is the total amount of cable needed to mount the post? Show or explain your thinking.

52 meters. ***Explanations vary.*** **First, I determined the length of one cable. Each cable represents the hypotenuse of a right triangle with legs that are 5 meters and 12 meters. Then, I used the Pythagorean theorem, $5^2 + 12^2 = x^2$, to determine one cable length of 13 meters. Finally, to determine the length for the four cables, I multiplied the length of one cable by four: $4 \cdot 13 = 52$.**



Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B.7
2	1	8.G.B.7
3	2	8.G.B.7
4	2	8.G.B.7
5	2	8.G.B.7
6	3	8.G.B.7

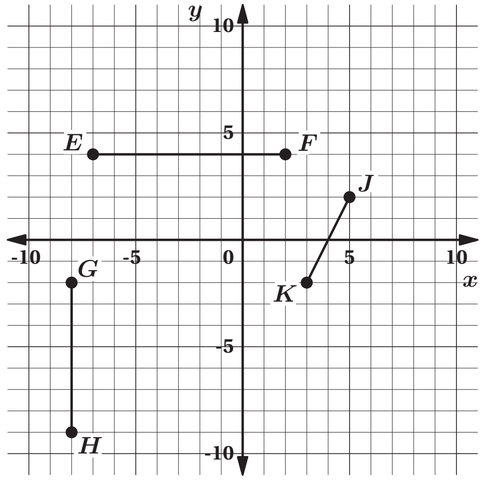
Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.9.11

1. Determine the exact length of each line segment.



a) Line segment EF

9 units

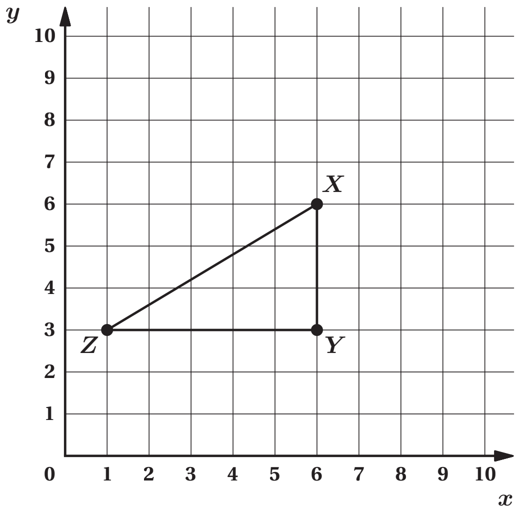
b) Line segment GH

7 units

c) Line segment JK

$\sqrt{20}$ (or equivalent) units

2. Determine the exact length of segment XZ. Show or explain your thinking.



$\sqrt{34}$ (or equivalent) units. ***Work varies.***

leg² + leg² = hypotenuse²

$$3^2 + 5^2 = x^2$$

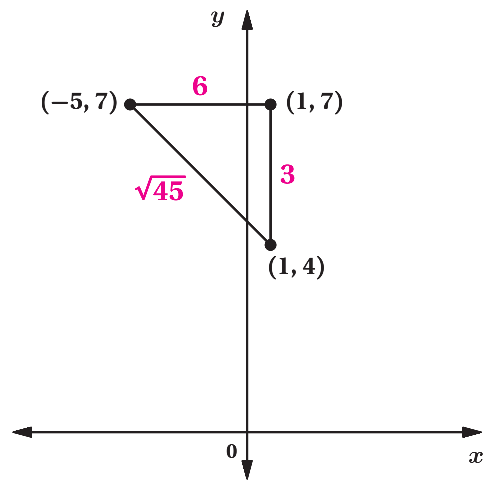
$$9 + 25 = x^2$$

$$34 = x^2$$

$$\sqrt{34} = x$$

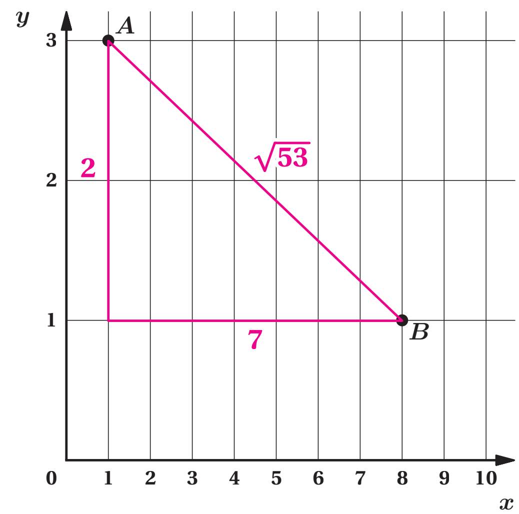
Name: .... Date: .... Period: ....

3. A right triangle is drawn on the coordinate plane, and the coordinates of its vertices are labeled. Label each side of the triangle with its exact length.



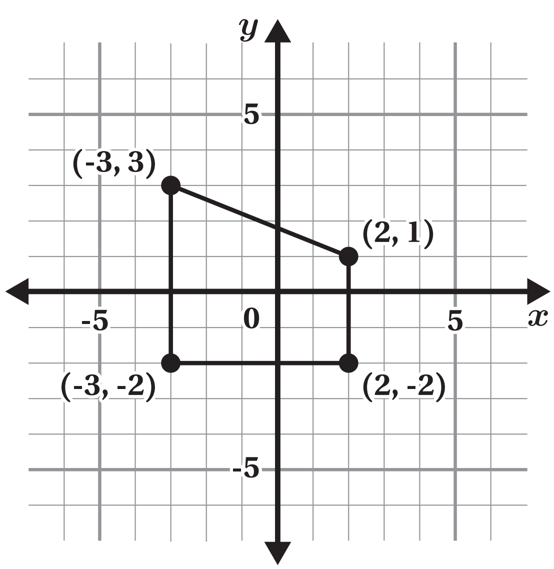
4. Determine the exact distance from point A to point B. Show or explain your thinking.

$\sqrt{53}$ units. *Work varies*. Sample shown on graph.



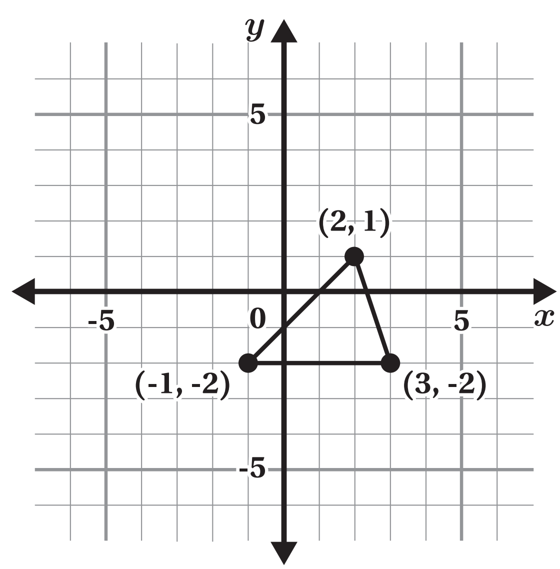
Problems 5–6: Calculate the perimeter of each polygon.

5.



$13 + \sqrt{29}$ or about 18.4 units

6.



$4 + \sqrt{10} + \sqrt{18}$ or about 11.4 units

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.11

1. Determine the exact length of each line segment.

   a. Line segment EF

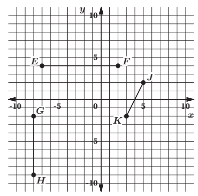
   9 units

   b. Line segment GH

   7 units

   c. Line segment JK

   $\sqrt{20}$ (or equivalent) units



2. Determine the exact length of segment XZ. Show or explain your thinking.

   $\sqrt{34}$ (or equivalent) units. *Work varies.*

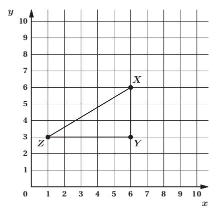
   $\text{leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$

   $3^2 + 5^2 = x^2$

   $9 + 25 = x^2$

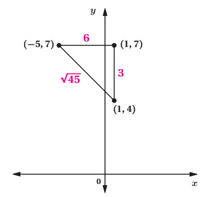
   $34 = x^2$

   $\sqrt{34} = x$



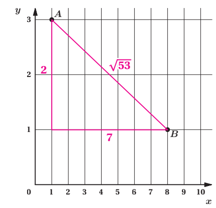
Name: .... Date: .... Period: ....

3. A right triangle is drawn on the coordinate plane, and the coordinates of its vertices are labeled. Label each side of the triangle with its exact length.



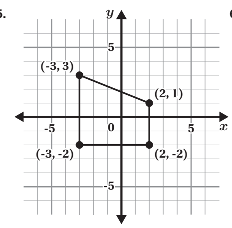
4. Determine the exact distance from point A to point B. Show or explain your thinking.

   $\sqrt{53}$ units. *Work varies.* Sample shown on graph.



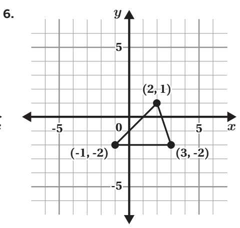
Problems 5–6: Calculate the perimeter of each polygon.

5.



$13 + \sqrt{29}$ or about 18.4 units

6.



$4 + \sqrt{10} + \sqrt{18}$ or about 11.4 units

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.G.B.8
2	1	8.G.B.8
3	1	8.G.B.8
4	2	8.G.B.8
5	2	8.G.B.8
6	2	8.G.B.8

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.12

1. Write each fraction as a decimal.

a $\frac{3}{5}$ **0.6**

b $-\frac{23}{100}$ **−0.23**

c $\frac{7}{45}$ **$0.1\overline{5}$**

2. Determine whether the decimal representation of each fraction will repeat or terminate. Circle your response.

a $\frac{1}{20}$ Terminate Repeat

b $\frac{1}{11}$ Terminate Repeat

c $\frac{1}{15}$ Terminate Repeat

3. Which decimal is equivalent to $\frac{4}{27}$?

A. $0.\overline{148}$

B. 0.148

C. $0.14\overline{8}$

D. $\overline{0.148}$

Name: .... Date: .... Period: ....

Problems 4–5: Determine whether the decimal representation of each fraction will repeat or terminate. Explain your thinking

4. $\frac{1}{8}$

Terminate. *Explanations vary.* $\frac{1}{8} = \frac{1}{2 \cdot 2 \cdot 2}$. The denominator only includes factors of 2, so the decimal will terminate.

5. $\frac{1}{18}$

Repeat. *Explanations vary.* $\frac{1}{18} = \frac{1}{2 \cdot 3 \cdot 3}$. The denominator includes factors other than 2 or 5, so it will repeat.

6. Order these numbers from *least* to *greatest*.

$0.\overline{4}$	$0.4\overline{1}$	$0.\overline{41}$	$4.\overline{04}$	0.44

$\mathbf{0.4\overline{1}}$	$\mathbf{0.\overline{41}}$	$\mathbf{0.44}$	$\mathbf{0.\overline{4}}$	$\mathbf{4.\overline{04}}$

Least **Greatest**

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.12

1. Write each fraction as a decimal.
   - a $\frac{3}{5}$ **0.6**
   - b $-\frac{23}{100}$ **-0.23**
   - c $\frac{7}{45}$ **$0.1\overline{5}$**

2. Determine whether the decimal representation of each fraction will repeat or terminate. Circle your response.
   - a $\frac{1}{20}$ Terminate Repeat
   - b $\frac{1}{11}$ Terminate Repeat
   - c $\frac{1}{15}$ Terminate Repeat

3. Which decimal is equivalent to $\frac{4}{27}$?

   A. $0.\overline{148}$ B. 0.148 C. $0.14\overline{8}$ D. $\overline{0.148}$

Name: .... Date: .... Period: ....

Problems 4–5: Determine whether the decimal representation of each fraction will repeat or terminate. Explain your thinking

4. $\frac{1}{8}$

   Terminate. *Explanations vary.* $\frac{1}{8} = \frac{1}{2 \cdot 2 \cdot 2}$. The denominator only includes factors of 2, so the decimal will terminate.

5. $\frac{1}{18}$

   Repeat. *Explanations vary.* $\frac{1}{18} = \frac{1}{2 \cdot 3 \cdot 3}$. The denominator includes factors other than 2 or 5, so it will repeat.

6. Order these numbers from *least* to *greatest*.

$0.\overline{4}$	$0.4\overline{1}$	$0.\overline{41}$	$4.\overline{04}$	0.44
$0.4\overline{1}$	**$0.\overline{41}$**	**0.44**	**$0.\overline{4}$**	**$4.\overline{04}$**

Least **Greatest**

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	8.NS.A.1
2	2	8.NS.A.1
3	2	8.NS.A.1
4	2	8.NS.A.1
5	2	8.NS.A.1
6	2	8.NS.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice ACC7.9.13

1. Write each decimal as a fraction.

a 0.4 $\frac{4}{10}$ (or equivalent)

b $0.\overline{4}$ $\frac{4}{9}$

2. Which fraction is equivalent to $0.\overline{23}$?

A. $\frac{2}{3}$

B. $\frac{23}{99}$

C. $\frac{23}{100}$

D. $\frac{2}{9}$

E. $\frac{3}{2}$

3. Complete the table.

Fraction	Decimal
$\frac{14}{5}$	**2.8**
$\frac{2}{9}$	$0.\overline{2}$
$-\frac{51}{99}$	$-0.\overline{51}$
$-2\frac{67}{100}$	-2.67
$\frac{82}{90}$ (or equivalent)	$0.9\overline{1}$

4. Here are the numbers 0.333 and $0.\overline{3}$.

a How are the numbers alike?

Responses vary. **They are both decimals between 0.3 and 0.4, and the first three digits in their decimal expansions are the same.**

b How are the numbers different?

Responses vary.

- **$0.\overline{3}$ is greater than 0.333 because it has a greater digit in the ten-thousandths place.**
- **0.333 is a terminating decimal, while $0.\overline{3}$ is an infinitely repeating decimal.**

Name: .... Date: .... Period: ....

5. Match each fraction with its decimal representation.

Decimal	Fraction
a. 0.07	a $\frac{7}{100}$
b. $0.0\overline{7}$	c $\frac{5}{9}$
c. $0.\overline{5}$	b $\frac{7}{90}$
d. $0.\overline{05}$	d $\frac{5}{99}$

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.

a. A repeating decimal is a rational number. Always Sometimes Never

b. If the digits in the decimal expansion do not repeat and do not terminate, the number is rational. Always Sometimes Never

c. A non-terminating decimal can be written as a fraction. Always Sometimes Never

7. Here are the fractions $\frac{3}{7}$ and $\frac{4}{7}$.

a. Calculate the sum $\frac{3}{7} + \frac{4}{7}$. **$\frac{7}{7}$ or 1**

b. Write $\frac{3}{7}$ as a decimal. **$0.\overline{428571}$**

c. Write $\frac{4}{7}$ as a decimal. **$0.\overline{571428}$**

d. Calculate the sum of the decimal representations. What do you notice?

$0.\overline{428571} + 0.\overline{571428} = 0.\overline{999999}$. *Responses vary.* I notice the decimal and fraction representations do not produce the same sum.

Name: ______ Date: ______ Period: ______

Additional Practice ACC7.9.13

1. Write each decimal as a fraction.
   - a. 0.4 $\frac{4}{10}$ (or equivalent)
   - b. $0.\overline{4}$ $\frac{4}{9}$

2. Which fraction is equivalent to $0.\overline{23}$?

   A. $\frac{2}{3}$ B. $\frac{23}{99}$ C. $\frac{23}{100}$

   D. $\frac{2}{9}$ E. $\frac{3}{2}$

3. Complete the table.

Fraction	Decimal
$\frac{14}{5}$	2.8
$\frac{2}{9}$	$0.\overline{2}$
$-\frac{51}{99}$	$-0.\overline{51}$
$-2\frac{67}{100}$	-2.67
$\frac{82}{90}$ (or equivalent)	$0.9\overline{1}$

4. Here are the numbers 0.333 and $0.\overline{3}$.
   - a. How are the numbers alike?

     Responses vary. They are both decimals between 0.3 and 0.4, and the first three digits in their decimal expansions are the same.
   - b. How are the numbers different?

     Responses vary.
     - $0.\overline{3}$ is greater than 0.333 because it has a greater digit in the ten-thousandths place.
     - 0.333 is a terminating decimal, while $0.\overline{3}$ is an infinitely repeating decimal.

Unit 9 Lesson 13 277

Name: ______ Date: ______ Period: ______

5. Match each fraction with its decimal representation.

Decimal	Fraction
a. 0.07	a $\frac{7}{100}$
b. $0.0\overline{7}$	c $\frac{5}{9}$
c. $0.\overline{5}$	b $\frac{7}{90}$
d. $0.\overline{05}$	d $\frac{5}{99}$

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.
   - a. A repeating decimal is a rational number. Always Sometimes Never
   - b. If the digits in the decimal expansion do not repeat and do not terminate, the number is rational. Always Sometimes Never
   - c. A non-terminating decimal can be written as a fraction. Always Sometimes Never

7. Here are the fractions $\frac{3}{7}$ and $\frac{4}{7}$.
   - a. Calculate the sum $\frac{3}{7}+\frac{4}{7}$. $\frac{7}{7}$ or 1
   - b. Write $\frac{3}{7}$ as a decimal. $0.\overline{428571}$
   - c. Write $\frac{4}{7}$ as a decimal. $0.\overline{571428}$
   - d. Calculate the sum of the decimal representations. What do you notice?

     $0.\overline{428571}+0.\overline{571428}=0.\overline{999999}$. *Responses vary.* I notice the decimal and fraction representations do not produce the same sum.

Unit 9 Lesson 13 278 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.NS.A.1
2	1	8.NS.A.1
3	2	8.NS.A.1
4	2	8.NS.A.1
5	2	8.NS.A.1
6	2	8.NS.A.1
7	3	8.NS.A.1

Notes:

Name: .... Date: .... Period: ....

Additional Practice

ACC7.9.14

1. Write three examples of *rational numbers*, including at least one number written as a square root or cube root.

   Responses vary. $\frac{1}{2}$, $\sqrt{25}$, 3.5

2. Write three examples of *irrational numbers*, including at least one number written as a square root or cube root.

   Responses vary. $\sqrt[3]{13}$, π, $\sqrt{2}$

3. Write each number as a fraction of two integers. If it is impossible, write "irrational number."

   a. 0.12 $\frac{12}{100}$ **(or equivalent)**

   b. $\sqrt[3]{27}$ $\frac{3}{1}$ **(or equivalent)**

   c. $\sqrt{73}$ **Irrational number**

   d. $-\sqrt{49}$ $-\frac{7}{1}$ **(or equivalent)**

   e. $\sqrt[3]{11}$ **Irrational number**

4. Han says that the solution to $x^2 = 50$ is a rational number. Is Han correct or incorrect? Explain your thinking.

   Han is incorrect. *Explanations vary.* The solution to the equation $x^2 = 50$ is $x = \sqrt{50}$, and 50 is not a perfect square, so the solution is an irrational number.

5. Sort the numbers based on whether they are rational or irrational.

$\sqrt{9}$	$\sqrt[3]{29}$	0.123	$-\frac{2}{13}$	$\sqrt{8}$	$\sqrt[3]{125}$

Rational	Irrational
$\sqrt{9}, 0.123, -\frac{2}{13}, \sqrt[3]{125}$	$\sqrt[3]{29}, \sqrt{8}$

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.

a. Perfect squares are irrational numbers. Always Sometimes Never

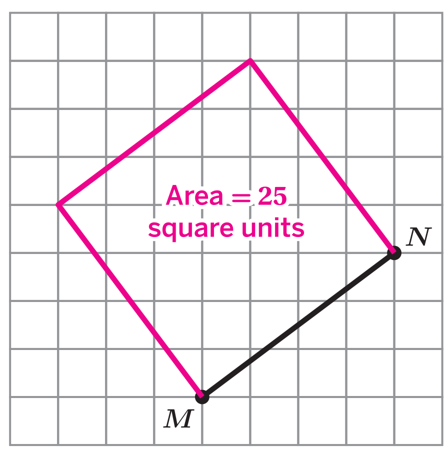
b. The square root of a number is an irrational number. Always Sometimes Never

c. Irrational numbers can be written as fractions. Always Sometimes Never

d. The cube root of a perfect cube is a rational number. Always Sometimes Never

7. Is the length of line segment MN a rational or irrational number? Explain your thinking.

Rational number. *Explanations vary.* I drew a square using the line segment as the side length of the square. I determined the area of the square, 25 square units, which means that the side length is $\sqrt{25}$ units. Because $\sqrt{25}$ can be written as the fraction $\frac{5}{1}$, it is a rational number.



Name: ______ Date: ______ Period: ______

Additional Practice ACC7.9.14

1. Write three examples of *rational numbers*, including at least one number written as a square root or cube root.

   Responses vary. $\frac{1}{2}, \sqrt{25}, 3.5$

2. Write three examples of *irrational numbers*, including at least one number written as a square root or cube root.

   Responses vary. $\sqrt[3]{13}, \pi, \sqrt{2}$

3. Write each number as a fraction of two integers. If it is impossible, write "irrational number."

   a. 0.12 — $\frac{12}{100}$ (or equivalent)

   b. $\sqrt[3]{27}$ — $\frac{3}{1}$ (or equivalent)

   c. $\sqrt{73}$ — Irrational number

   d. $-\sqrt{49}$ — $-\frac{7}{1}$ (or equivalent)

   e. $\sqrt[3]{11}$ — Irrational number

4. Han says that the solution to $x^2 = 50$ is a rational number. Is Han correct or incorrect? Explain your thinking.

   Han is incorrect. *Explanations vary.* The solution to the equation $x^2 = 50$ is $x = \sqrt{50}$, and 50 is not a perfect square, so the solution is an irrational number.

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5. Sort the numbers based on whether they are rational or irrational.

   $\sqrt{9}$ $\sqrt[3]{29}$ 0.123 $-\frac{2}{13}$ $\sqrt{8}$ $\sqrt[3]{125}$

Rational	Irrational
$\sqrt{9}, 0.123, -\frac{2}{13}, \sqrt[3]{125}$	$\sqrt[3]{29}, \sqrt{8}$

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.

   a. Perfect squares are irrational numbers. Always Sometimes Never

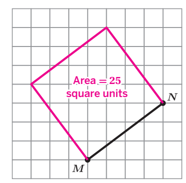
   b. The square root of a number is an irrational number. Always Sometimes Never

   c. Irrational numbers can be written as fractions. Always Sometimes Never

   d. The cube root of a perfect cube is a rational number. Always Sometimes Never

7. Is the length of line segment MN a rational or irrational number? Explain your thinking.

   Rational number. *Explanations vary.* I drew a square using the line segment as the side length of the square. I determined the area of the square, 25 square units, which means that the side length is $\sqrt{25}$ units. Because $\sqrt{25}$ can be written as the fraction $\frac{5}{1}$, it is a rational number.



Unit 9 Lesson 14 280 Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	8.NS.A
2	1	8.NS.A
3	2	8.NS.A.1
4	2	8.NS.A.1
5	1	8.EE.A.2, 8.NS.A
6	3	8.NS.A
7	3	8.EE.A.2, 8.NS.A.1

Notes: